

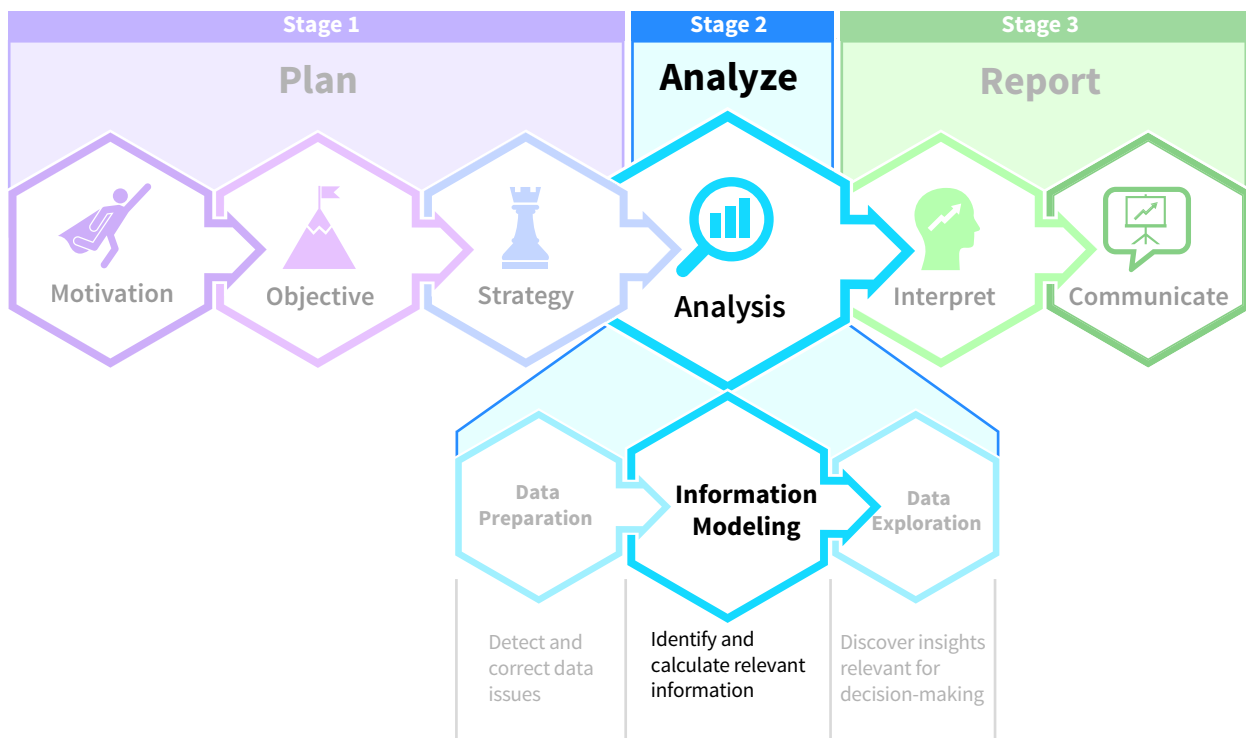
Analysis: Information Modeling

Chapter Preview

After planning the project and completing the data preparation, you are probably excited to finally explore the data and generate insights. While the data may be ready for analysis, there is still a missing piece—the information necessary to perform the analysis. For example, the data selected and prepared for analysis might include a set of transactions, but without knowing the total revenue or profit generated, they cannot be analyzed to determine which products are profitable.

There are two important information modeling tasks to complete before data exploration. First, you must understand what information is required for analysis and what data are needed to produce that information. You did this in the planning stage when you developed a strategy to select the best data and analysis to answer accounting questions. The second task is identifying the specific measures you want to use for analysis based on the prepared data set, and then writing the code that generates this information.

Information modeling is a process. This chapter will help you develop the skills to do it successfully by offering a menu of patterns for common coding structures and for identifying information used in accounting analytics.



Professional Insight: Why is Information Modeling Important?

After earning his accounting degree, Dan started working as a tax consultant for one of the Big Four public accounting firms, which is where he fell in love with data and data analytics. More recently, he moved to a multinational telecommunications corporation where he works as a data analyst.

As a data analyst, I must be versatile. I process raw data sets from multiple sources and am heavily involved in ETL (extract-transform-load). I also explore data to generate insights for stakeholders across multiple departments and develop dashboards. **Continuously facing new challenges is what I really like about my job.** Two days ago, I worked on processing unstructured data (text), yesterday I coded a new algorithm, and today I am working on a dashboard that presents an interrelated set of KPIs.

While I spend most of my time on ETL, I think **information modeling is the heart of data analytics**. It involves coding the calculations, testing whether they are correct, and making sure the information necessary for the different business cases is available. **This is where my accounting degree pays off, as I can understand the business context.**

Chapter Roadmap

LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 6.1 Describe the foundational concepts of information modeling.	- The Information Modeling Process - A Structured Approach	Complete a Star Schema (Example: Accounting Information Systems and Financial Accounting)
LO 6.2 Apply common information modeling algorithms.	Seven Information Modeling Patterns for Algorithms	Use Algorithms to Calculate Net Revenue (Example: Financial Accounting)
LO 6.3 Develop and implement information models for common accounting data structures.	Six Information Modeling Patterns for Common Accounting Data Structures	Answer Questions with a Star Schema (Example: Managerial Accounting)

Data The Data tag appears in the chapter when the data for an example, illustration, or application are available on Wiley's online learning platform.

Data analytics software is continuously changing, and there may be more recent versions of the software referenced in this chapter. For more information access the accompanying video on Wiley's online learning platform.

6.1 What is Information Modeling?

LEARNING OBJECTIVE 1

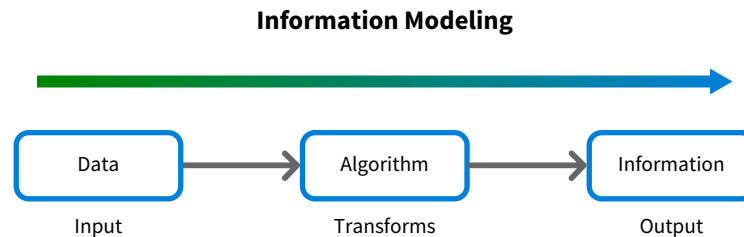
Describe the foundational concepts of information modeling.

Information modeling is the process of generating additional knowledge from data that is relevant for analysis purposes. Next, we examine how this is performed in an accounting context.

The Information Modeling Process

In information modeling, data are the input. They are the raw figures and facts. **Algorithms** are sets of instructions that transform the data into **information**, which is the output of additional knowledge gained from the data (**Illustration 6.1**).

ILLUSTRATION 6.1 The Information Modeling Process



Algorithms are the link between the input of facts and the output of useful information. Keep in mind that data (input) for one application can also be information (output) for another application. One example is how financial statements are information (output) for financial statement preparers, but they are data (input) for financial analysts.

Algorithms are intrinsic to the work of accountants. These instructions calculate depreciation, cost, financial ratios, and more. They range from simple to complex and can be coded with a host of languages, including Excel functions, Power BI's Data Analysis Expressions (DAX) language, and the Python programming language. In this chapter we will rely on general descriptions of the algorithms.

The first seven columns in **Illustration 6.2** are data fields. They are factual descriptions about sales transactions, such as a transaction's identification number and the date it occurred. In accounting these descriptions can range from the social security numbers of employees to the names of customers who have made payments. The last two columns in **Illustration 6.2** contain information that was calculated with an algorithm.

ILLUSTRATION 6.2 Data, Information, and Algorithms for Sales Transactions

IF the NetAmount of a transaction is smaller than \$1,000,
THEN the TransactionSize is S (small),
ELSE
IF the NetAmount of a transaction is at least \$1,000 but smaller than \$2,500,
THEN the TransactionSize is M (medium),
ELSE the transaction size is L (large).

Algorithm for TransactionSize

Amount - Discount

Algorithm for NetAmount

ID	Date	Product	Amount	Discount	Customer	SalesPerson	NetAmount	TransactionSize
1	1/3/2022	Dishwasher	\$ 825	0	Jeremy	Ed	\$ 825	S
2	1/3/2022	Refrigerator	\$ 2,500	100	Jane	Juana	\$ 2,400	M
3	1/3/2022	Refrigerator	\$ 2,500	250	Molly	Hakeem	\$ 2,250	M
4	1/4/2022	Cooktop	\$ 1,250	75	Whitney	Ed	\$ 1,175	M
5	1/4/2022	Dishwasher	\$ 900	125	Zuzu	Juana	\$ 775	S
6	1/4/2022	Cooktop	\$ 1,500	250	Carmen	Cindy	\$ 1,250	M
7	1/4/2022	Oven	\$ 1,500	150	Ben	Hakeem	\$ 1,350	M
8	1/4/2022	Cooktop	\$ 1,000	50	Lebron	Juana	\$ 950	S
9	1/4/2022	Cooktop	\$ 900	0	Trisha	Hakeem	\$ 900	S
10	1/5/2022	Dishwasher	\$ 500	50	Xuomin	Cindy	\$ 450	S

Data

Information

There are two types of information fields:

- In Illustration 6.2, both NetAmount and TransactionSize are a **calculated column** in the SalesTransactions table because a value is calculated for each cell in the columns. They are integral parts of the table.
- The second type of information field is a **measure**, which is an aggregate, or total, that can be used in reports, and thus for analytical purposes. Measures are created by algorithms, but they are not integral parts of a table.

Two measures that could be calculated for the table in Illustration 6.2 are the total net amount generated by sales and the number of large transactions. The algorithms for these measures are shown in **Illustration 6.3**.

ILLUSTRATION 6.3 Algorithms to Calculate Total Net Amount and Number of Large Transactions

Measure	Algorithm
TotalNetAmount	Sum of all the values in the NetAmount column.
NumberOfLargeTransactions	Count the number of cells in the TransactionSize column that have the value "L."

Measures are at the center of data analytics. They can be calculated and then sliced in many ways during data exploration. Data analytics software, such as Power BI and Tableau, make it easy to create and slice them.

A Structured Approach

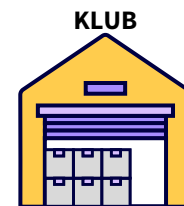
Performing data analysis requires determining the numbers to analyze (these are the measures) and how to analyze them (these are the **dimensions**, or the fields that can slice them). This chapter describes a structured approach to developing accounting information models by defining dimensions of *who*, *what*, and *when* in transactions to help make accounting sense of data. A measure such as total revenue can be broken down by the following dimensions:

- Customer (*who?*).
- Product category (*what?*).
- Year (*when?*).

Dimensions like these, and the measures they can slice, are typically configured in a star schema. (As you learned earlier in this course, star schemas are the preferred data structures for data analysis.) Let's use an example to help illustrate this. **Data** KLUB is a retail store located in Interlochen, MI:

- KLUB's owner uses business connections in different retail spaces to negotiate bulk purchases at heavily discounted prices and signs no-return policies with all vendors to further lower the prices.
- The company sells the items at prices lower than its competitors, but still with a substantial profit margin.
- The owner handles all negotiations, contracts, and deliveries with vendors.
- Salespeople are contractors who are paid a commission on sold goods.

KLUB's website shows currently available items, but sales are exclusively made over the phone. The company regularly sends marketing emails to a list of 100,000 potential customers. When customers want to buy something, they email the company and a salesperson contacts them.

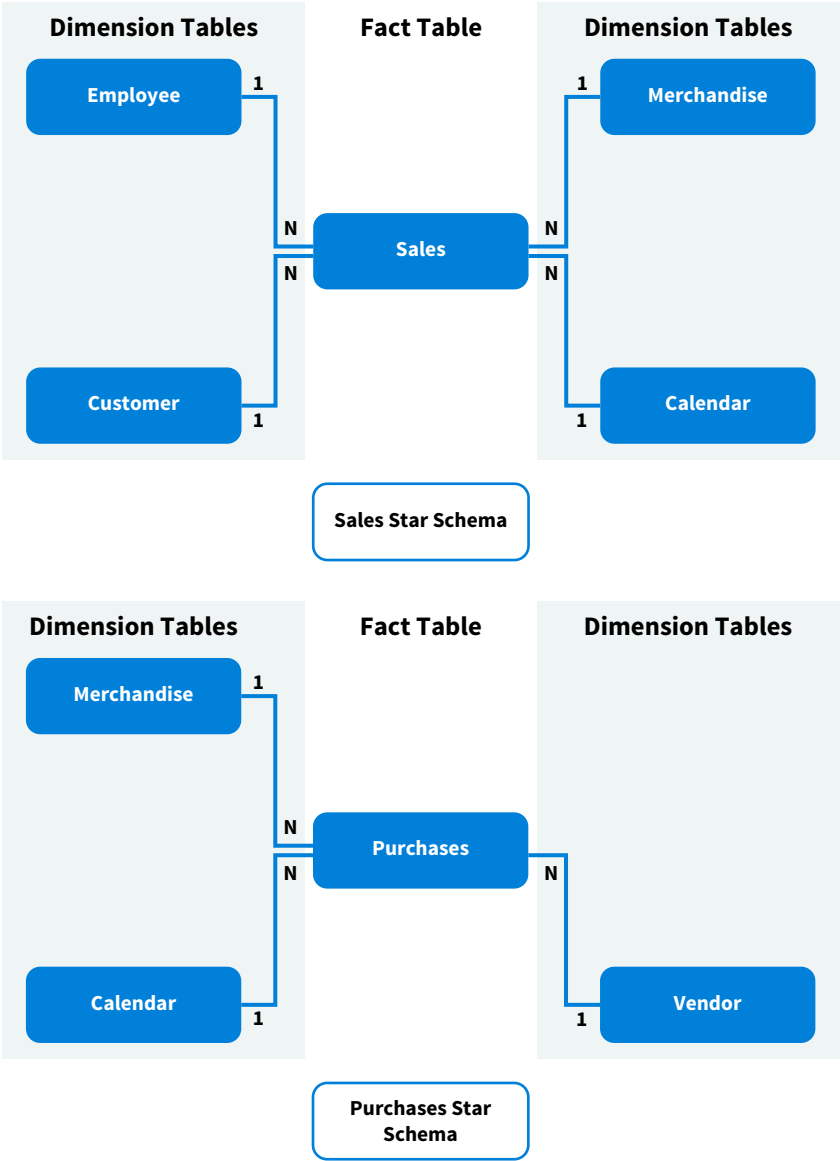


After six months in business, the owner is curious where the company stands from a sales and profit perspective and hires you to provide some insights. You are given the company’s available data. The data set is clean and structured as star schemas.

From Data Model to Information Model

Illustration 6.4 shows the two star schemas for the KLUB data set. The information model has not been created yet.

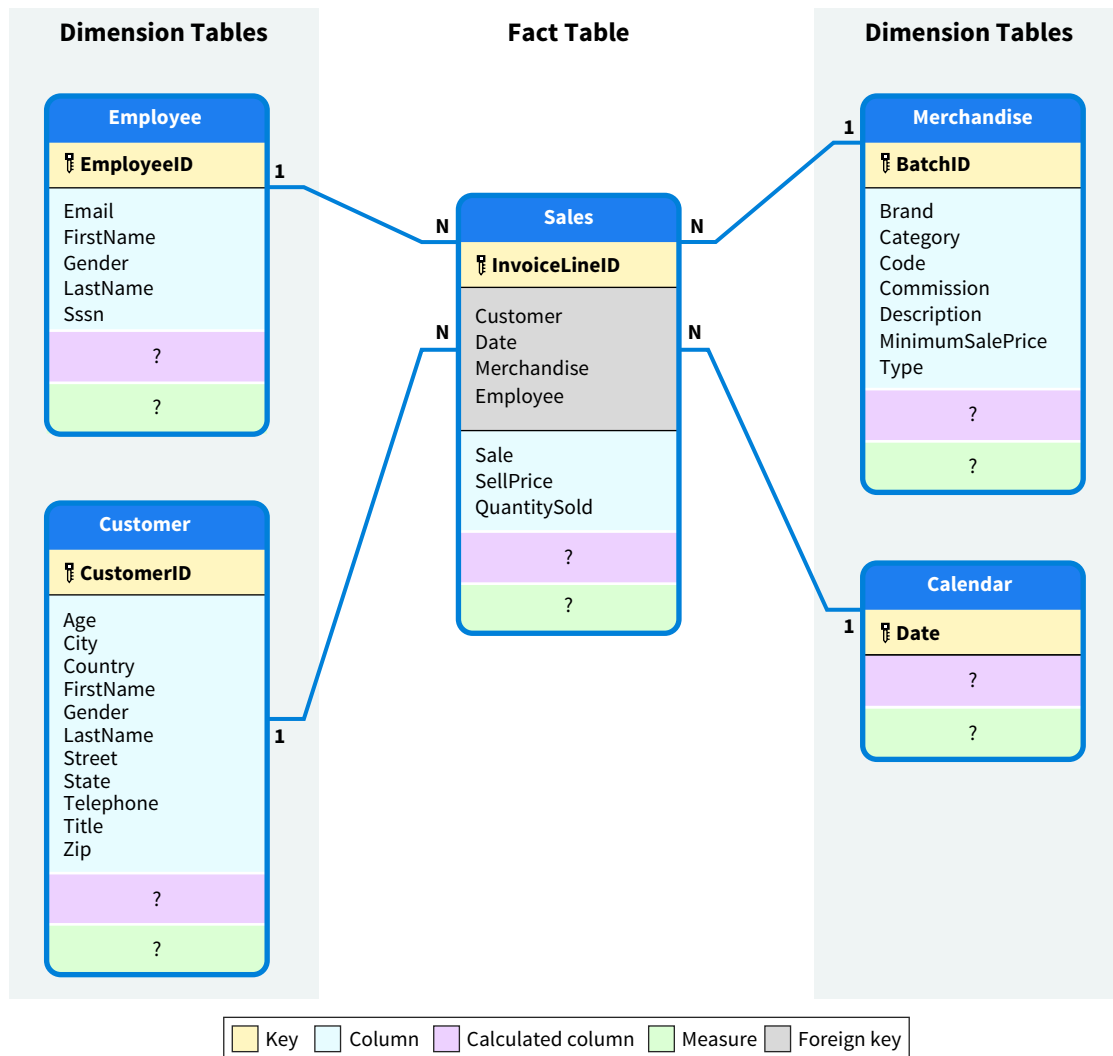
ILLUSTRATION 6.4 KLUB’s Sales and Purchases Star Schemas



A star schema is a data model. A **data model** shows the structure of a data set. It shows the concepts being described, the tables, and the fields used to describe the concepts. An **information model** extends the data model because it also includes calculated columns and measures. It has additional information that is calculated from the data set, which can be used for analysis purposes.

Illustration 6.5 shows the extended Sales star schema with field names such as Brand, Country, and Gender. In this illustration, the fact table describes sales transactions. The fields with question marks indicate the information model’s calculated columns and measures that will be developed in this chapter.

ILLUSTRATION 6.5 Extended Sales Star Schema



APPLYING CRITICAL THINKING 6.1: Same Data, Different KPIs

Measures feed into key performance indicators (KPIs), which are the numbers businesses want to track and then analyze. Developing an information model requires talking with different people across your company:

- People who are impacted by the analysis all likely want their own KPIs included. Consider sales data—the marketing manager might be interested in analyzing churn rates, while the accountant might be interested in analyzing accounts receivable and bad debt expenses. The same data is being analyzed, but different KPIs must be applied (**Stakeholders**).
- One KPI might mean different things to different people. What if you are asked to analyze profit margin? Is it the gross profit margin or the net profit margin? Make sure you understand the algorithm (**Knowledge**)!

Create Measures and Dimensions

There are two goals for building an information model for a star schema. The first is to create a rich set of measures for the fact table. Recall that in a star schema, fact tables typically hold business transactions. For example, in Illustration 6.2 the NetAmount column was created

and then used to create the TotalNetAmount measure, which is an aggregate. The calculated fields in the fact table are often used as building blocks for the measures.

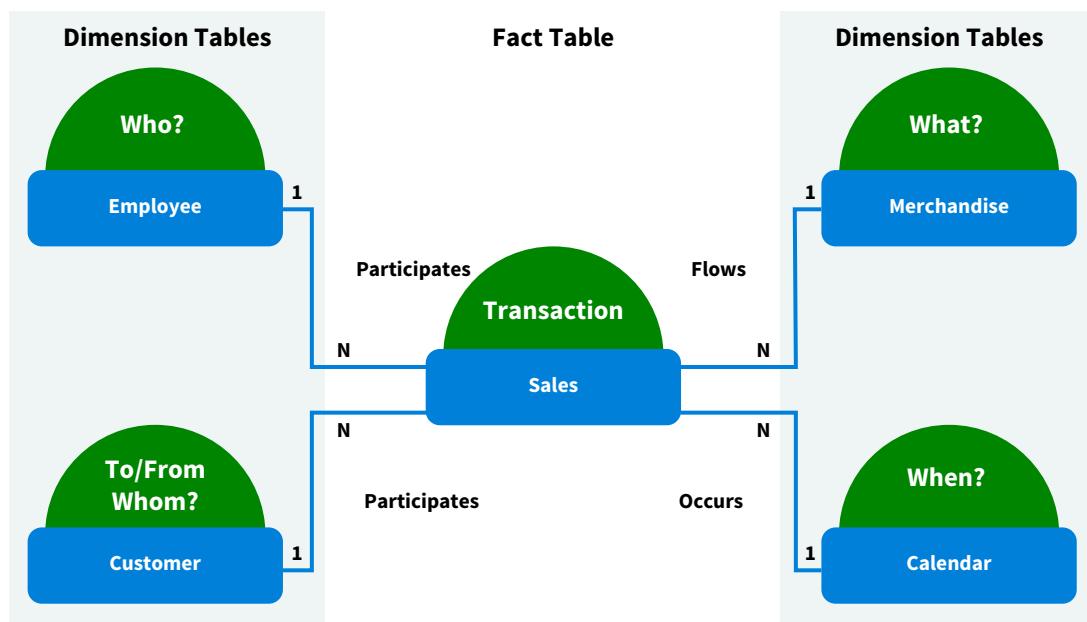
The second goal for building an information model is a rich set of dimensions that can break down, or slice, measures in many ways. Helpful dimensions in Illustration 6.5 include Gender, State, Country, Brand, Type, and Category. In the KLUB example, it may be useful to compare profit across product brands, types, categories, and countries. More dimensions can be created through calculated columns. An example of a dimension we will develop with calculated columns for KLUB's Sales star is AgeCategory, which differentiates between junior and senior customers.

Dimension tables can describe specific characteristics of transactions: who participated, what was involved, and when they occurred¹:

- A dimension table that describes the agents involved in the transactions is a **who table**. Internal agents are the employees who participate in the transaction. External agents, such as customers and vendors, are parties external to the organization involved in the accounting transactions. A **participates relationship** links agents to specific transactions.
- A dimension table that describes the resources that were given up or acquired as part of a transaction is a **what table**. In the KLUB example, purchases increase merchandise, while sales decrease merchandise. A **flows relationship** links resources to specific transactions.
- A dimension table that describes when a transaction occurs is a **when table**. Such tables often take the form of a Calendar table. An **occurs relationship** links the calendar to specific transactions.

Illustration 6.6 is KLUB's Sales star schema labeled with who, what, and when dimension tables and their relationships.

ILLUSTRATION 6.6 Sales Star with Who, What, and When Dimension Tables



Stars like the one shown in Illustration 6.6 are designed for accounting analytics. They make it easy to answer transaction questions that include who, what, and when characteristics. For example, an accountant at KLUB could ask: How much revenue (the measure) was

¹Who-what-when dimensions of accounting transactions were first recognized in McCarthy, William E. (1982) The REA accounting model: A generalized framework for accounting systems in a shared data environment. *Accounting Review*: 554-578.

generated by selling garden items (what) to customers living in Texas (who) during the 2023 spring season (when)?

When developing stars like this, it is critical to determine:

- The accounting measures to break down.
- The who, what, and when dimensions for analysis purposes.

This chapter explains how to do both. Information modeling is the heart of data analytics. Rich, robust information models lead to rich and robust analyses. Creating measures and dimensions that are relevant for analysis requires coding skills. But once these measures and dimensions are in place, software, such as Power BI or Tableau, makes it easy to do the analysis. Whatever software and language you use, invest some time into learning how to code. It will pay off!

Accounting Information Systems **Financial Accounting** Abroad is a store in the King of Prussia mall near Philadelphia, PA that sells stylish suitcases and bags. Sales are spectacular, but management believes data analytics are the key for more growth. The company is in the process of building their application and the owner asks you for help.

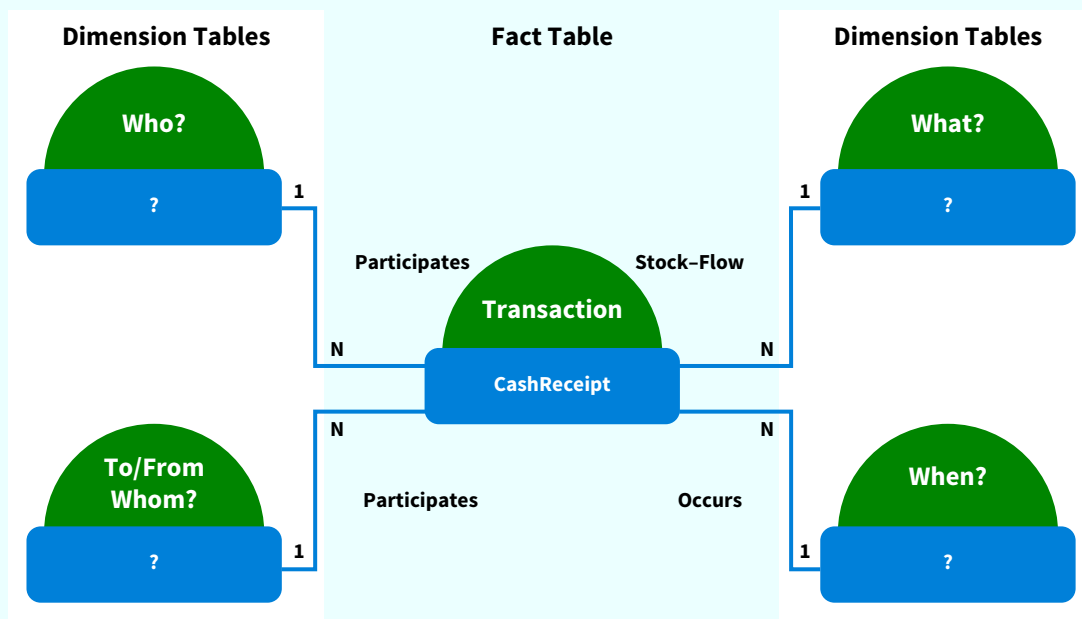
The illustration shows an empty star schema for cash receipts transactions, or payments made by a customer.

Apply It 6.1

Complete a Star Schema



Draft Cash Receipt Star Schema



Complete the star schema by answering these questions.

1. What labels would you use for the who (external and internal), what, and when tables that are represented by empty boxes with question marks?
2. Indicate two relevant measures for cash receipts.
3. For both measures, identify a relevant question that shows how the measure can be broken down by the information in the who, what, and when tables.

SOLUTION

(Solutions will vary. What follows are examples.)

1. Who Table External: Customer (Who makes the payments?)
Who Table Internal: Employee/Cashier. (Who is handling the payments by the customers?)

What Table: Account (A payment from a customer increases an account's balance. An account can be cash, a bank, etc.)
When Table: Calendar (The calendar contains information regarding the date the payment was made.)

- 2. Measure 1: Total number of cash receipt transactions.
Measure 2: Total amount of dollars received.
- 3. Measure 1: How do the number of cash receipt transactions (measure) in the first quarter of 2025 (when) compare across employees (who)?
Measure 2: What are the monthly trends (when) for the amounts (measure) received for the different accounts (what), and are there any fluctuations?

6.2 Which Patterns Implement Information Modeling Algorithms?

LEARNING OBJECTIVE 2

Apply common information modeling algorithms.

This section presents seven implementation patterns that each represent a different type of algorithm. The first four create calculated columns, while the final three patterns create measures. (Data How To 6.1 at the end of the chapter walks through how to create calculated columns and measures with Power BI.) The details of each pattern are illustrated with the KLUB example and data set. (Data Download the available data set to create the information model as you work through the chapter.

Information Modeling Pattern 1: Within-Table Numeric Calculation

The within-table numeric calculation pattern creates a new field (a calculated column) from one or more numeric columns, or fields in the same table. Basic arithmetic operations of addition, subtraction, multiplication, and division can do this. A typical example is determining the dollar total for an order line or invoice line by multiplying price and quantity. Illustration 6.7 (A) shows how to create a Revenue field in KLUB's sales

ILLUSTRATION 6.7 Within-Table Numeric Calculation to Determine Revenue

Table: Sales

(A)

Calculated Column	Algorithm
Revenue	$SellPrice \times QuantitySold$

(B)

InvoiceLineID	Sale	Date	Merchandise	Customer	Employee	SellPrice	QuantitySold	Revenue
1	1	1/5/23	8	223	7	60	12	720
2	1	1/5/23	9	223	7	20	4	80
3	1	1/5/23	16	223	7	320	1	320
4	2	1/6/23	19	80968	635	9	40	360
5	2	1/6/23	20	80968	635	20	40	800

Result of Algorithm

$SellPrice \times QuantitySold$

table with such a calculation. Illustration 6.7 (B) shows the result: the Revenue field that is part of the Sales table. Later, we will use the Revenue field as a building block for the TotalRevenue measure.

Information Modeling Pattern 2: Within-Table Text Calculation

The second implementation pattern also involves creating a new calculated column from one or more fields in the same table. However, the information in the new column is created from text fields. For example, different pieces of location information could be linked, or concatenated, into an address so it can be read by a map service such as Bing Maps or Google Maps.

Illustration 6.8 (A) shows an algorithm that creates an address field in KLUB's Customer table. (The ampersand symbol represents concatenation.) The new field in the Customer table in Illustration 6.8 (B) is the result of applying this calculation.

ILLUSTRATION 6.8 Within-Table Text Calculation to Determine Customer Address

Table: Customer

(A)

Calculated Column

Algorithm

Address

Street & “,” & City & “,” & State & “,” & Zip & “,” & Country

Street + City + State + Zip + Country

Result of Algorithm

(B)

CustomerID	Street	City	State	Zip	Country	Address
1	3462 MacLaren Street	Ottawa	ON	K1P 5M7	CA	3462 MacLaren Street,Ottawa,ON,K1P 5M7,CA
2	2608 Laurel Lane	Terminal	TX	79703	US	2608 Laurel Lane,Terminal,TX,79703,US
3	4679 Byrd Lane	Albuquerque	NM	87102	US	4679 Byrd Lane,Albuquerque,NM,87102,US
4	975 Armbrester Drive	Malibu	CA	90265	US	975 Armbrester Drive,Malibu,CA,90265,US
5	3251 St. John Street	Bruno	SK	S4P 3Y2	CA	3251 St. John Street,Bruno,SK,S4P 3Y2,CA

Once implemented, the Address field can be used to create maps.

Using similar formulas, KLUB's information model can be extended with Name fields for both the Customer and Vendor tables (**Illustration 6.9**). If you are following along building KLUB's information model, make sure to create both columns.

ILLUSTRATION 6.9 Within-Table Text Calculations to Determine Customer and Vendor Names

Table: Customer	
Calculated Column	Algorithm
Name	LastName & " " & FirstName
Table: Vendor	
Calculated Column	Algorithm
Name	Lastname & " " & Firstname

Information Modeling Pattern 3: Within-Table Classification

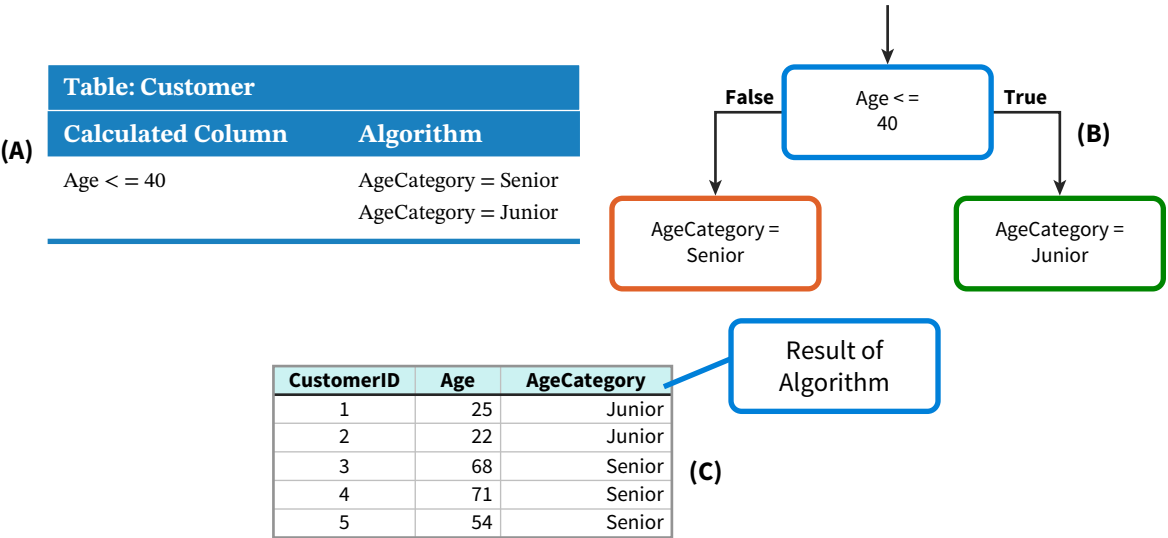
The number of classification algorithms that can be applied to a data set is, in essence, unlimited. You may be familiar with airlines' tiered loyalty programs classifications (such as silver, gold, platinum) where an algorithm determines your status based on miles flown or money spent. The statistical technique of data binning is another example, where numbers are

grouped into smaller numbers of “bins,” such as age groups. In fact, classifications are omnipresent in accounting, such as classifying accounts receivable by age to determine bad debt expense.

The third implementation pattern is used when classifications are determined based on data that are part of the same table. Most classification algorithms rely on Boolean logic, so it is necessary to understand Boolean operators like AND, OR, NOT, and IF functions and how to create them.

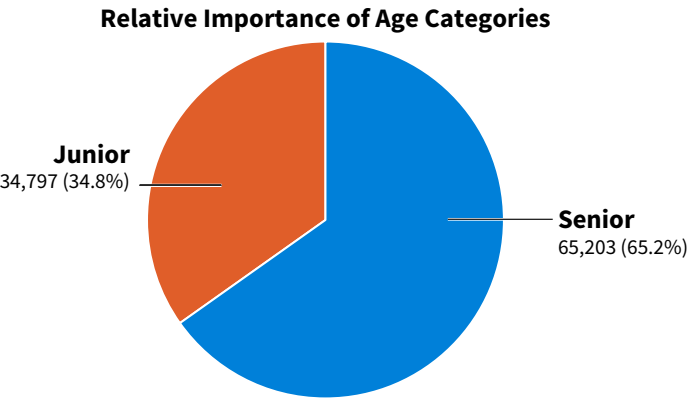
With both code (A) and a flowchart (B) **Illustration 6.10** shows how an IF function can split customers into two categories: junior (up to 40 years old), and senior (older than 40). Illustration 6.10 (C) is the result of applying this algorithm. The new AgeCategory field is now part of the Customer table.

ILLUSTRATION 6.10 Within-Table Classification Using an IF function to Determine Customer Age Category



Once implemented, we can use the AgeCategory field for analysis. **Illustration 6.11** shows the proportional number of KLUB’s junior and senior customers, respectively.

ILLUSTRATION 6.11 Pie Chart Created With AgeCategory Field



Information Modeling Pattern 4: Across-Table Calculation

The calculations for this pattern are like the first three—a new column is created from existing data using arithmetic, text calculation, or classification. But this pattern is different from the earlier patterns because data from different tables are used to create a new column. This means the tables must be linked. How this is accomplished depends on the software being

used. Excel uses a VLOOKUP, in a relational database this is done with a join, and in DAX it is accomplished with the RELATED function. Regardless of the software, there are multiple questions to consider when linking tables:

- Are there properly defined relationships between the tables?
- What is the nature of the relationship, and what type of join should be considered?
- What cardinalities apply to the relationship: 1-1, 1-N, N-1, or N-N? For example, calculations become more complex when navigating a 1-N relationship.

Let's use an example of an across-table calculation that illustrates some of these issues. Recall that KLUB buys products in batches:

- The buy price, or cost per unit, in a batch is recorded in the Purchase table.
- The revenue of sold items is determined by multiplying the SellPrice and QuantitySold fields in the Sales table, which results in the Revenue field (which is a calculated column) discussed earlier.
- The matching cost is determined by multiplying the BuyPrice in the Purchase table with QuantitySold in the Sales table.

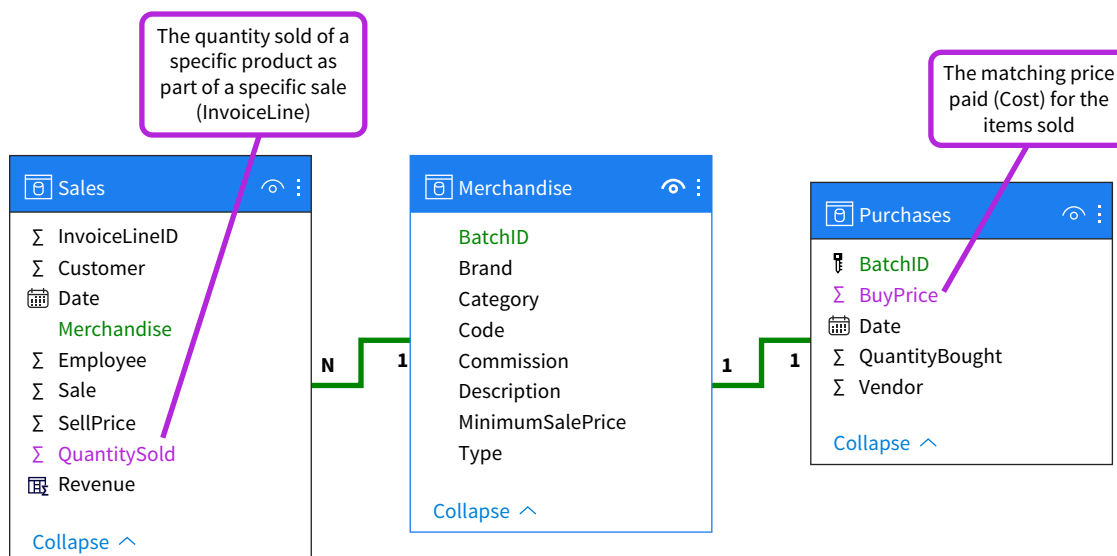
Illustration 6.12 is an algorithm that calculates cost of goods sold (COGS). Since data from different tables are used, table names are now included in the algorithm as TableName.FieldName. For example, Sales.QuantitySold refers to the QuantitySold field in the Sales table.

ILLUSTRATION 6.12 Across-Table Calculation to Determine COGS

Table: Sales	
Calculated Column	Algorithm
COGS	<code>Sales.QuantitySold × Purchases.BuyPrice</code>

Illustration 6.13 is the data model on which the implementation of the COGS column relies.

ILLUSTRATION 6.13 COGS as an Across-Table Calculation



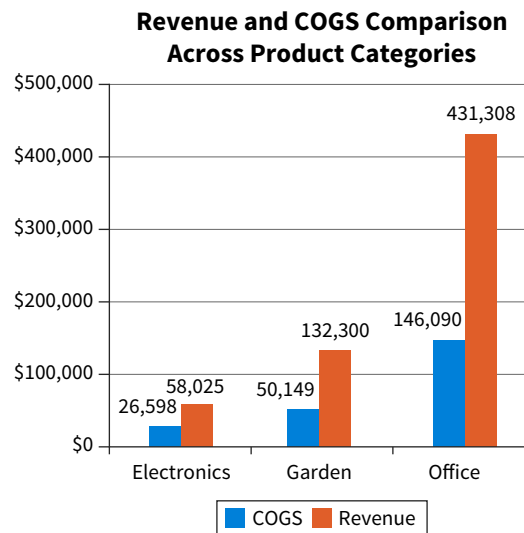
The values in the COGS column in the Sales table are calculated by multiplying the QuantitySold field in the Sales table by the corresponding BuyPrice field in the Purchase table. Corresponding values are determined by following the relationships, which is known as the

navigation path. Here are some characteristics of the navigation process for the execution of this formula:

- A green line indicates the navigation path in Illustration 6.13. The Merchandise field in the Sales table is linked to the BatchID field in the Merchandise table based on matching values. A similar relationship is created between the BatchID field in the Merchandise table and the BatchID field in the Purchases table. If KLUB sells a hairdryer to a customer, the navigation path will reveal which batch it belongs to and thus from whom they bought it, when, and for what price (cost).
- The navigation path has 1-1 cardinalities. There is one merchandise item (Merchandise) per invoice line (Sales), and one purchase (Purchase) for a merchandise item (Merchandise).
- There are no join issues. There is exactly one merchandise for each invoice line, and there is exactly one purchase for each merchandise.

Once implemented, the COGS field can be used for analysis. The clustered column chart created with the COGS field in [Illustration 6.14](#) can be used by KLUB management to compare revenue and COGS across three product categories: electronics, garden, and office.

ILLUSTRATION 6.14 Column Chart Created with COGS and Revenue Fields

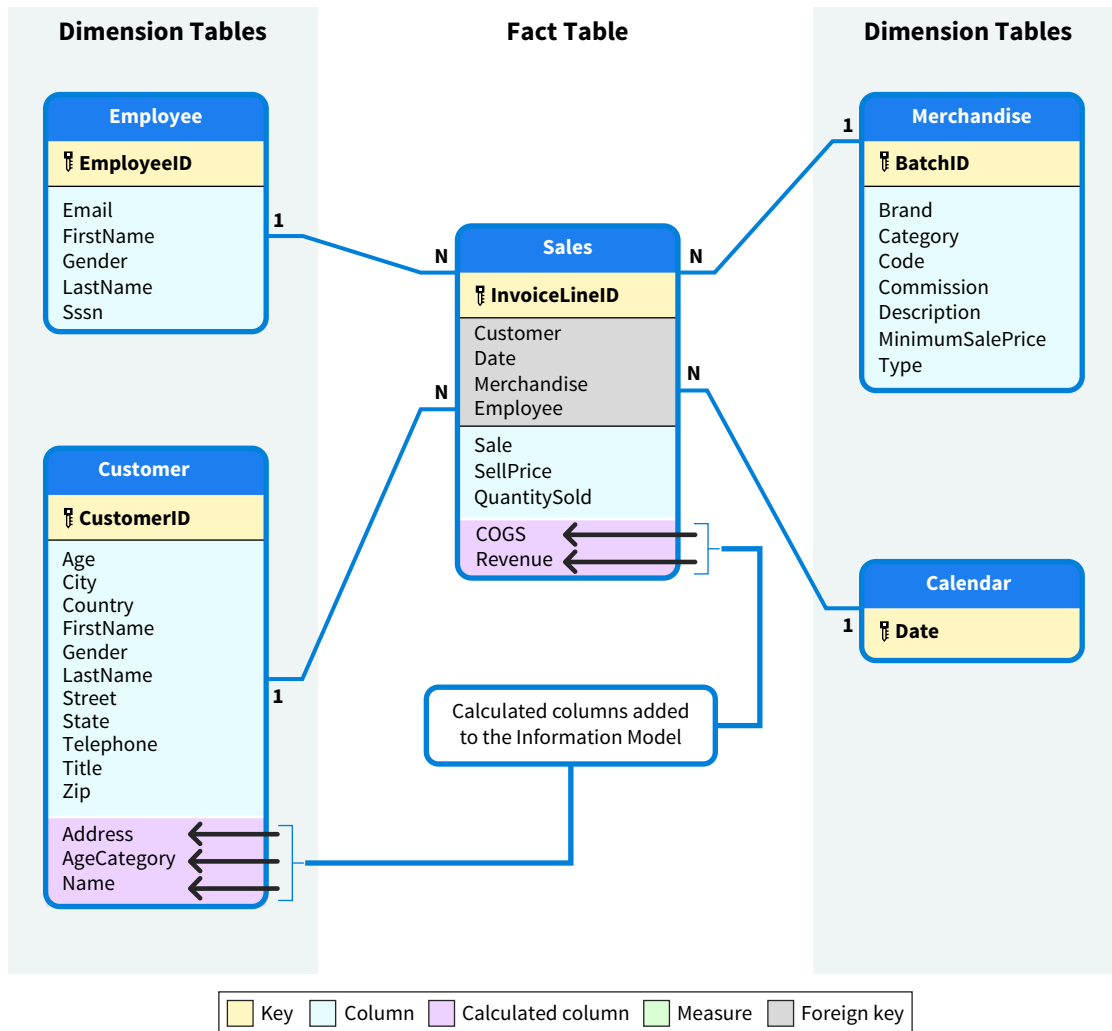


Applying the first four implementation patterns has extended KLUB's Sales star schema ([Illustration 6.15](#)) with the following calculated columns:

- Address (Customer)
- AgeCategory (Customer)
- COGS (Sales)
- Name (Customer)
- Revenue (Sales)

In addition, a Name calculated field was added to the Vendor table in KLUB's Purchases star schema.

ILLUSTRATION 6.15 KLUB's Information Model Extended With Calculated Columns



Information Modeling Pattern 5: Single-Column Aggregation

The single-column aggregation pattern is the first to focus on measures, which, as you have learned, are the heart of data analytics. Measures calculate aggregates that can be sliced, or broken down, in many ways.

In its simplest form, a measure applies a mathematical operation to *all* the values of a single column. Operations that can be applied include aggregate functions such as SUM, AVERAGE, COUNT, MIN, and MAX. Each of these functions generate a single value. For most applications, single-column measures make up a significant part of the information model. A few more things about aggregation functions:

- Except for aggregation functions that count, most require a numeric data type.
- Be sure to understand how the aggregate functions deal with specific values, such as null, as this can differ depending on the implementation platform.

Three measures can be created using single-column aggregations for KLUB's data set:

1. **TotalQuantitySold:** Calculates the total number of units (aggregate) sold by KLUB by using the SUM function to add all values in the QuantitySold field in the Sales table.
2. **TotalRevenue:** Calculates the total revenue (aggregate) generated by KLUB thus far. Notice that the SUM aggregation function uses the calculated field Revenue as an argument.
3. **NumberOfCustomers:** Calculates the number of KLUB's customers (aggregate) by counting all cells in the CustomerID field of the Customer table that are not blank. Because CustomerID is a primary key, there are no empty cells, and all rows in the table are being counted.

Illustration 6.16 lists these three measures and their algorithms.

ILLUSTRATION 6.16 Single-Measure Aggregation to Create TotalQuantitySold, TotalRevenue, and NumberOfCustomers Measures

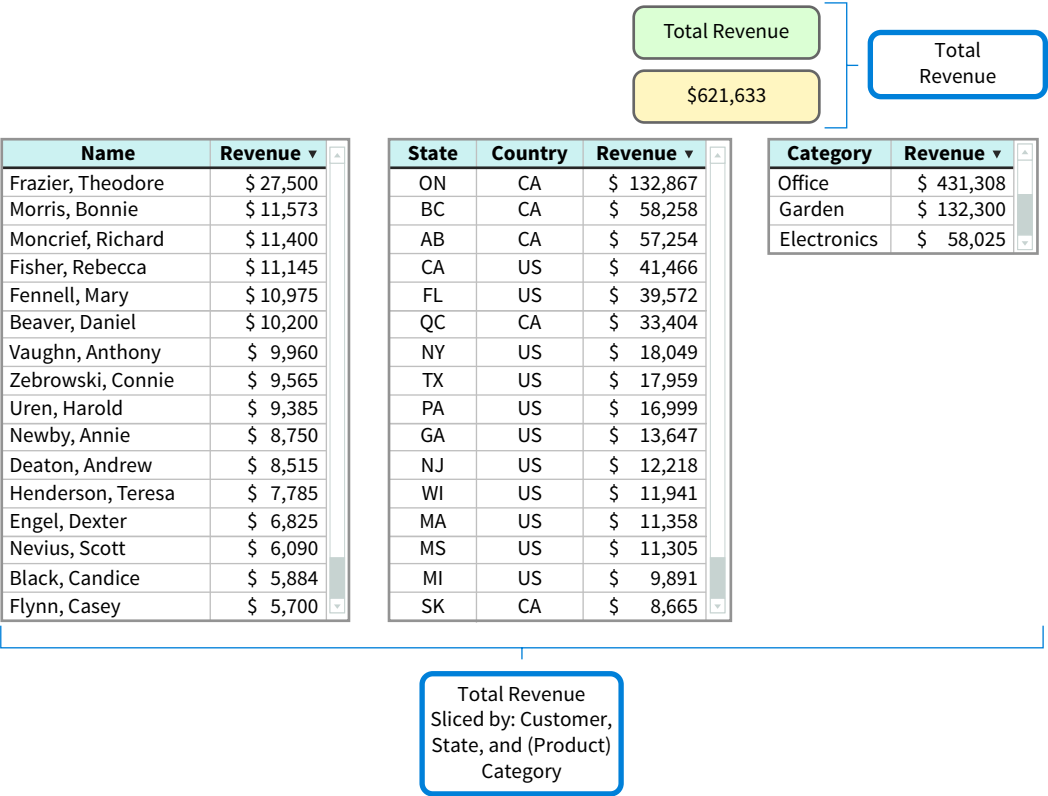
Table: Sales	
Measure	Algorithm
TotalQuantitySold	SUM(QuantitySold)

Table: Sales	
Measure	Algorithm
TotalRevenue	SUM(Revenue)

Table: Customer	
Measure	Algorithm
NumberOfCustomers	COUNT(CustomerID)

These measures calculate aggregates that can be sliced in many ways. The report in Illustration 6.17 is an example of this **one measure, multiple analyses principle**.

ILLUSTRATION 6.17 The One Measure, Multiple Analyses Principle



It breaks revenue down by customer, state, and product category. There are many more types of analysis that could be performed with TotalRevenue, including a break down by the AgeCategory field we calculated (which is a classification).

APPLYING CRITICAL THINKING 6.2: Evaluate Algorithms

Implementing an algorithm (or writing a program) is not an exact science. There are often multiple appropriate solutions, but also some bad ones. The most important criterion for a good program is that the output is correct. But it should also be fast, and this is especially true for large data sets. An algorithm should use the least amount of memory possible and be readable. **(Alternatives)**.

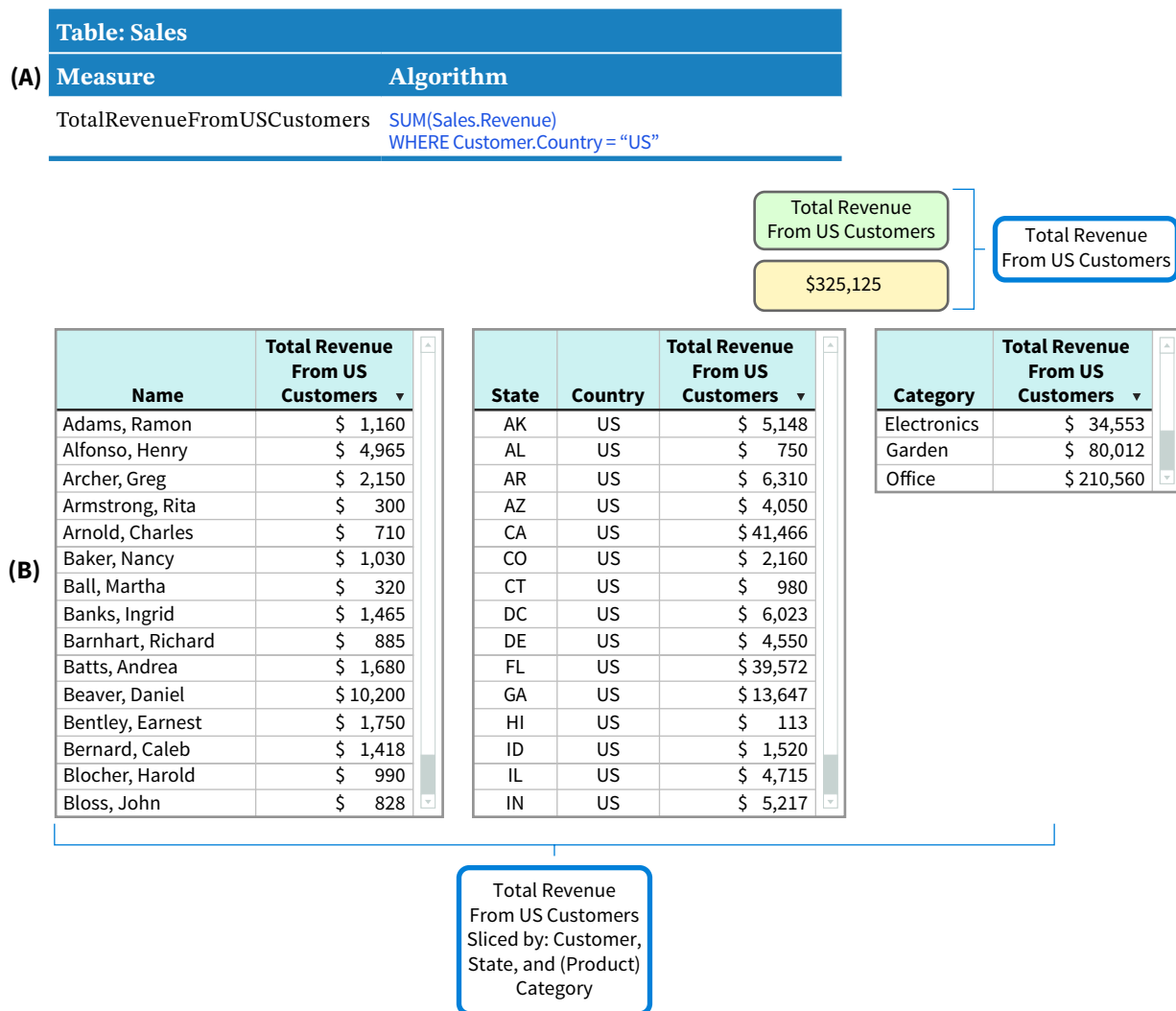
Information Modeling Pattern 6: Filtered Aggregation

It is often necessary to analyze aggregates based on filtered data sets. One example is analyzing the total revenue generated by sales to U.S. customers. There are different ways to implement filtered aggregations. The method often depends on the software being used.

- In Excel, use the SUMIF, COUNTIF, AVERAGEIF, etc., functions.
- In SQL, use the WHERE clause (**Data** How To 6.2 demonstrates how to implement a filtered aggregation with SQL).
- In Power BI, create filters in the Filters pane, or use the CALCULATE function.

Illustration 6.18 (A) is an example of a filtered aggregation that determines the total revenue generated by KLUB from sales to its U.S. customers. Illustration 6.18 (B) recreates the report from Illustration 6.17 using the total revenue from U.S. customers measure.

ILLUSTRATION 6.18 Filtered Aggregation to Calculate Total Revenue From U.S. Customers



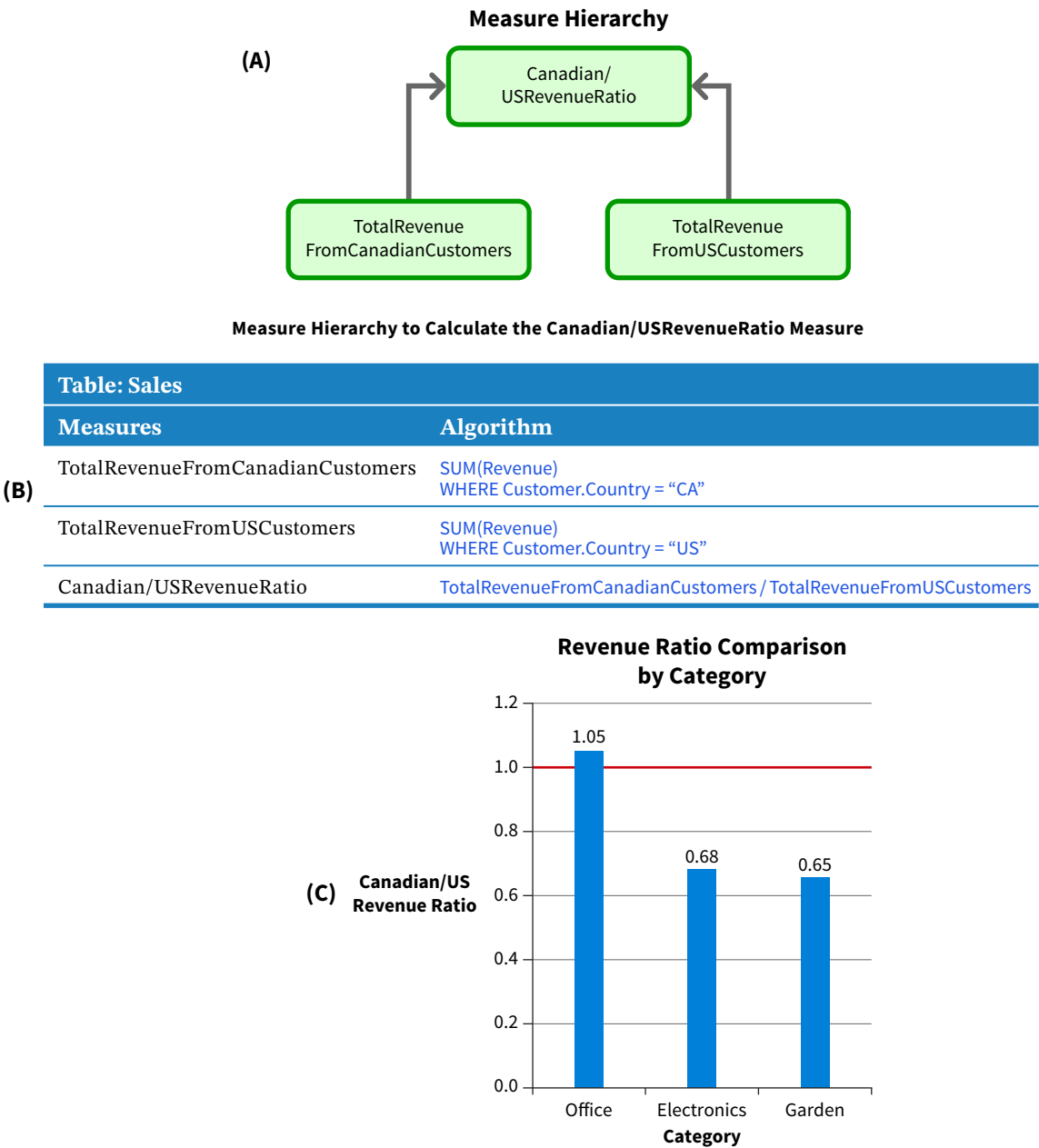
Information Modeling Pattern 7: Measure Hierarchies

A **measure hierarchy** is used when a new, more complex measure is created using existing measures. In fact, complex problems can sometimes be solved by dividing the problem into smaller problems, then combining the different solutions. Measure hierarchies are especially useful when calculating ratios and benchmarks.

Illustration 6.19 (A) visualizes a measure hierarchy. The first measure, TotalRevenueFromCanadianCustomers, determines the total revenue generated by KLUB’s customers in Canada. The second measure, TotlRevenueFromUSCustomers, determines the total revenue generated by KLUB’s U.S. customers. The third measure reuses the first two measures to calculate a ratio that compares Canadian revenue with U.S. revenue, using the latter as the benchmark.

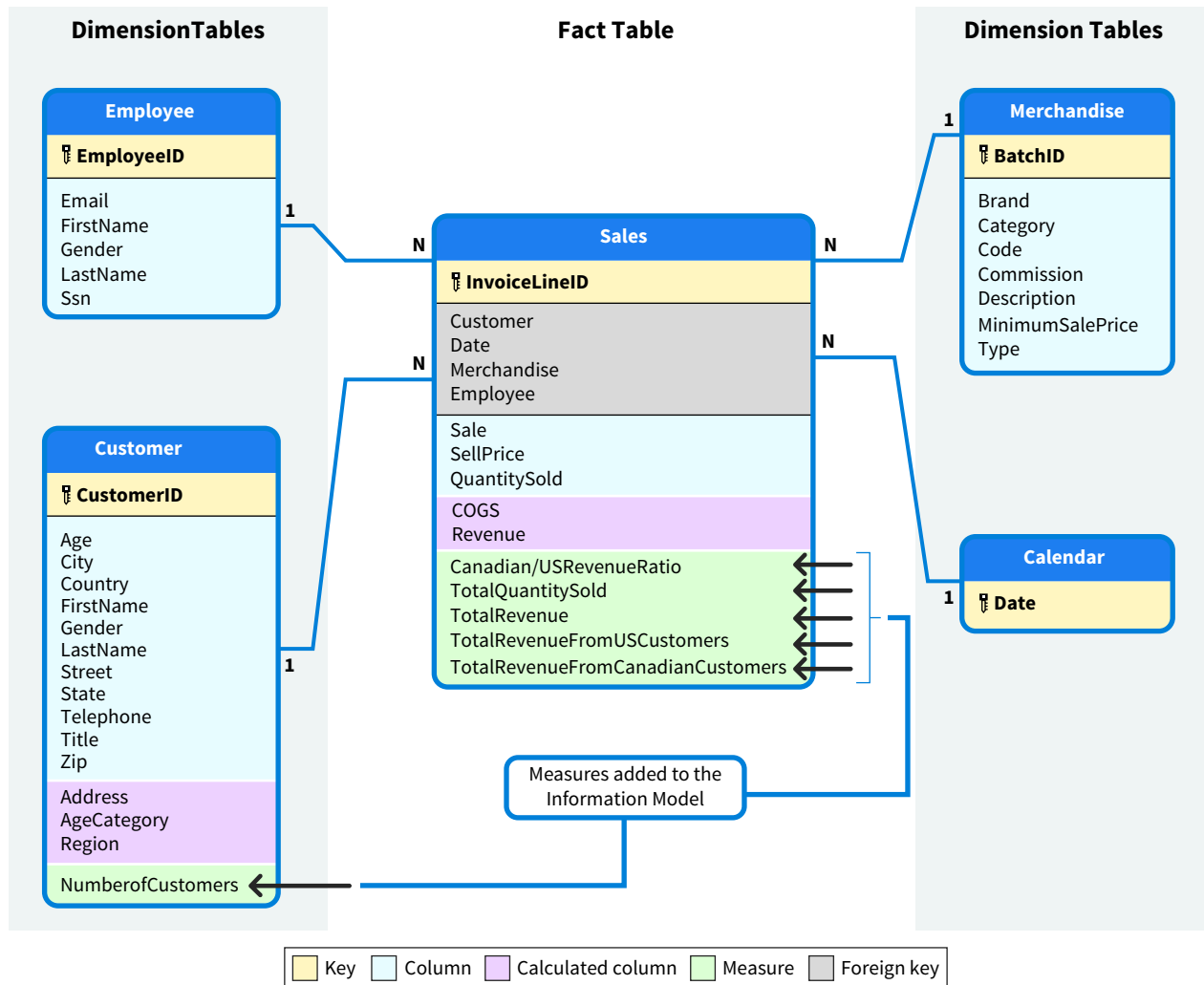
Illustration 6.19 (B) shows a possible implementation for the view hierarchy. The slash symbol represents division. Finally, Illustration 6.19 (C) shows one way to use the Canadian/USRevenueRatio measure. The red line, a ratio with value 1, indicates the equal contribution to revenue by both countries. With this analysis, management at KLUB can see that while Canada generates slightly more revenue for office products, most of the revenue for Electronics and Garden products are generated in the United States.

ILLUSTRATION 6.19 Measure Hierarchy for Canadian/U.S. Revenue Ratio: Visualization and Implementation²

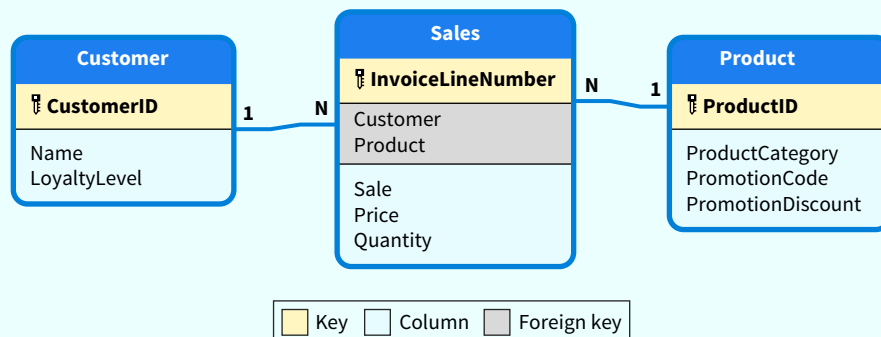


After creating the NumberOfCustomers, TotalQuantitySold, TotalRevenue, TotalRevenueFromUSCustomers, TotalRevenueFromCanadianCustomers, and Canadian/USRevenueRatio measures, KLUB’s information model is growing ([Illustration 6.20](#)). Remember, the richer your information model is, the richer your analysis will be!

²The TotalRevenueFromUSCustomers measure is the same as the measure in Illustration 6.18.

ILLUSTRATION 6.20 KLUB's Information Model Extended With Measures

Financial Accounting The owner at luggage retailer Abroad gives you the following data model, which shows the sales-related data that Abroad collects. Each row in the Sales table represents an invoice line.



Apply It 6.2

Use Algorithms to Calculate Net Revenue



Some products have a promotion code and a promotion discount. For products with no promotion code, the discount is 0. Abroad has three loyalty levels: Fish, Eagle, and Lion. The discounts given to each of the loyalty levels is shown in the table.

Level	Discount
Fish	0
Eagle	3
Lion	5

Abroad's owner asks you to calculate the total net revenue generated thus far. When calculating the total net revenue, be sure to consider both the loyalty and promotion discounts. Assume that both discounts are recorded as percentages, e.g., 5%. For each of the following, indicate which pattern(s) you applied, then show your algorithms.

1. Loyalty discount
2. Net revenue
3. Total net revenue

SOLUTION

1. Pattern 2.

Algorithm (calculated column): LoyaltyDiscount =

IF Level = "Fish" THEN 0, ELSE

IF Level = "Eagle" THEN 3, ELSE 5

2. Pattern 3 and Pattern 4.

Algorithm (calculated column): NetRevenue =

IF Customer.LoyaltyDiscount > 0 and Product.PromotionDiscount > 0

THEN

Sales.Price × Sales.Quantity × (1 - ((Customer.LoyaltyDiscount / 100) + (Product.PromotionDiscount/100)))

ELSE

IF Customer.LoyaltyDiscount = 0 and Product.PromotionDiscount > 0

THEN Sales.Price × Sales.Quantity × (1 - Product.PromotionDiscount/100)

ELSE

IF Customer.LoyaltyDiscount > 0 and Product.PromotionDiscount = 0

THEN Sales.Price × Sales.Quantity × (1 - Customer.LoyaltyDiscount/100)

ELSE Sales.Price × Sales.Quantity

3. Pattern 5.

Algorithm (measure): Total Net Revenue = SUM(Amount)

6.3 Which Patterns Help Develop and Implement Accounting Information Models?

LEARNING OBJECTIVE 3

Develop and implement information models for common accounting data structures.

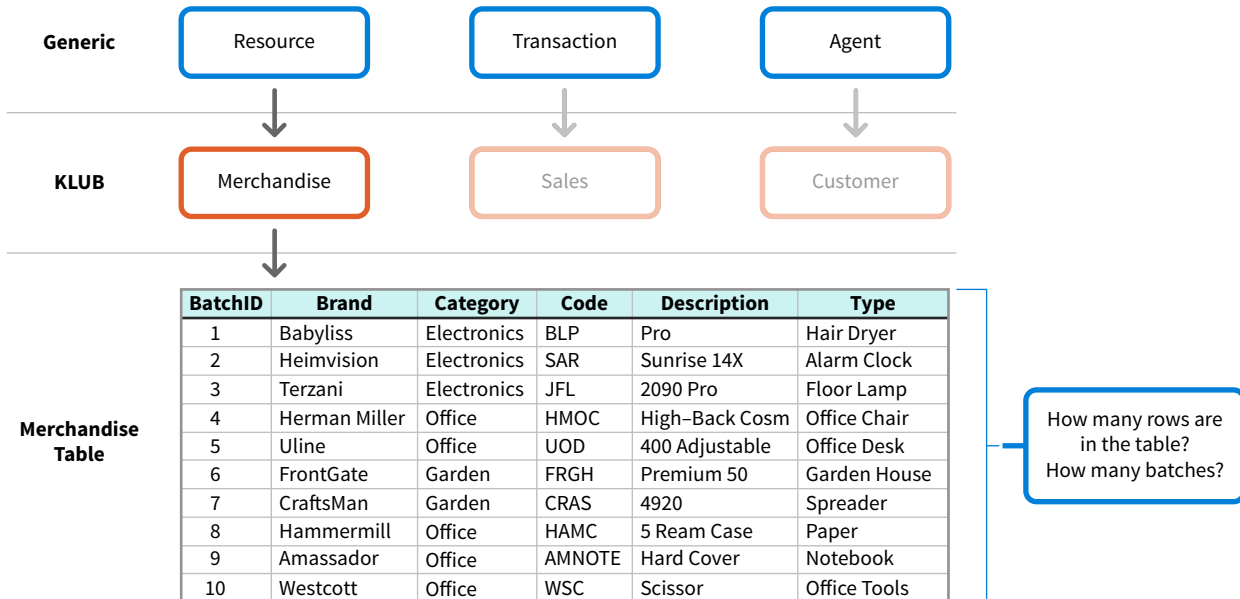
Earlier in the course you learned the importance of applying dimensional modeling to accounting data. Organizing data properly in specific ways makes analysis easier. Information modeling, or identifying and implementing measures and dimensions, also benefits from having similar structures in place. The previous section covered seven common patterns for implementing calculated columns and measures. Here, we present six more patterns that help identify measures to analyze who, what, and when characteristics of accounting transactions.

Information Modeling Pattern 8: How Many

Tables usually describe concepts like customers, vendors, sales, and products. The next pattern creates a measure that counts the number of rows, or instances, of a concept within a table. This pattern can count resources, transactions, and agents, which means it can be used for both fact (transactions) and dimension (resources and agents) tables.

The example in [Illustration 6.21](#) determines the number of rows in the Merchandise table, which reveals how many batches KLUB has purchased.

ILLUSTRATION 6.21 The How Many Pattern



In the KLUB example, we can create “how many” measures for:

- Resources: Merchandise.
- Transactions: Sales and purchases.
- Agents: Customer, vendor, and employee.

Information Model

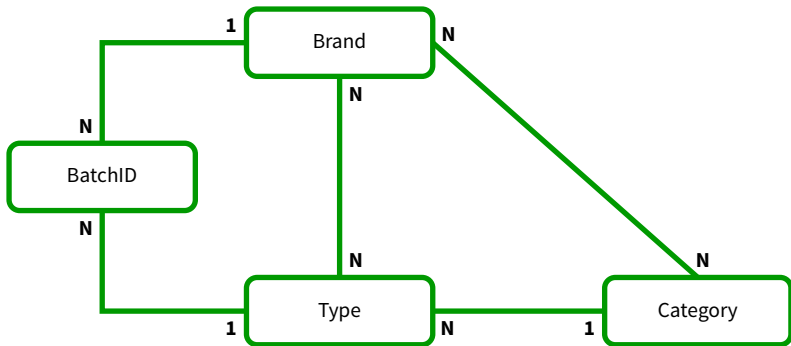
These measures can be broken down further. Using the KLUB data set, we could explore questions like the following by applying these measures to transactions, who, and what tables:

- Transactions: How many sales have occurred, and how many purchases?
- Agents (Who): How many customers does KLUB have, how many employees, and how many vendors do they work with? It is also possible to drill down further to reveal how many customers or vendors KLUB has per country, per state, or per province.
- Resources (What): How many products does KLUB have? It is then possible to find the number of products per brand, per product type, and per product category.

When answering questions about agents and resources, we can analyze relationships between dimensions in the same table, such as the number of customers for a state or province, or the number of product types per category. [Illustration 6.22](#) shows how some dimensions in the Merchandise table are related. Remember the N-1 cardinalities for the relationship between Batch ID and Brand:

- 1: One brand is specified for a batch.
- N: There can be many batches of the same brand.

ILLUSTRATION 6.22 Relationships Between the Merchandise Dimensions



The algorithms in [Illustration 6.23](#) show that implementing measures to answer questions about the number of transactions, agents (who), and resources (what) is straightforward. Implementation is performed with a single-column aggregation—Count. A similar formula was used for the NumberOfCustomers measure earlier in this chapter.

ILLUSTRATION 6.23 How Many Measures for KLUB’s Information Model

Table: Vendor	
Measure	Algorithm
NumberOfVendors	COUNT(VendorID)

Table: Sales	
Measure	Algorithm
NumberOfSales	COUNT(InvoiceLineID)

Table: Purchase	
Measure	Algorithm
NumberOfPurchases	COUNT(BatchID)

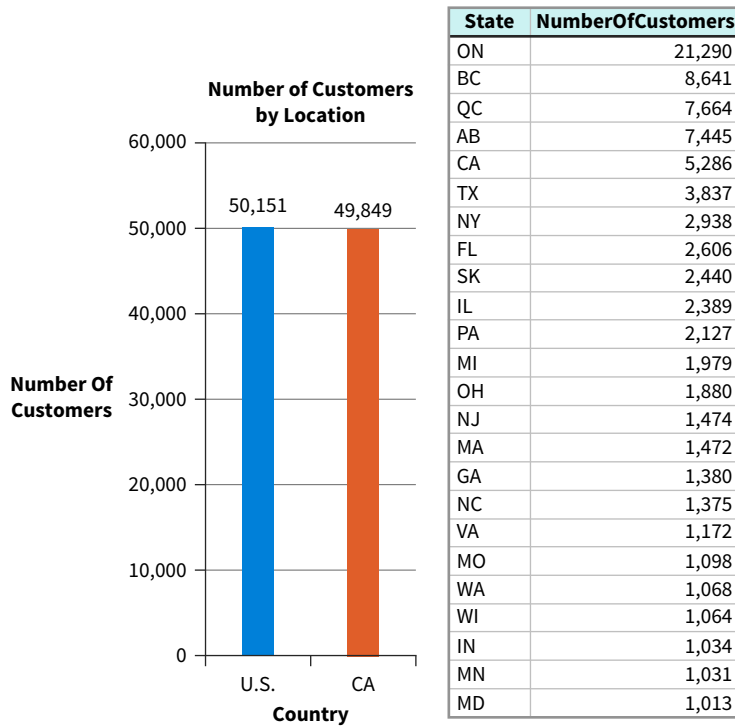
Table: Employee	
Measure	Algorithm
NumberOfEmployees	COUNT(EmployeeID)

Table: Merchandise	
Measure	Algorithm
NumberOfBatches	COUNT(BatchID)

Once in place, filters can be applied to the measures. For example, tables that describe agents (who) often contain location information, so it is also possible to determine the number of KLUB customers living in California, or any other location. On the other hand, tables that describe resources (what) often contain information about categories that can be used as filters. For example, it is possible to determine how many electronic batches KLUB bought.

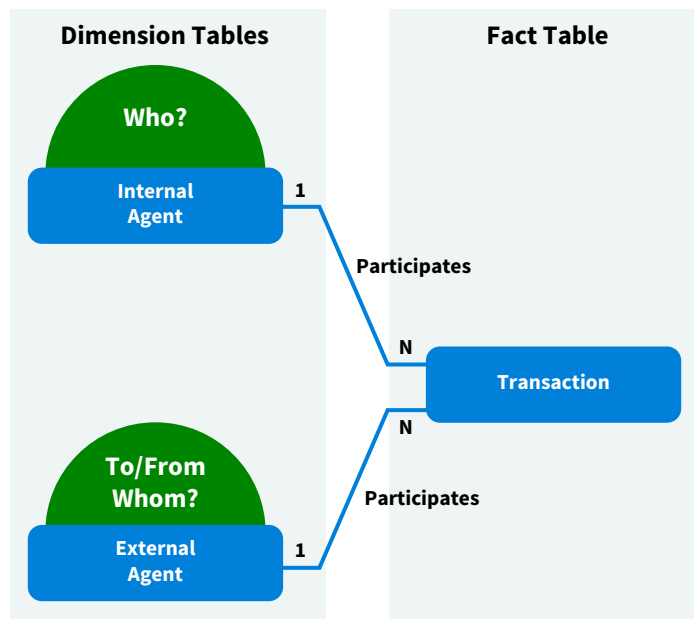
Analysis

The NumberOfCustomers measure can determine how potential customers are spread out across countries by provinces in Canada and states in the U.S. ([Illustration 6.24](#)). Clicking a bar in the column chart filters the state/provinces in the table, so management at KLUB could select the Canadian provinces only.

ILLUSTRATION 6.24 Analysis with the Number of Customers Measure

Information Modeling Pattern 9: Participates | Transaction-Who

The participates relationship describes *who* is involved in a transaction ([Illustration 6.25](#)).

ILLUSTRATION 6.25 The Participates Relationship

In KLUB's data model there are three participates relationships:

- From Sales to Customer (the customer is an external agent).
- From Sales to Employee (the employee is an internal agent).
- From Purchases to Vendor (the vendor is an external agent).

There are also other types of participates relationships that do not appear in the KLUB data set:

- CashReceipts to Customer.
- CashReceipts to Employee.
- CashDisbursements to Vendor.
- CashDisbursements to Employee.

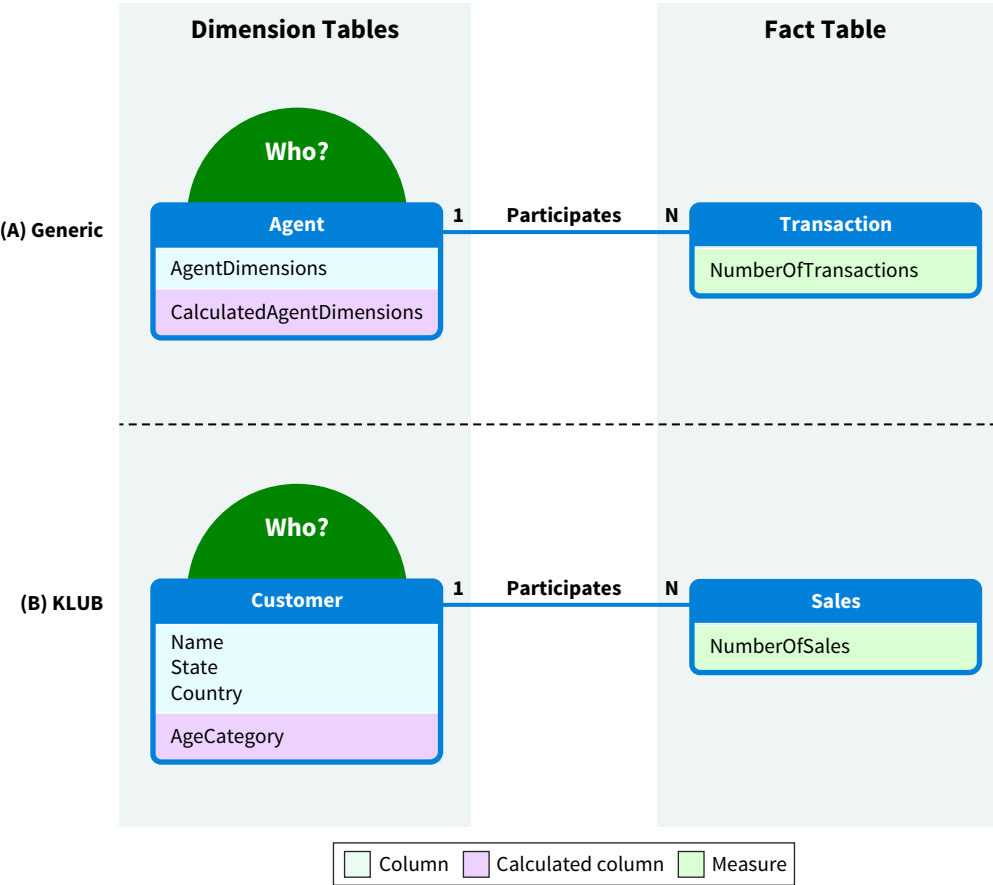
Because the information captured by these relationships is similar, the same types of analyses apply to all of them.

Information Model

Following the principles of dimensional modeling, measures must be developed for the sales transactions and broken down by agent dimensions, such as customer tables:

- What are relevant measures for transactions that can be broken down by agents? **Illustration 6.26** (A) shows that the number of transactions is a key measure for the participates relationship information model. For example, how many flights (transaction) has an airline customer taken (agent)?
- What dimensions can be defined for agents? Analysis can be performed for individual customers, but typically customers are grouped in a variety of ways for analysis purposes, including by location, membership group, or performance category.

ILLUSTRATION 6.26 Information Model Pattern for the Participates Relationship



To implement the participates information model, create relevant measures and calculated columns in the Transaction (fact) and Agent (dimension) tables. The actual measures and calculated fields will depend on a business’s specific needs. Illustration 6.26 (B) applies the participates model to KLUB’s Customer-Sales relationship. The additional calculated

columns and measures create a richer information model and therefore, a richer analysis. The algorithms to implement the NumberOfSales measure and the AgeCategory calculated field in Illustration 6.26 were discussed earlier in this chapter.

Analysis

KLUB could use the participates relationship to answer several questions using data analysis (**Illustration 6.27**).

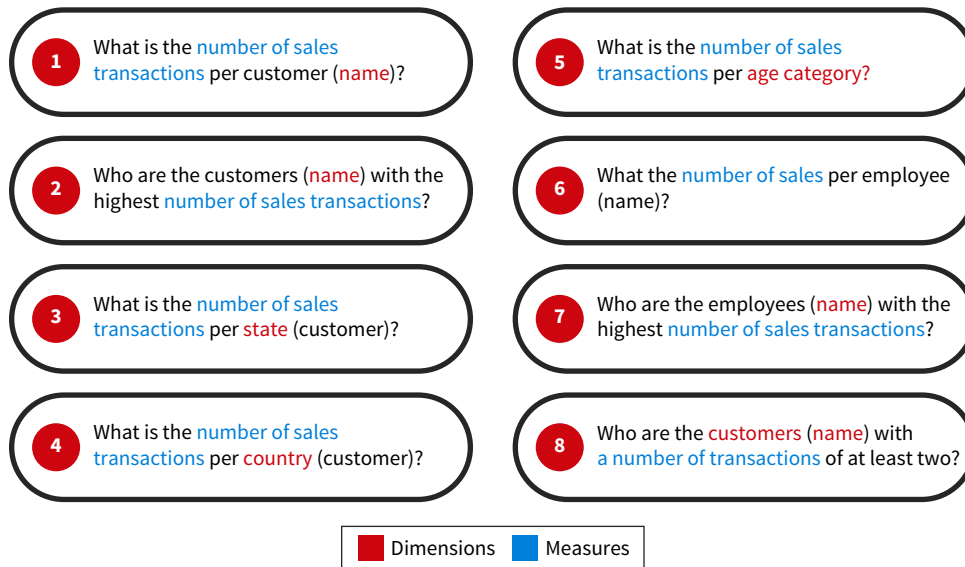


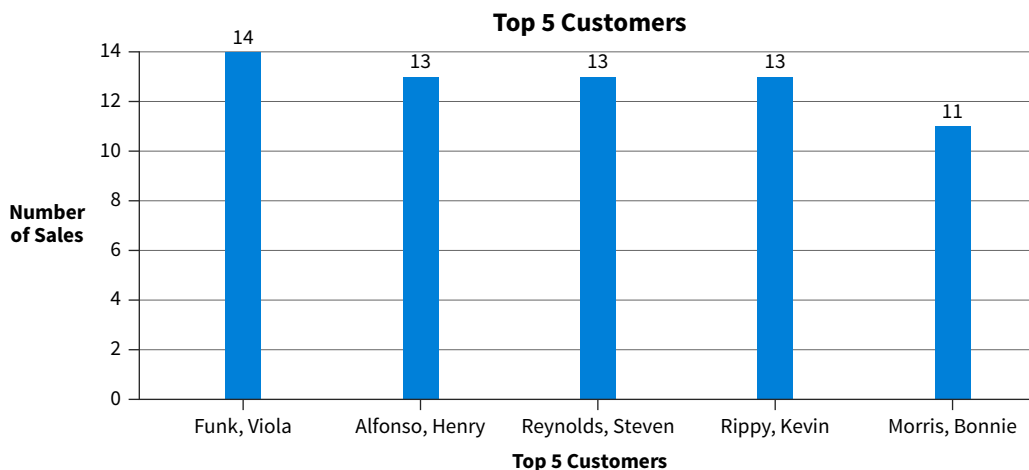
ILLUSTRATION 6.27 Answering Questions with the Participates Relationship

Next, let's use KLUB's information model to further explore two of the questions posed in Illustration 6.27.

- **Question 2**: Who are the customers (name, which is a dimension) with the highest number of sales transactions (measure)?

In **Illustration 6.28** the number of sales transactions is broken down by customer name, and the top five are shown in descending order.

ILLUSTRATION 6.28 Number of Sales Transactions Per Customer: Top Five



- Question 8:** Who are repeat customers? That is, who are the customers (name, which is a dimension) with a number of transactions (measure) of at least two?

Illustration 6.29 is a partial list of customers who have bought at least twice from KLUB.

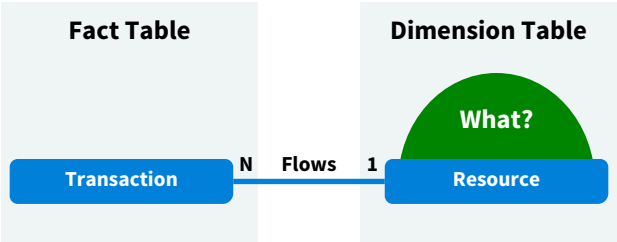
ILLUSTRATION 6.29 List of Repeat Customers

Name	NumberOfSales
Funk, Viola	14
Alfonso, Henry	13
Reynolds, Steven	13
Rippy, Kevin	13
Morris, Bonnie	11
Moore, Steven	9
Williams, Mary	9
Zebrowski, Connie	9
Garza, Jeffrey	8
Uren, Harold	8
Barnhart, Richard	7
Edens, Blaine	7
Jenkins, Toni	7
Ward, Jenny	7
Younce, Stacey	7

Information Modeling Pattern 10: Flows | Transaction-What

The flows relationship describes *what* is involved in a transaction, such as goods and services (**Illustration 6.30**). A transaction results in either an increase or a decrease of the resources.

ILLUSTRATION 6.30 The Flows Relationship



In KLUB’s data model there are two flows relationships:

- From Sales to Merchandise.
- From Purchases to Merchandise.

There are also other types of flows relationships that do not appear in the KLUB data set:

- Cash Receipts to Cash.
- Cash Disbursements to Cash.

Because the information captured by these relationships is similar, the same types of analysis apply to all of them.

Information Model

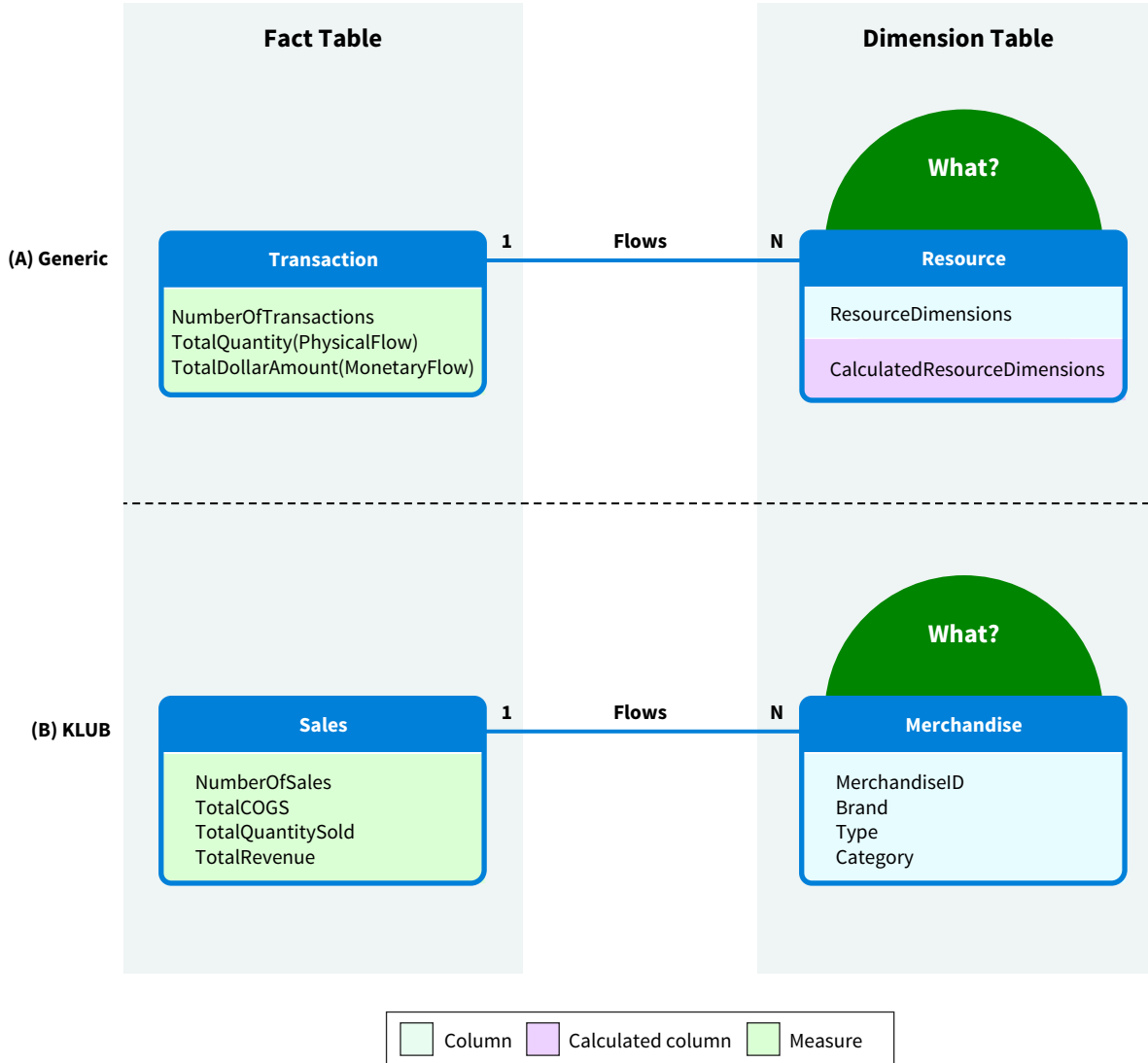
The first step is to develop measures for the transactions, such as sales, that can be broken down by dimensions that are part of the Resource tables, such as merchandise. For flows relationships, consider measures that summarize the physical and monetary flows:

- The physical flows describe *how many*. How many units were bought? How many units were sold?
- The monetary flows describe *how much*? What is the value of the units sold, and for how much have they been sold?

In addition, we can use the number of transactions and number of resources measures discussed earlier.

Illustration 6.31 (A) shows the generic information model for flows relationships. Panel (B) in the same illustration applies this pattern to KLUB's Sale-Merchandise flows relationship, which results in four measures: NumberOfSales, TotalCOGS, TotalQuantitySold, and TotalRevenue.

ILLUSTRATION 6.31 Information Model Pattern for the Flows Relationship



Implementing the flows information model involves creating measures in the Transaction table and creating Resource dimensions through algorithms. All dimensions in Illustration 6.31 are given; that is, additional calculations are unnecessary. How to implement the NumberOfSales, TotalQuantitySold, and TotalRevenue measures was discussed earlier. TotalCOGS, a measure in the Sales table, is a single-column aggregation that sums all the values in the COGS column—SUM(COGS).

There are two additional flows-related measures that should be added to the information model (Illustration 6.32). The TotalQuantityBought measure uses the single-column aggregation pattern to determine the total units bought by KLUB, and it is added to the Purchases table. The TotalQuantityOfPrintersSold measure uses the filtered aggregation pattern to determine the total quantity of printers sold.

ILLUSTRATION 6.32 Additional Measures for KLUB’s Flows Information Model

Table: Purchases	
Measure	Algorithm
TotalQuantityBought	SUM(QuantityBought)

Table: Sales	
Measure	Algorithm
TotalQuantityOfPrintersSold	SUM(QuantitySold) WHERE Type = “Printer”

Analysis

In the KLUB flows information model, the TotalQuantityBought measure in the Purchases table and the TotalQuantitySold measure in the Sales table summarize the physical flows, while the TotalCost measure in the Purchases table and the TotalRevenue and TotalCOGS measures in the Sales table summarize the monetary flows. Brand, Type, and Category are resource dimensions.

What questions can the data described by a flows relationship answer? **Illustration 6.33** presents some that can be analyzed using KLUB’s flows information model.

ILLUSTRATION 6.33 Answering Questions with the Flows Relationship

1

What is the number of sales transactions per batch (MerchandiseID)?

2

What is the number of units bought per batch (MerchandiseID)?

3

What is the number of units sold per batch (MerchandiseID)?

4

What is the number of items sold per brand?

5

What is the number of items sold per type?

6

What is the number of items sold per category?

7

What is the relative contribution of each category to the total revenue generated?

8

What is the total cost per batch (MerchandiseID)?

9

What is the total revenue generated per batch (MerchandiseID)?

10

What is the total revenue generated per brand?

11

What is the total revenue generated per type?

12

What is the total revenue generated per category?

13

What is the total number of printers (type) sold?

■ Dimensions
■ Measures

Next, use KLUB’s information model to explore two of the questions posed in this illustration.

- Question 7:** What is the relative contribution of each category (dimension) to the Total-Revenue (measure)?

In **Illustration 6.34**, the TotalRevenue measure in the Sales table is broken down by the category dimension in the Merchandise table. Pie chart slices represent the relative importance of each category. The pie chart shows that KLUB generates most of its revenue from its office products.

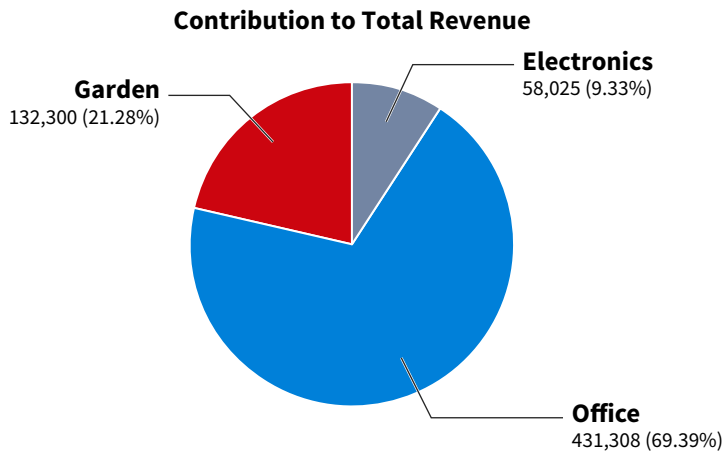


ILLUSTRATION 6.34 Relative Contribution of Categories to Total Revenue

Now, explore another question.

- **Question 13:** What is the total number (measure) of printers (type, which is a dimension) sold?

Illustration 6.35 shows the total quantity of printers sold and breaks that number down by brand using a column chart.

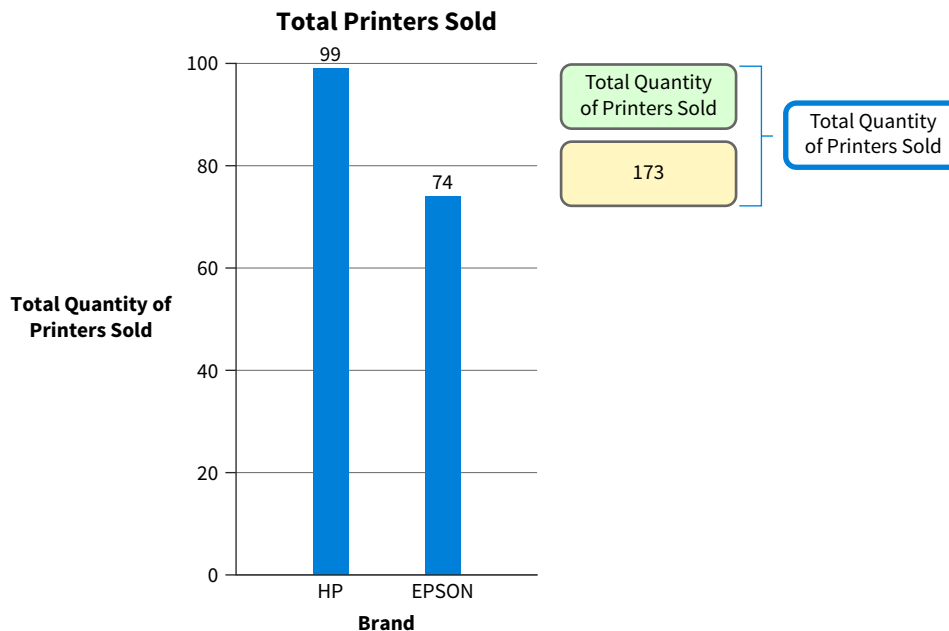


ILLUSTRATION 6.35 Total Number of Printers Sold

Information Modeling Pattern 11: Occurs | Transaction-When

The occurs relationship describes *when* a transaction takes place by connecting it to a Calendar table (**Illustration 6.36**).

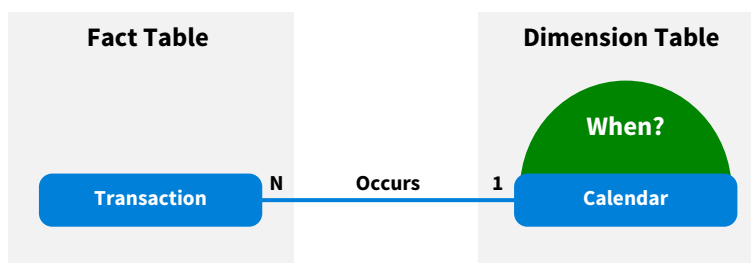


ILLUSTRATION 6.36 The Occurs Relationship

There are two occurs relationships in KLUB’s data model:

- Sales to Calendar.
- Purchases to Calendar.

There are other occurs relationships that do not appear in the KLUB data set:

- CashReceipts to Calendar.
- CashDisbursements to Calendar.

Because the information captured by these relationships is similar, the same types of analyses apply to all of them.

Information Model

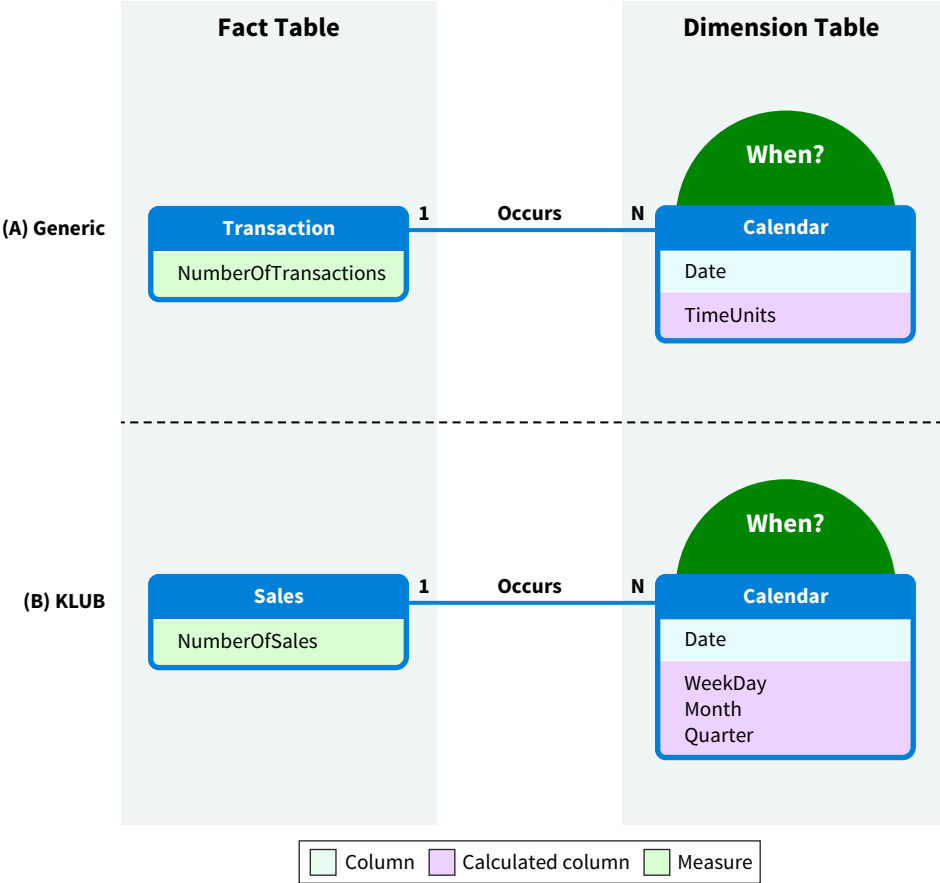
Following the principles of dimensional modeling, measures must be developed for the transactions (such as Sales) and broken down by dimensions that are part of the Calendar table:

- The measures developed earlier for the Transaction tables, such as NumberOfSales, can also be used here.
- In addition to dates, a Calendar table contains a series of time units as dimensions for analysis, including weekday, week number, month, quarter, and year. Because there are only six months of data for KLUB, year dimensions are not relevant.

Illustration 6.37 (A) shows the generic information model for occurs relationships. Panel (B) of the same illustration applies this pattern to KLUB’s Sales-Calendar occurs relationship. NumberOfSales can be analyzed by weekday, month, and quarter.

A calendar table contains all dates between a start date and an end date. Software such as Power BI makes it easy to create a Calendar table. Once that is done, time unit dimensions such as month, quarter, and year can be added and used for analysis. Creating these dimensions is usually easy. An example of this is Excel’s year function, which returns the year component of a date.

ILLUSTRATION 6.37 Information Model Pattern for the Occurs Relationship



Analysis

Illustration 6.38 presents some example questions that can be analyzed using KLUB’s occurs relationships.

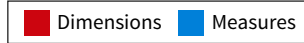
1 How do the **number of sales transactions** compare across **weekdays**?

3 What is the trend for the **number of sales transactions** during the first six **months** of 2023?

2 How do the **number of purchase transactions** compare across **weekdays**?

4 How do the **number of sales transactions** for the **first and second quarters** compare?

ILLUSTRATION 6.38 Answering Questions With the Occurs Relationship



Once again, we can use the information model to explore two of these questions. **Illustration 6.39** explores the first question with KLUB's occurs information model.

- **Question 1:** How do the number of sales transactions (measure) compare across weekdays (dimension)?

The column chart breaks down the number of sales transactions by weekday.

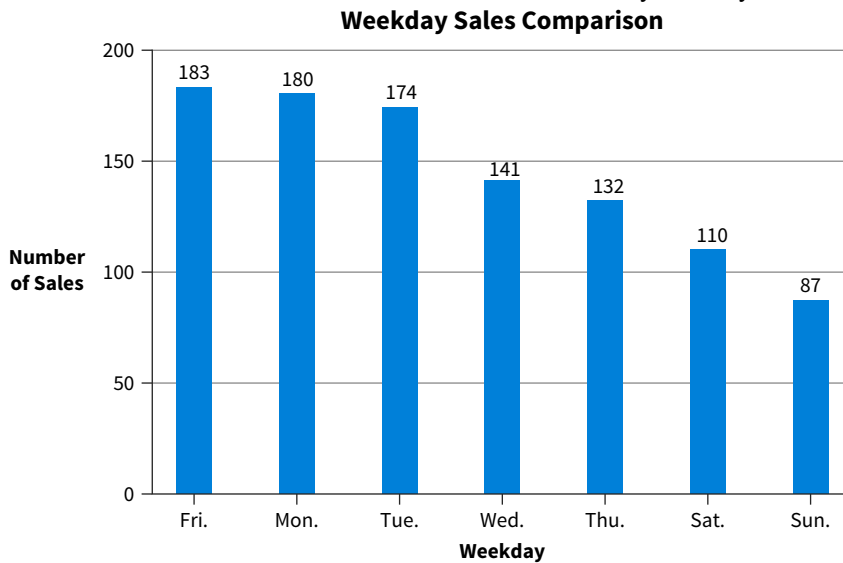


ILLUSTRATION 6.39 Comparing Number of Sales Transactions Across Weekdays

Illustration 6.40 explores another question.

- **Question 3:** What is the trend for the number of sales transactions (measure) during the first six months (dimension) of 2023?

The line chart breaks down the number of sales transactions by month and helps identify trends.

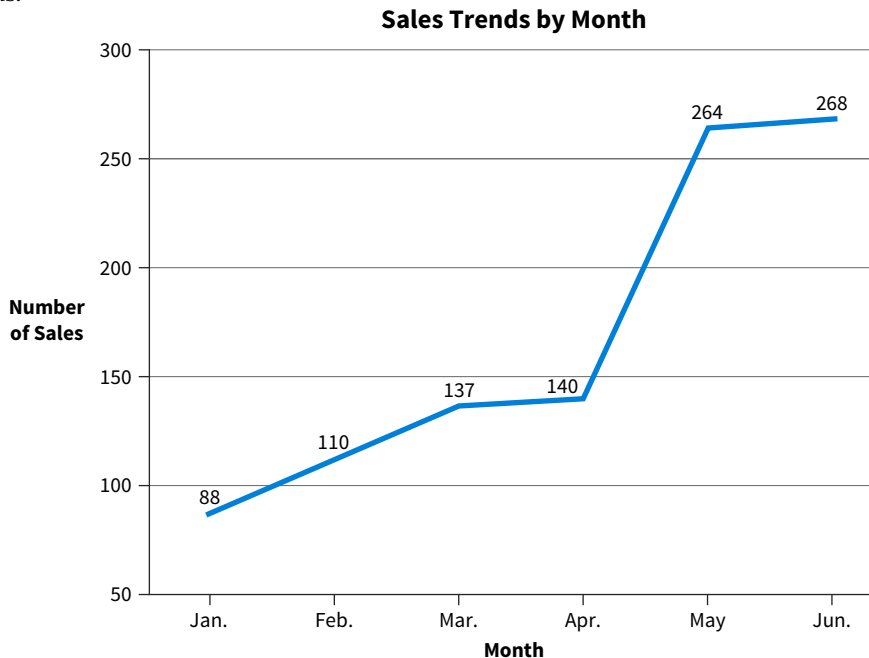
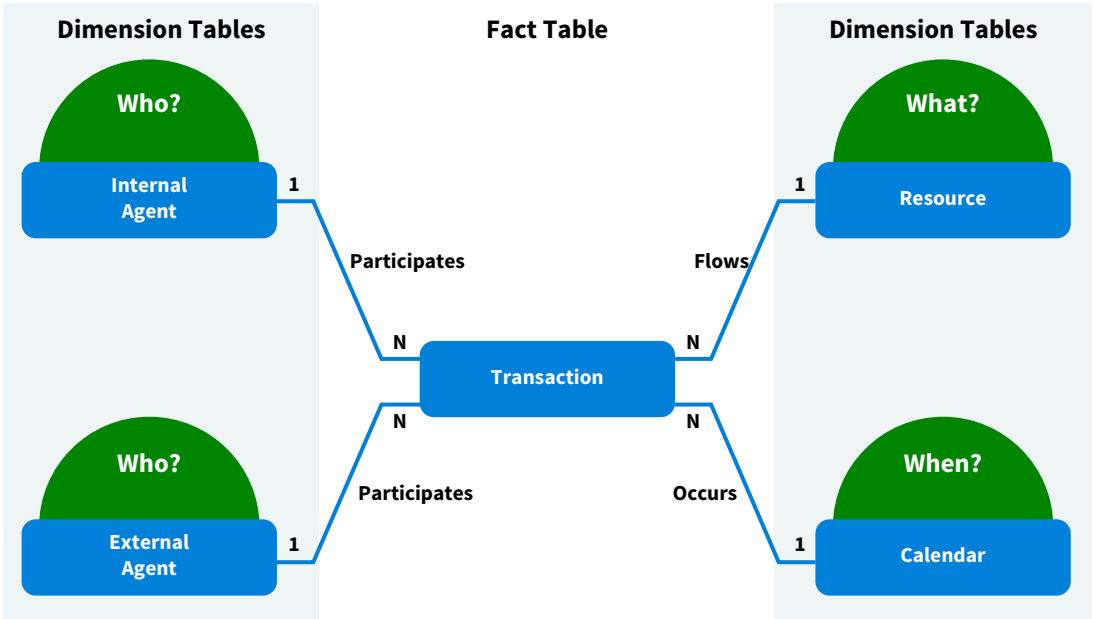


ILLUSTRATION 6.40 Identifying Trends for the Number of Sales Transactions Across Months

Information Modeling Pattern 12: Who-What-When Star Schema

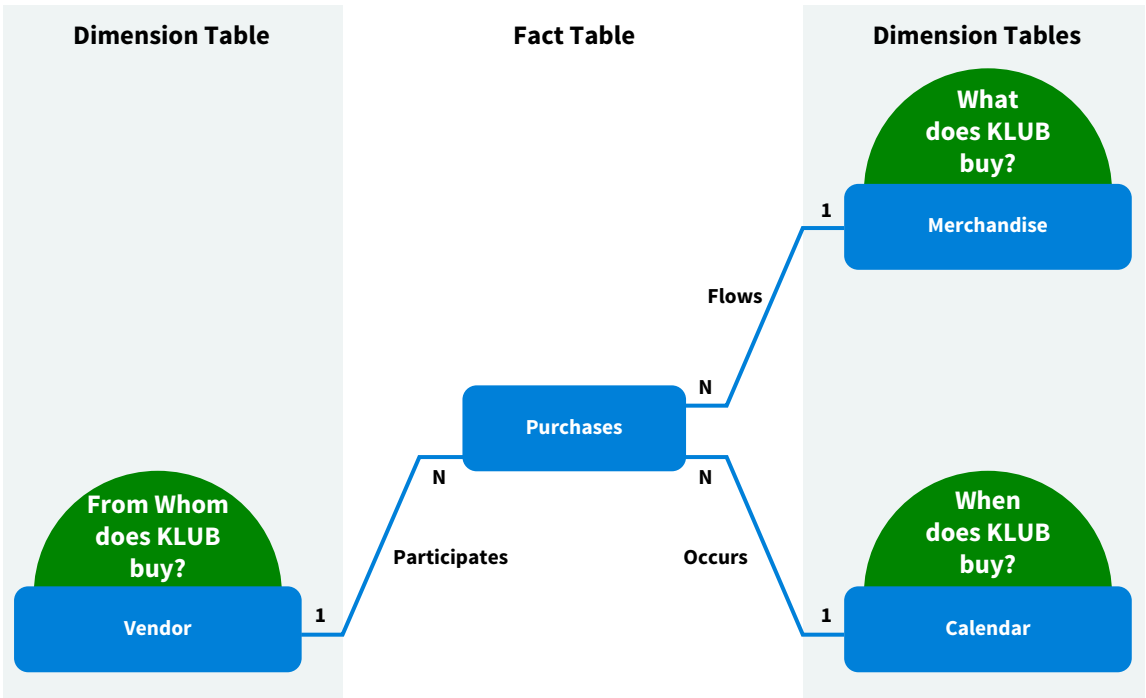
While patterns 9, 10, and 11 analyze the who, what, and when dimensions of accounting transactions, this pattern integrates these three relationships ([Illustration 6.41](#)).

ILLUSTRATION 6.41 The Who-What-When Star Schema Pattern



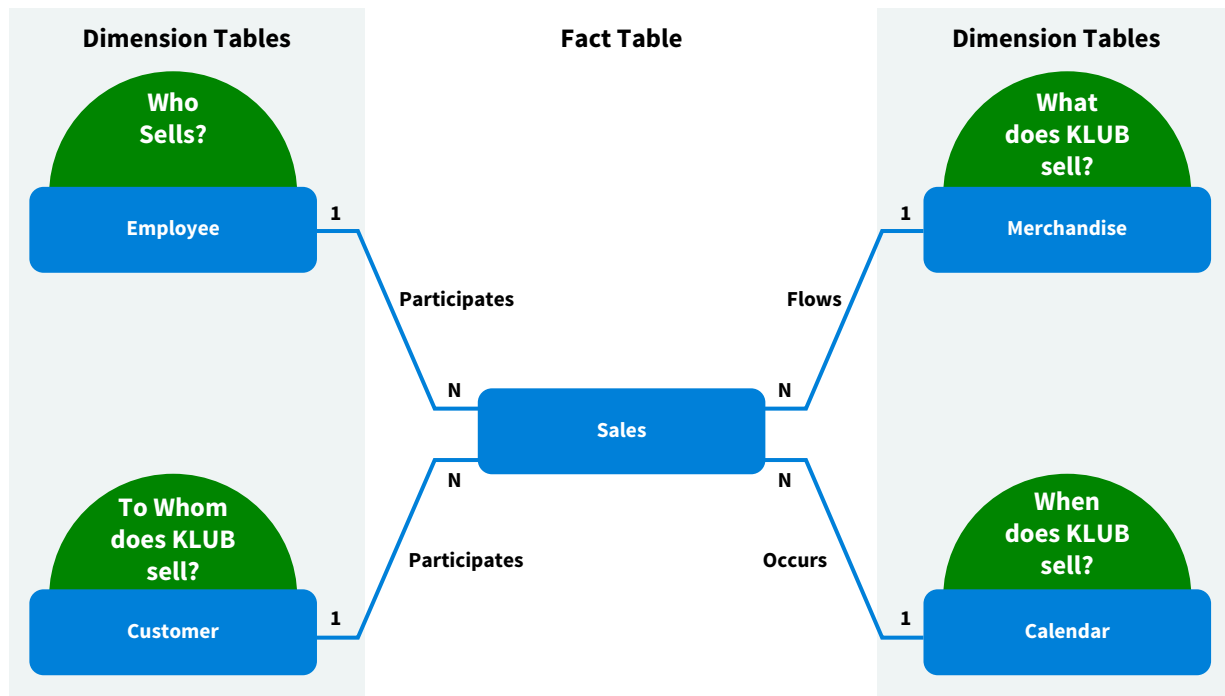
KLUB's data model integrates two instances of the who-what-when pattern. The first is for the acquisition of items, which are purchases ([Illustration 6.42](#)). There is no employee information to record for purchases because the owner of the company oversees the acquisition process.

ILLUSTRATION 6.42 The Purchases Who-What-When Star Schema Pattern



The second is for selling the items, or sales ([Illustration 6.43](#)).

ILLUSTRATION 6.43 The Sales Who-What-When Star Schema Pattern



Information Model

Recall that a data model contains all fields, while the information model also contains calculated columns and measures created for analysis purposes. Information models can be extended depending on business' specific needs. [Illustration 6.44](#) is the data model and information model for KLUB's Sales star schema.

To create a richer information model for KLUB, we added an extra calculated column called *TargetAmount*, as well as the three extra measures of *TotalTargetAmount*, *Discrepancy*, and *Discrepancy%* to the Sales star schema:

- For each batch, KLUB has a minimum target price per unit called *MinimumSalePrice*.
- The *Discrepancy* measure determines the difference between the actual price charged to the customer and the target price.
- *Discrepancy%* results from dividing the sales price by the target price. It indicates by what percentage the minimum target price is exceeded.

Among others, these measures can be used to analyze the negotiation skills of KLUB's salespeople. [Illustration 6.45](#) shows formulas to create the new calculated column (*TargetAmount*) and measures (*TotalTargetAmount*, *Discrepancy*, and *Discrepancy%*).

The calculated column and measures determine specific pieces of information:

- The *TargetAmount* column determines the dollar amount for an invoice line if the minimum sale price had been charged to the customer. Its algorithm follows Pattern 4: Across-Table Calculation.
- The *TotalTargetAmount* measure determines the total amount KLUB would have received if the company had charged the target prices for all its sales. Its algorithm follows Pattern 5: Single-Column Aggregation.
- The *Discrepancy* measure determines the discrepancy between total revenues and total target amounts. The “–” symbol represents subtraction. This is an aggregate that can be sliced in many ways. Its algorithm follows Pattern 7: Measure Hierarchies.
- The *Discrepancy%* measure compares actual revenues with targeted revenues as a percentage. Its algorithm also follows Pattern 7: Measure Hierarchies.

ILLUSTRATION 6.44 Data and Information Models for the Who-What-When Sales Star Schema

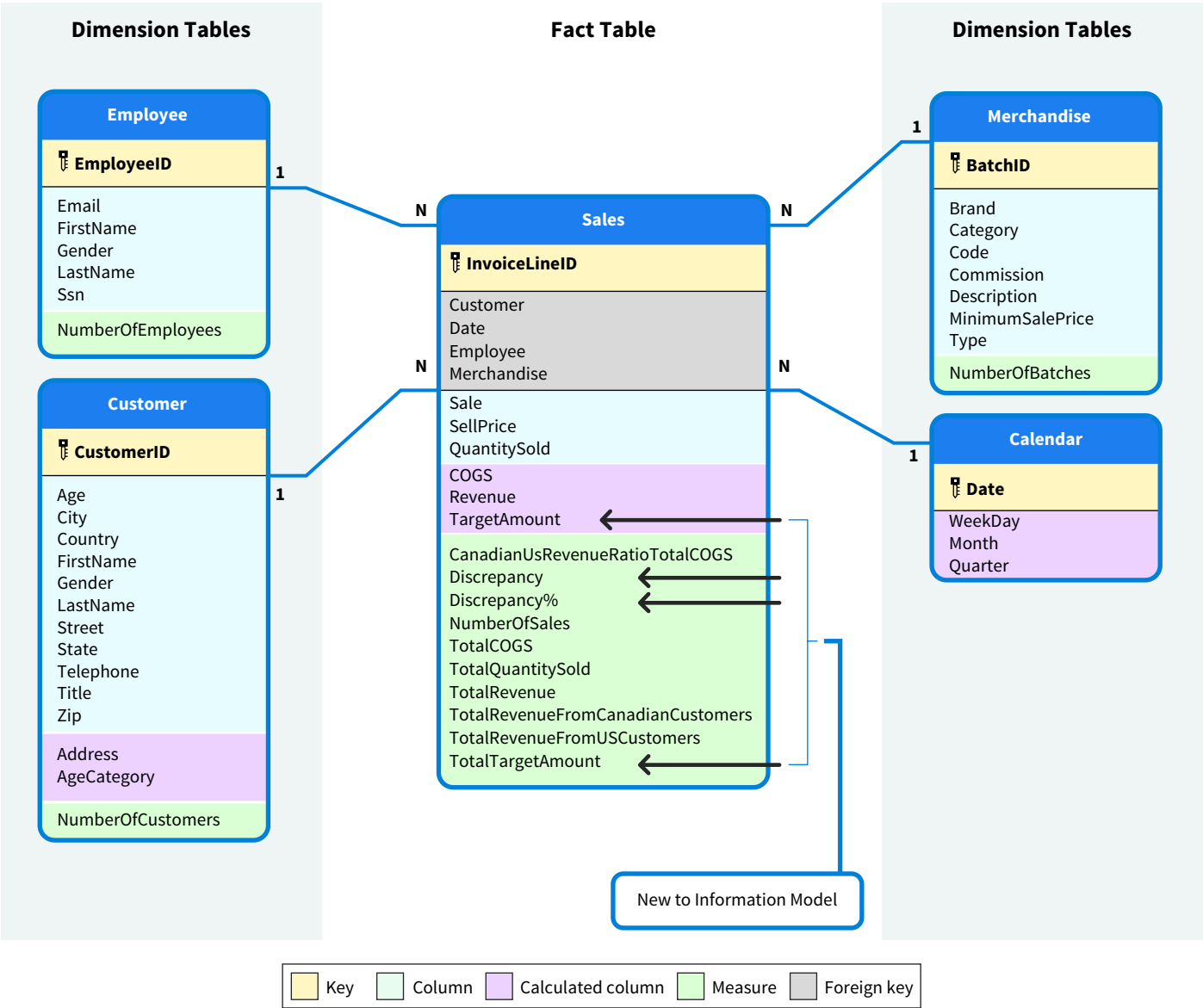


ILLUSTRATION 6.45 Additional Calculated Columns and Measures for the Who-What-When Sales Star Schema

Table: Sales	
Calculated Column	Algorithm
TargetAmount	$\text{Sales.QuantitySold} \times \text{Merchandise.MinimumSalePrice}$

Table: Sales	
Measures	Algorithm
TotalTargetAmount	$\text{SUM}(\text{TargetAmount})$
Discrepancy	$[\text{TotalRevenue}] - [\text{TotalTargetAmount}]$
Discrepancy%	$[\text{TotalRevenue}] / [\text{TotalTargetAmount}]$

Analysis

A who-what-when star schema is a powerful structure for the analysis of accounting data. Any transaction measures in the fact table can be broken down by any combination of dimensions (who, what, and/or when), and these breakdowns can be shaped through a variety of visualizations.

The information models for the Sales and Purchase star schemas have been gradually developed in this chapter as we created the necessary measures and calculated columns. A who-what-when star schema is an effective way to perform an analysis because it highlights *what* to break down (the measure) and *how* to break it down, which is an intersection of dimensions. Star schemas can be developed independently of the software being used. **Illustration 6.46** contains just a few examples of questions that can be analyzed using the sales star schema shown in Illustration 6.44.

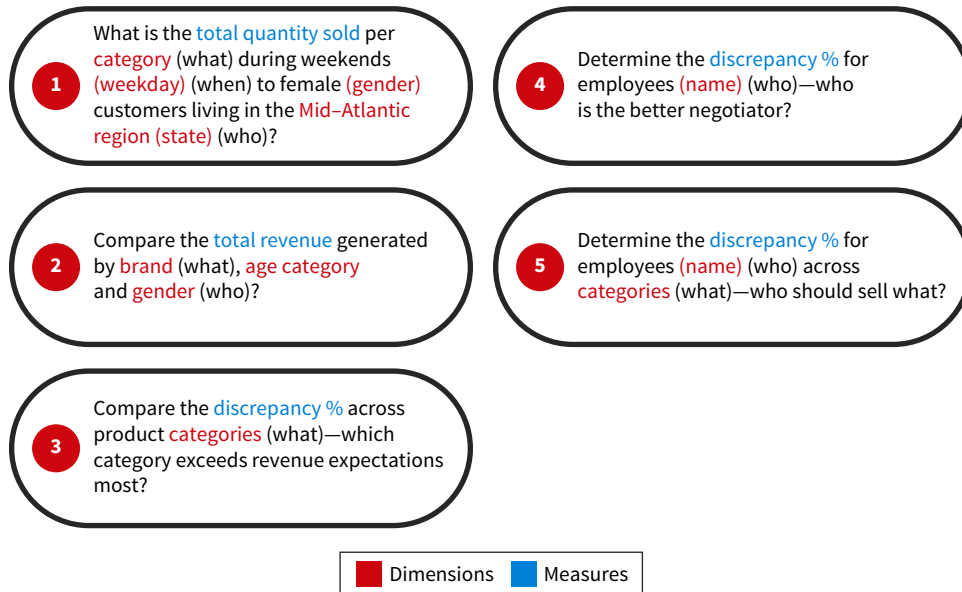


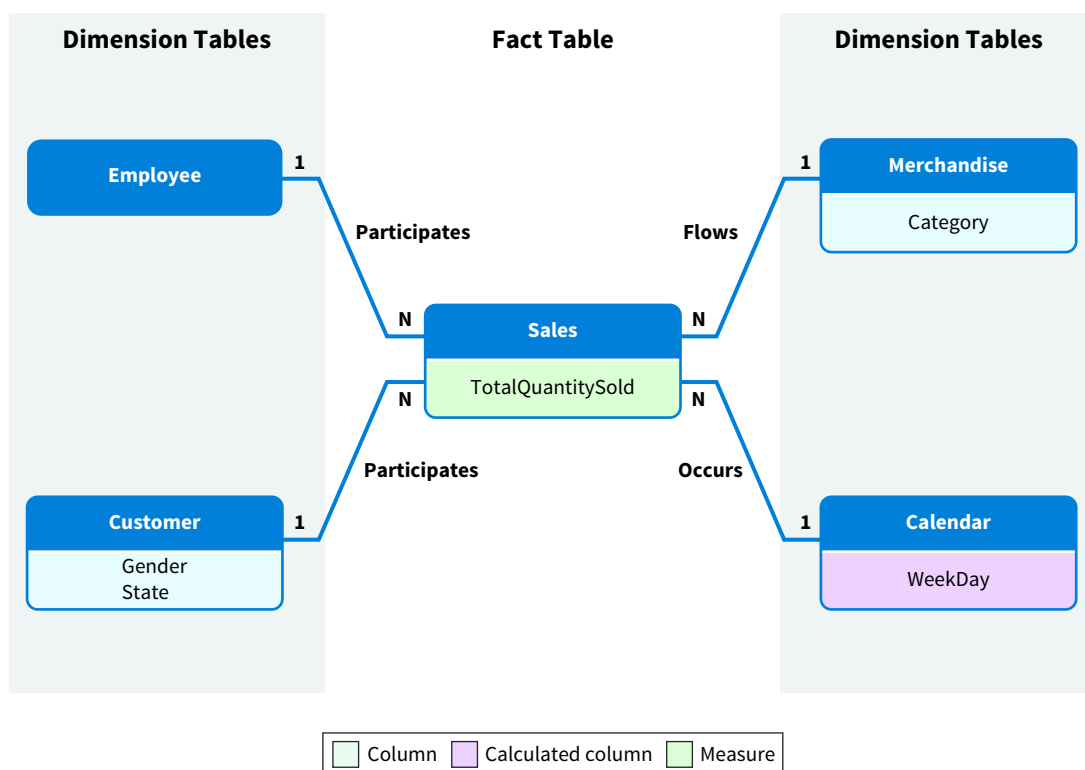
ILLUSTRATION 6.46 Answering Questions with the Sales Star Schema

Next, we explore two of these questions.

- **Question 1:** What is the total quantity sold (measure) per category (what dimension) during weekends (weekday) (when dimension) to female (gender) customers living in the Mid-Atlantic region (who dimensions)?

Illustration 6.47 shows how the who-what-when star schema can answer this question.

ILLUSTRATION 6.47 Who-What-When Star Schema for Quantity Sold Analysis



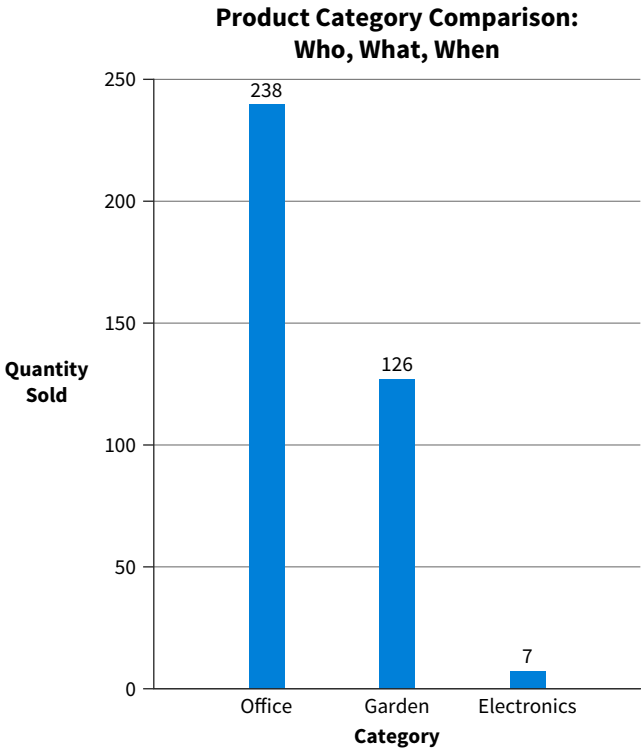
This is a query-type question that uses filters, so the next step is to create the algorithm that integrates these filters ([Illustration 6.48](#)).

ILLUSTRATION 6.48 Implementation of the QuantitySoldOnWeekdaysToFemaleMidAtlanticCustomers Measure

Table: Sales	
Measure	Algorithm
QuantitySoldOnWeekdaysToFemaleMidAtlanticCustomers	<code>SUM(QuantitySold)</code> <code>WHERE Calendar.Weekday = 1 or 7, AND</code> <code>Customer.Gender = "Female", AND</code> <code>Customer.State = "PA" OR "MD" OR "DC" OR "VA" OR "WV" OR "DE" OR "NJ" OR "NY"</code>

Finally, the number generated by QuantitySoldOnWeekdaysToFemaleMidAtlanticCustomers must be sliced by product category. The column chart in [Illustration 6.49](#) shows this slicing.

ILLUSTRATION 6.49 Product Category Comparison: Quantity Sold to Mid-Atlantic Female Customers on Weekdays



Next, use the information model to explore another question.

- **Question 4:** Determine the discrepancy % (measure) for employees (name, which is a who dimension).

The answer to this question can reveal which employees are the better negotiators. This is an exploration-type of question. [Illustration 6.50](#) shows how the who-what-when star schema answers this question.

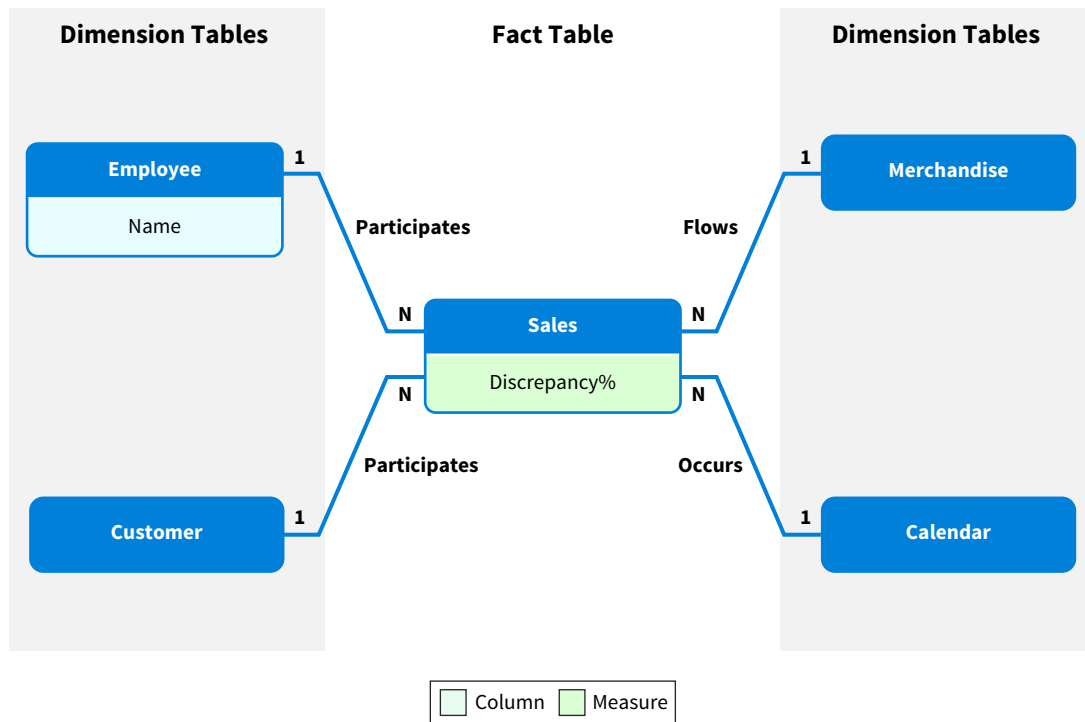
ILLUSTRATION 6.50 Who-What-When Star Schema for Discrepancy% Analysis

Illustration 6.51 slices the Discrepancy% measure by employee (name), which reveals the negotiation power for each employee. To generate more robust insights, sort the percentages in descending order.

EmployeeName	Discrepancy%
Gianni, Kobierski	12.06
Jade, Steeden	4.80
Arlyn, McCreery	3.43
Rosabel, Gabbitus	2.94
Dalenna, Patillo	2.86
Valery, Bayles	2.86
Catherin, Tolliday	2.66
Haze, Coomer	2.59
Florella, McAllan	2.56
Hermina, Jeandillou	2.47
Padriac, Parlour	2.44
Alyda, Greenhall	2.40
Tamra, Lambdin	2.38
Berti, Pridding	2.36
Loreen, Polye	2.36
Catharine, Laimable	2.35
Audie, Braniff	2.33
Charlie, Libri	2.27
Clim, Orhtmann	2.25
Chancey, Donke	2.25
Aarika, Bircher	2.20
Edik, Knowles	2.19
Silvain, MacTague	2.07
Sally, Blizard	1.94
Timmie, Brookzie	1.93

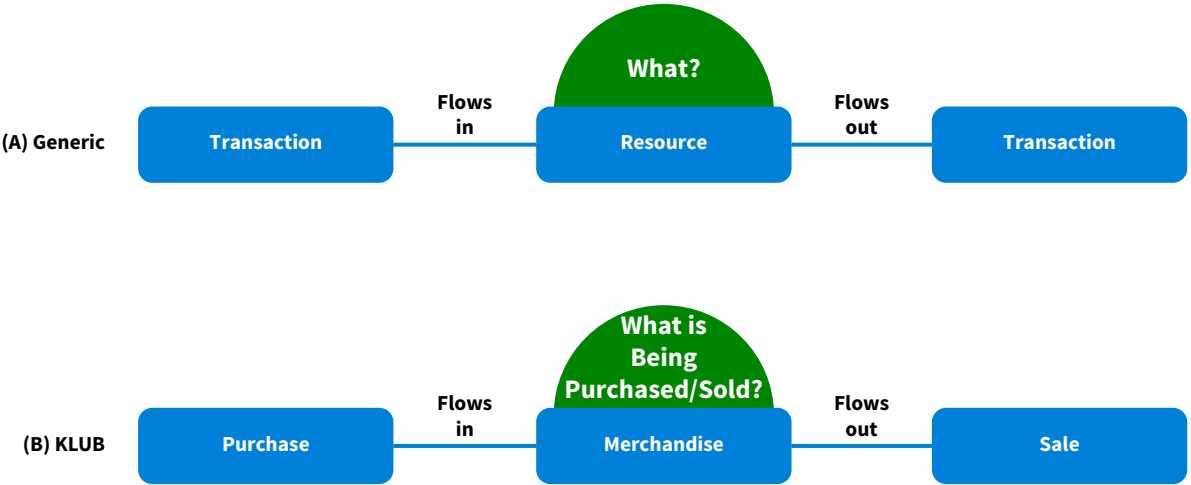
ILLUSTRATION 6.51 Comparing Employees Based on Discrepancy%

Information Modeling Pattern 13: Integrated Star Schemas

Illustration 6.52 (A) shows how who-what-when stars are connected through resources. Resources are being acquired for a purpose—to sell or to use them for manufacturing, for example. Illustration 6.52 (B) applies the pattern to KLUB. Because KLUB buys items in batches

and sells them to make a profit, the Sales and Purchases star schemas are connected through the Merchandise table.

ILLUSTRATION 6.52 Integrated Star Schemas Pattern



Additional examples of integrated stars that are not part of the KLUB data set are:

- Cash Receipts to Cash to Cash Disbursements.
- Fixed Asset Acquisitions to Fixed Asset to Depreciation.

Information Model

This pattern describes the flow of resources in terms of quantity and value. It helps answer questions about how many resources have been acquired, which is the inflow, how many are in stock, and how many have been used and how (such as sales). [Illustration 6.53](#) provides an overview of accounting concepts, the information model, that can be derived from integrating KLUB’s Sale and Purchase star schemas as shown in [Illustration 6.52](#).

ILLUSTRATION 6.53 Information Model for KLUB’s Integrated Star Schemas

Concept/Field	Field Type	Description
Revenue	Calculated Column	The dollar amount paid by the customers for items sold.
COGS	Calculated Column	The dollar amount paid by KLUB for the items they sold.
Profit	Calculated Column	The profit KLUB made on the items they sold.
NetProfit	Calculated Column	The net profit KLUB made on the items sold after paying a commission (additional cost) to their employees (salespeople).
TotalRevenue	Measure	Total revenue generated from sales.
TotalCOGS	Measure	Total dollar amount paid by KLUB for items they sold.
TotalProfit	Measure	Total profit generated from sales.
TotalNetProfit	Measure	Total net profit generated from sales. Commissions for employees are considered.
OverallNetProfit Margin	Measure	The overall net profit margin for KLUB’s sales. Total net profit / total revenue.

Again, there are many ways to implement the enhanced information model associated with the integrated star schemas. The formulas for the additional columns and measures are shown in [Illustration 6.54](#).

Table: Sales	
Calculated Columns	Algorithm
Profit	Revenue – COGS
NetProfit	Sales.Profit – (Sales.Revenue × (Merchandise.Commission)/100)

Table: Sales	
Measure	Algorithm
TotalProfit	SUM(Profit)
TotalNetProfit	SUM(NetProfit)
OverallNetProfitMargin	[TotalNetProfit] / [TotalRevenue]

The Profit column is implemented with the within-table numeric calculation implementation pattern (Pattern 1). The implementation of the NetProfit column is more complex, given that information from both the Sales table and the Merchandise table is needed. This is an application of the across-table calculation implementation pattern (Pattern 4).

The implementation of the TotalProfit and TotalNetProfit measures are both applications of the single-column aggregation implementation pattern (Pattern 5), summing the values of a calculated column. Finally, the OverallNetProfitMargin measure is an application of the measure hierarchy implementation pattern (Pattern 7).

The information model can be further expanded with more specific measures. For example, we could calculate the total profits resulting from electronics sales in the U.S. A possible implementation for this measure is shown in [Illustration 6.55](#), which is an application of the filtered aggregation implementation pattern (Pattern 6).

Table: Sales	
Measure	Algorithm
ProfitsForUSElectronicSales	TotalProfit Where Customer.Country = “US” AND Merchandise.Category = “Electronics”

ILLUSTRATION 6.54 Extending KLUB’s Information Model

ILLUSTRATION 6.55 Implementation of the ProfitsForUSElectronicSales Measure

Analysis

This extended information model can answer a variety of questions regarding KLUB’s financial situation ([Illustration 6.56](#)).

- 1

What is the **overall net profit margin** for the different product **categories**?
- 2

Who are the 25 **customers** (name) with the **highest net profit margin**?
- 3

Which printer **brand**, HP or Epson, has the highest **profit margin**?
- 4

Which employees (**name**) have generated at least \$1,000 in **profit**?
- 5

How do Canadian provinces (**state**) rank based on **total profit** generated?
- 6

What is the trend (**month**) in **profits** generated by selling electronics (**category**) to U.S. (**state**) customers?

■ Dimensions ■ Measures

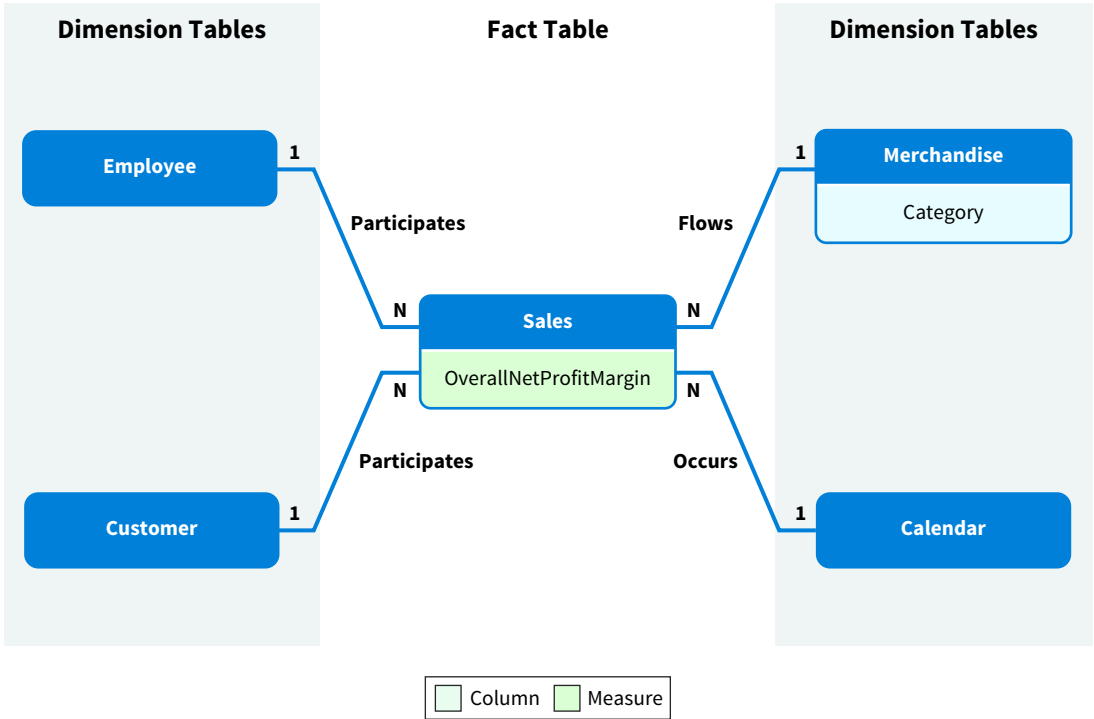
ILLUSTRATION 6.56 Answering Questions with KLUB’s Integrated Star Schemas

Next, we use KLUB’s integrated star schemas to further explore two of these questions.

- **Question 1:** What is the overall net profit margin (measure) for the different product categories (dimension)?

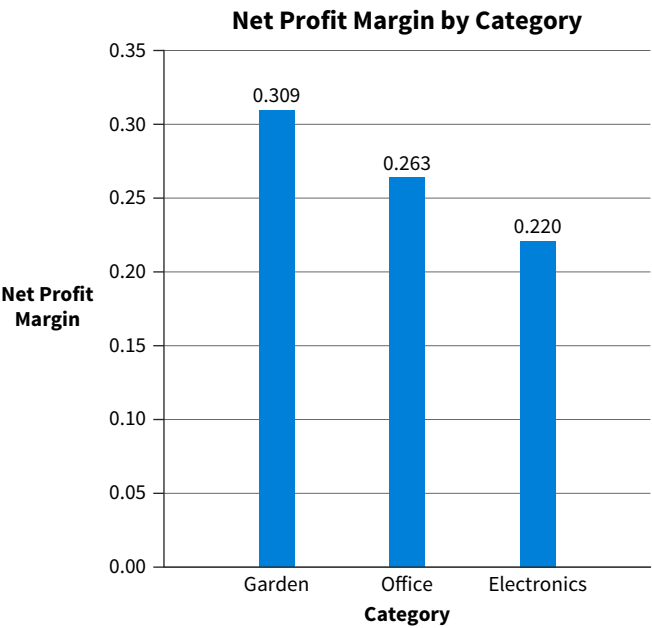
Illustration 6.57 shows how the star schema can be used to answer this question.

ILLUSTRATION 6.57 Star Schema for Net Profit Margin Analysis



The corresponding report is shown in **Illustration 6.58**.

ILLUSTRATION 6.58 Comparing Product Categories by Net Profit Margin

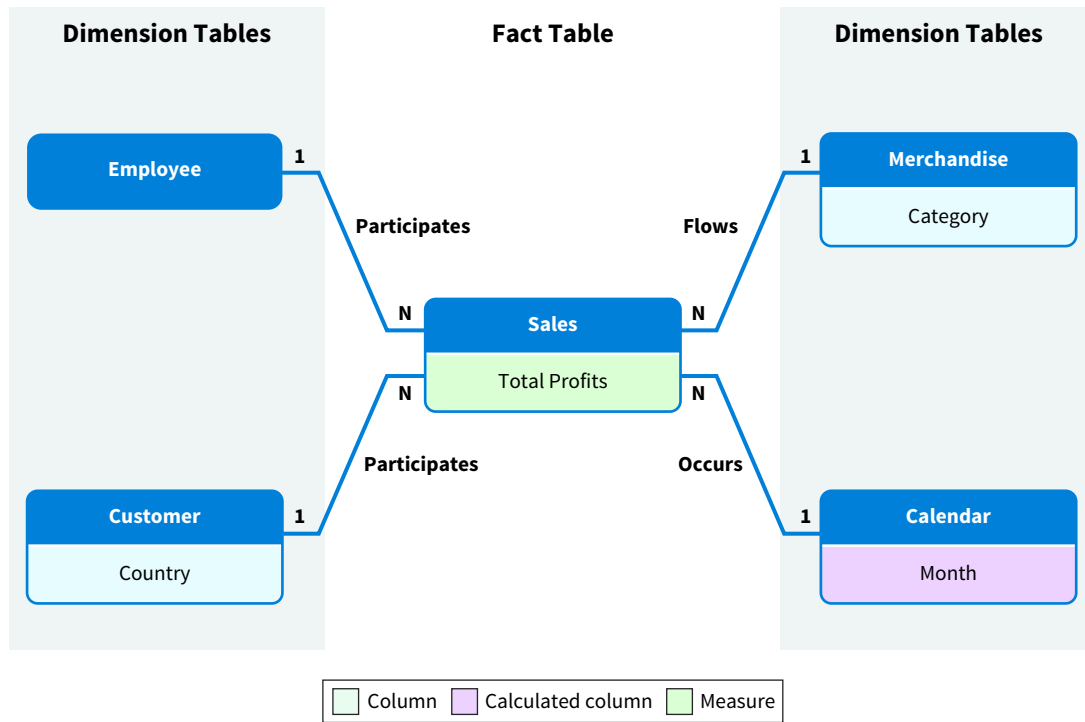


Category is used as a dimension, and net profit margin is calculated for each category (garden, office, and electronics) and represented as a bar. KLUB’s negotiation power seems to be strongest for garden items.

- **Question 6:** What if we wanted to know the trend (month, which is a dimension) in profits (measure) generated by selling electronics (category, which is a dimension) to U.S. (country, which is a dimension) customers?

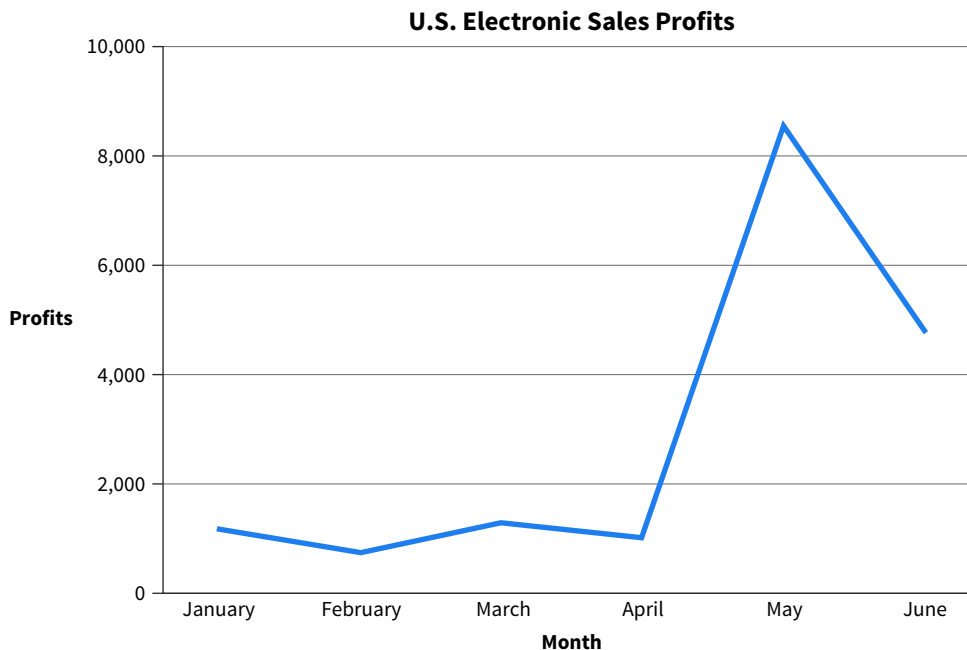
Illustration 6.59 shows the star schema for this question. It should be noted that an alternative star schema could be created where the ProfitsForUSElectronicSales measure is sliced by month.

ILLUSTRATION 6.59 Star Schema for Profit Analysis



The corresponding report is shown in **Illustration 6.60**. Profits for U.S. electronics sales are sliced by month, which is accomplished with a line chart.

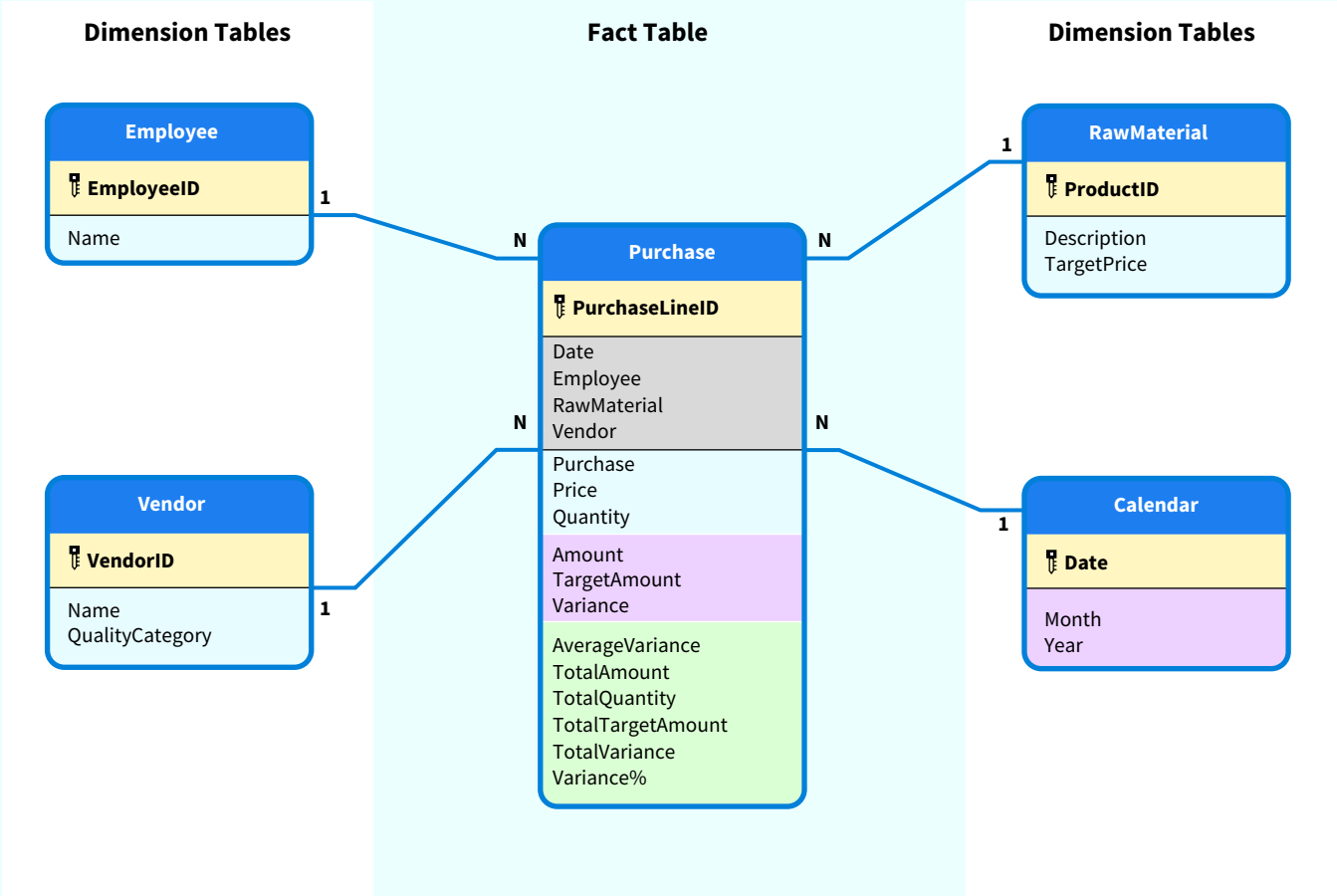
ILLUSTRATION 6.60 Trend Analysis of Profits Generated From U.S. Electronics Sales



Thirteen different patterns were applied to complete KLUB’s information model. Keep in mind that patterns are best practices. They are a guide for determining what information, or content, is relevant and how to implement it. These thirteen patterns will help you identify a range of relevant accounting information and code the algorithms that can generate such information.

Apply It 6.3
Answer Questions
with a Star Schema

Managerial Accounting The CEO at Abroad is positive that data analytics can also help on the production side of the business. The star schema summarizes the data and information captured about the acquisition of raw materials.



You are given the following table that has additional information about some of the calculated columns and measures.

Table	Field	Description/Algorithm
RawMaterial	TargetPrice	What Abroad is expected to pay for one unit of the raw material.
Purchase	Amount	Price × Quantity.
	TargetAmount	TargetPrice × Quantity.
	AverageVariance	TotalVariance/TotalQuantity
	Variance	TargetAmount – Amount
	Variance%	TotalAmount/TotalTargetAmount
Vendor	QualityCategory	Assessment of the vendor’s quality on a 1 to 5 scale, with 1 being the worst and 5 being the best.

You are asked to develop three questions that can be explored with this information model. Abroad's owner is especially interested in questions related to variances. Identify three potential questions and the dimensions and measures you would use to answer them.

SOLUTION

There are many different questions that can be answered using the star schema. Following are five examples.

1. Per raw material, which of the vendors with a quality category of at least four has the highest average variance?
 - Description (RawMaterial), Name (Vendor), and QualityCategory (Vendor) are dimensions.
 - AverageVariance (Purchase) is a measure.
2. Per raw material how has the overall average variance changed during the last 12 months?
 - Description (RawMaterial) and month (Calendar) are dimensions.
 - AverageVariance (Purchase) is a measure.
3. Who are the top three employees with the strongest negotiation skills, i.e., with the lowest variance percentage?
 - Name (Employee) is a dimension.
 - Variance% (Purchase) is a measure.
4. Who are the top three vendors with the strongest negotiation skills, i.e., with the highest variance percentage?
 - Name (Vendor) is a dimension.
 - Variance% (Purchase) is a measure.
5. Determine the raw materials for which the target price might have to be reviewed; i.e. the raw materials with a variance percentage of greater than 1.2 or less than 0.8.
 - Description (RawMaterial) is a dimension.
 - Variance% (Purchase) is a measure.

Chapter Review and Practice

Learning Objectives Review

1 Describe the foundational concepts of information modeling.

Successful information modeling requires an in-depth understanding of its foundational concepts:

- Information modeling is a process. Algorithms transform data into information.
- Information models are built using calculated columns and measures. Calculated columns are new columns with values that are calculated, row-by-row, from existing data. Measures are aggregates resulting from vertical calculations that can be broken down in many different ways.
- Star schemas are the preferable data structure for data analytics. Measures are created for the fact table and are sliced, or broken down, with the fields in the dimension tables. A robust set of measures and dimensions results in a rich information model that enables analytics.
- Who-what-when stars structure accounting data and make accounting analytics easier.

2 Apply common information modeling algorithms.

Seven foundational algorithms provide a strong base for developing information models:

- Pattern 1 is within-table numeric calculation. It adds a new column to a table by applying arithmetic operators such as addition, subtraction, multiplication, and division to columns in the same table.
- Pattern 2 is within-table text calculation. It adds a new column to a table by applying text manipulation operators such as concatenation, extraction, and replacement to columns in the same table.
- Pattern 3 is within-table classification. It adds a new column to a table by applying a classification algorithm to data in other columns in the same table.
- Pattern 4 is across-table calculation. It creates a new column by using data from different tables. Tables are linked through relationships, and algorithms are required to navigate these relationships.

- Pattern 5 is single-column aggregation. It aggregates all the values in a single column using mathematical operators such as SUM AVERAGE, and COUNT.
- Pattern 6 is filtered aggregation. It filters the data in a single column based on one or more criteria and then aggregates the remaining values.
- Pattern 7 creates a measure hierarchy. It creates a new, more complex measure using existing measures.

3 Develop and implement information models for common accounting data structures.

Six common accounting data structures can help identify an information model's measures and dimensions:

- Pattern 8 is the how many pattern. It counts existing agents, transactions, and resources.

- Pattern 9 is the participates | transaction-who pattern. It captures information about who is involved in the transactions. Agents participating in an accounting transaction can be external to an organization, e.g., customer, or internal, e.g., salesperson.
- Pattern 10 is the flows | transaction-what pattern. It captures information about what resources are involved in the transaction. A transaction can result in an inflow or outflow of resources.
- Pattern 11 is the occurs | transaction-when pattern. It captures information about when a transaction occurs.
- Pattern 12 is the who-what-when star schema pattern. It integrates the who, what, and when dimensions of transactions and supports a wide range of accounting analyses.
- Pattern 13 is the integrated star schemas pattern. It combines multiple star schemas and enables integrated analyses of resource inflows and outflows.

Illustration 6.61 is an overview of the thirteen patterns.

ILLUSTRATION 6.61 Overview of Information Modeling Patterns

ID	Algorithm	Description	Implementation
Implementation Patterns			
1	Within-Table Numeric Calculation	Creates a new calculated column by applying arithmetic operators to columns in the same table.	Basic arithmetic operators: addition, subtraction, multiplication, division, or exponentiation.
2	Within-Table Text Calculation	Creates a new calculated column by applying text manipulation operators to columns in the same table.	Text manipulation operators such as concatenate, extract, or replace.
3	Within-Table Classification	Creates a new calculated column that contains new knowledge by applying conditional logic.	Boolean expressions and Boolean operators such as AND, OR, and NOT.
4	Across-Table Calculation	Data from different, linked tables are used to create a new calculated column.	Navigation operators such as joins.
5	Single-Column Aggregation	Creates a new measure that aggregates all the values of a single column.	Aggregate functions such as SUM, COUNT, AVERAGE, MIN, and MAX.
6	Filtered Aggregation	Creates a new measure that aggregates a subset of the values of a single column.	Apply filters. Examples are SUMIF, COUNTIF, and AVERAGEIF functions in Excel or the WHERE clause in SQL.
7	Measure Hierarchies	Creates a new, more complex measure using existing measures.	Apply arithmetic operators to measures such as addition, subtraction, multiplication, or division.

(Continued)

ID	Information	Description	Implementation
Content Patterns			
8	How Many	Analyzes how many of a concept exists.	Create measures that counts rows in tables.
9	Participates Transaction-Who	Analyzes who is involved in a transaction.	Create measures for transactions and calculated columns for agents.
10	Flows Transaction-What	Analyzes what is involved in a transaction.	Create measures for transactions and calculated columns for resources.
11	Occurs Transaction-When	Analyzes when a transaction occurred.	Create measures for transactions and calculated columns for calendars.
12	Who-What-When Star Schemas	Integrates analysis of the who, what, and when transactions.	Create measures for transactions and calculated columns for agents, resources, and calendars.
13	Integrated Star Schemas	Analyze how inflows and outflows affect resources.	Create measures for transactions and calculated columns for resources.

Key Terms Review

Algorithms 6-2	Information model 6-4	Participates relationship 6-6
Calculated columns 6-3	Information modeling 6-1	What table 6-6
Data model 6-4	Measure 6-3	When table 6-6
Dimensions 6-3	Measure hierarchy 6-15	Who table 6-6
Flows relationship 6-6	One measure, multiple analysis principle 6-14	
Information 6-2	Occurs relationship 6-6	

How To Walk-Throughs



HOW TO 6.1 Create Calculated Columns and Measures with Power BI

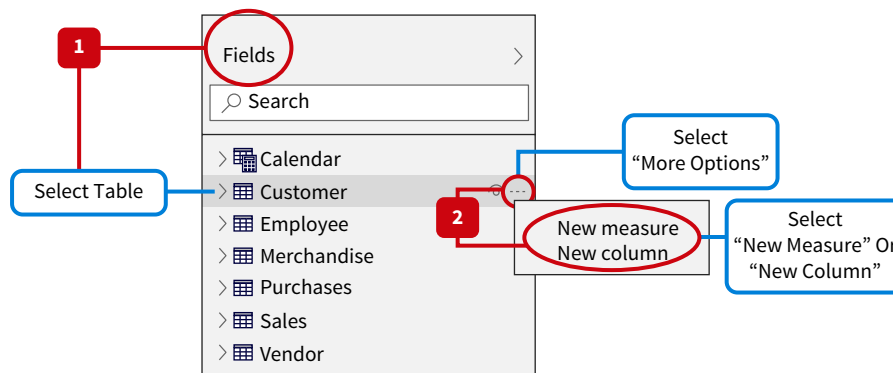
Calculated columns and measures are the building blocks of information models. Here, we illustrate how they can be specified with Power BI.

What You Need: **Data** The How To 6.1 data file.

STEP 1: In the **Fields** pane, select the table to which you want to add a column or a measure. In **Illustration 6.62**, the Customer table is selected.

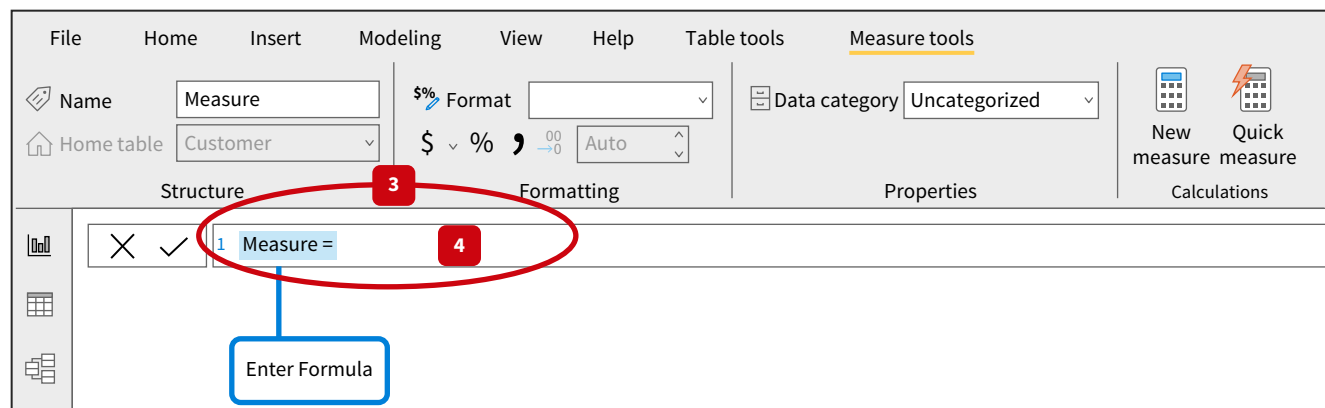
STEP 2: Next to the table name, click **More Options** and select either **New Measure** or **New Column**.

ILLUSTRATION 6.62 Creating Columns and Measures



STEP 3: An area in which to enter the formula that creates the new column, or measure, will appear (**Illustration 6.63**).

ILLUSTRATION 6.63 Entering a Formula



STEP 4: Enter a formula. For example, enter the following formula to create a measure which determines the number of customers:

NUMBER OF CUSTOMERS = COUNT(Customer[CustomerID]).

Then, press the **Enter** key.

HOW TO 6.2 Implement a Filtered Aggregation With SQL

The patterns discussed in this chapter are not specific to a software or language. Here, you can practice implementing filtered aggregation with relational database software (e.g., Microsoft Access) using the Structured Query Language (SQL).

What You Need: **Data** The How To 6.2 data file.

STEP 1: Open the file using Microsoft Access and select the **Create** tab in the Main Menu. Click **Query Design** in the **Queries** group of the ribbon.

STEP 2: When a new ribbon appears in the **Results** group, select **SQL View**.

STEP 3: Enter the SQL command shown in **Illustration 6.64**.

ILLUSTRATION 6.64 SQL Query for Determining the Total Revenue Generated from Sales to U.S. Customers by Product Category

Query: Total Revenue from US Customers by Product Category

```
SELECT Category, SUM([SellPrice] * [QuantitySold]) AS [TotalrevenueFromUSCustomers]
FROM Merchandise INNER JOIN (Customer INNER JOIN Sales ON Customer.CustomerID = Sales.Customer)
ON Merchandise.BatchID = Sales.Merchandise
WHERE Country="US"
GROUP BY Category
```

Detailed Descriptions Code

Code

Description

SUM([SellPrice] * [QuantitySold])

Calculates the total revenue generated from sales.

WHERE Country="US"

A filter defined through a Boolean expression. Only sales from U.S. customers are considered.

GROUP BY Category

The total revenue is broken down by category. There are three values for CATEGORY: Electronics, Garden, and Office.

FROM Merchandise INNER JOIN (Customer INNER JOIN Sales ON Customer.CustomerID = Sales.Customer) ON Merchandise.BatchID = Sales.Merchandise

Defines the joins between the tables and thus the navigation path.

STEP 4: Select the Exclamation Mark (!) in the **Results** group of the ribbon. The table shown in **Illustration 6.65**, which summarizes revenues generated from sales to U.S. customers by (product) category, will appear.

ILLUSTRATION 6.65 Query Results

Category	Total Revenue From US Customers
Electronics	34,553
Garden	80,012
Office	210,560

Data The Data tag appears when the data required to answer a question or complete an exercise are available on Wiley's online learning platform.

Multiple Choice Questions

1. (LO 1) Which of the following statements is *false*?
 - a. Calculated columns are the center of data analytics. Once calculated, they can be sliced in many ways during data exploration.
 - b. A measure hierarchy can be built by applying arithmetic operators to existing measures, such as dividing existing measures.
 - c. A calculated column is an integrated part of a table.
 - d. Calculated columns and measures are both implemented using algorithms.
2. (LO 1) Information modeling is the process of
 - a. defining relationships between tables and the constraints that apply to them.
 - b. generating additional knowledge from data that is relevant for analysis purposes.
 - c. creating star and/or snowflake schemas.
 - d. creating data structures that make analytics easy.
3. (LO 1) Which of the following does *not* describe an accounting algorithm?
 - a. Calculating depreciation expenses using the straight-line method.
 - b. Creating a line chart that shows changes in an organization's net profit margin during the last five years.
 - c. Determining last year's net profit margin.
 - d. Calculating how much overhead should be allocated to different product categories.
4. (LO 1) Assume a grocery store recorded the following information for each transaction.

TransactionID	Date	Amount	Loyalty	LoyaltyDiscount
1	2/1/2025	\$ 144.17	Y	1.44
2	2/1/2025	\$ 35.80	N	0
3	2/2/2025	\$ 289.09	N	2.89
4	2/2/2025	\$ 134.88	Y	1.35
5	2/3/2025	\$ 99.50	N	1

The information model consists of three measures and a calculated column. Which of the following is the calculated column?

- a. Calculation of the loyalty discount. Members of the loyalty program get a 1% discount on all transactions.
 - b. Calculation of the % of transactions from members of the loyalty program.
 - c. Calculation of the total dollar amount generated on a specific day.
 - d. Calculation of the total loyalty discount amount paid on a specific day.
5. (LO 1) Star schemas represent data structures that help determine which numbers to break down (facts) and how (what, who, when) to break them down (dimensions). Which of the following scenarios is *not* compliant with such a structure?

Scenario	Facts	Dimensions		
		What	Who	When
a.	Total revenue	Product category	Customer	Month
b.	Total revenue		Region	
c.	Total profit	Product type	Region	Weekday
d.	Total number of customers		Region	
e.	Total number of sales	Product category	Customer	
f.	Total number of sales			Month
g.	Total cost	Product type		

6. (LO 2) Which of the following statements regarding algorithms would you disagree with?
- Boolean expressions are an integral part of classification algorithms.
 - Analytics based on data from multiple tables heavily rely on properly defined relationships among the tables.
 - A measure hierarchy defines a relationship between measures from two different tables.
 - SUM, AVERAGE, and COUNT are mathematical operations often used to define measures.
7. When applying the following classification algorithm to the data set, which of the four customers (CustomerID) is assigned the value B?

Algorithm

```
IF ((Age < 30 AND LoyaltyProgram = "Yes") OR (Sales > 10,000 AND NumberOfIssues = 0))
THEN "A"
ELSE "B"
```

Data set

CustomerID	Name	Age	LoyaltyProgram	Sales	NumberOfIssues
1	Kinsun	32	Yes	\$ 13,833	0
2	Barbara	29	No	\$ 15,000	2
3	Jon	29	Yes	\$ 15,000	2
4	Gabriella	25	Yes	\$ 20,000	0

- 1
 - 2
 - 3
 - 4
8. (LO 2) Which of the following is *not* a within-table text calculation?
- Combining two text fields, such as combining a customer's first name John, and last name Doe, resulting in John Doe.
 - Replacing a number of characters in a text field, such as replacing "str" with "street" in an address field.
 - Extracting a substring for a text field. For example, extracting the area code 302 from telephone number 302-200-5731.
 - Counting the number of different cells in a text field. For example, counting the different values in the Country field.
9. (LO 2) When applying the following algorithm to the data set, which of the following implementation patterns are used?

Algorithm

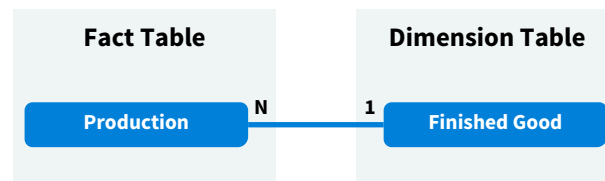
```
PRICE TYPE =
IF Price < 25,000 THEN "LOW-END" ELSE
IF Price >= 25,000 AND PRICE < 50,000 THEN "MID-RANGE" ELSE
"HIGH-END"
```

Data set

ProductID	Description	Year	Price
1	VW Beetle	2019	\$ 14,322
2	Honda Pilot	2018	\$ 25,119
3	BMW X7	2019	\$ 50,432
4	Chevrolet Cru..	2019	\$ 11,654
5	Mercedes ML	2017	\$ 29,750
6	Ford Explorer	2019	\$ 32,881
7	Honda Civic	2017	\$ 9,811
8	Chevrolet Tah..	2016	\$ 42,889
9	Jeep Wrangler	2007	\$ 9,999
10	Maserati Leva..	2020	\$ 69,922

- Within-table numeric calculation
- Within-table text calculation
- Within-table classification
- Filtered aggregation

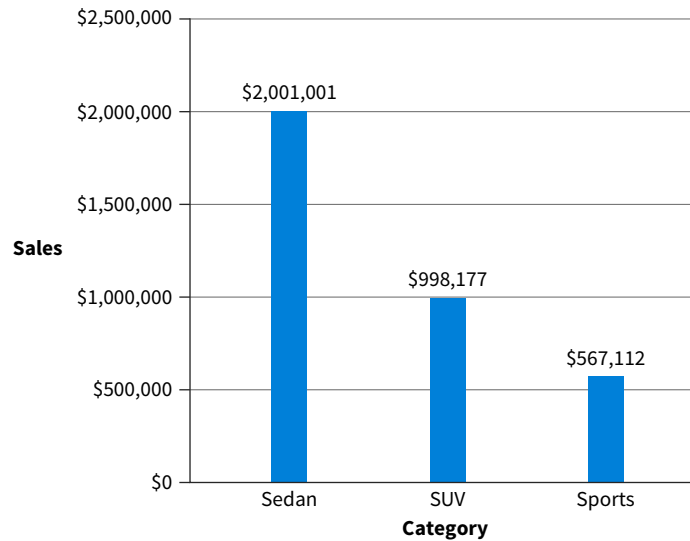
10. (LO 2) Which of the following statements regarding the coding patterns is *incorrect*?
- Ratios are most often implemented as measure hierarchies.
 - Across-table calculations strongly rely on matching primary key and foreign key values.
 - Single-column aggregations (measures) generate a singular value.
 - Often, the same formula combines numeric and textual calculations.
11. (LO 3) For what type of table or tables is the how many pattern most relevant?
- Fact tables.
 - Dimension tables.
 - Fact and dimension tables.
 - Neither fact nor dimension tables.
12. (LO 3) Which of the following statements regarding dimensional modeling is *incorrect*?
- Dimensional modeling aims to create data structures that make data analytics easier.
 - Dimensions can be used for filtering purposes.
 - Calculated columns can function as dimensions.
 - How many measures can be specified for dimension tables only.
13. (LO 3) The diagram is part of a star schema that describes a manufacturing process.



How would you describe the nature of the relationship between production and finished goods?

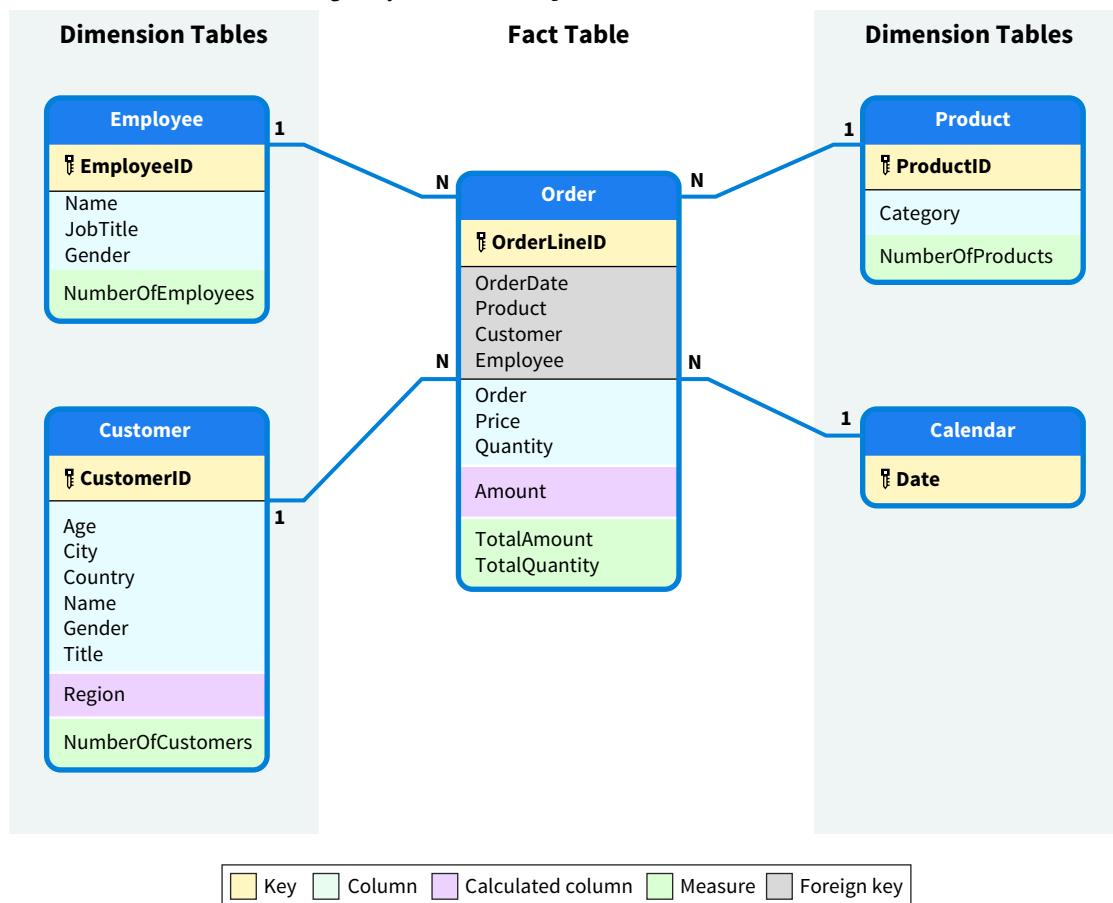
- A participates relationship.
 - An occurs relationship.
 - A flows relationship resulting in a decrease or outflow of finished goods.
 - A flow relationship resulting in an increase or inflow of finished goods.
14. (LO 3) Which of the following questions does *not* rely on a participates relationship?
- Which vendor offers the lowest price for a specific product?
 - What is the total units sold for each product category?
 - For each product category, who is the customer that bought the highest quantity last year?
 - Which employee has the worst return ratio?
15. (LO 3) Which of the following questions does *not* rely on a flows relationship?
- How many payments did John, an accounting clerk, make in January?
 - How many units did we produce in January?
 - How much money was deposited in the Fulton bank account in January?
 - Which customer bought the most electronics products?
16. (LO 3) Who-what-when stars are connected through
- transactions (e.g., sales).
 - resources (e.g., finished goods).
 - internal agents (e.g., employees).
 - external agents (e.g., customers).

17. (LO 3) This column chart was generated by a car manufacturer.



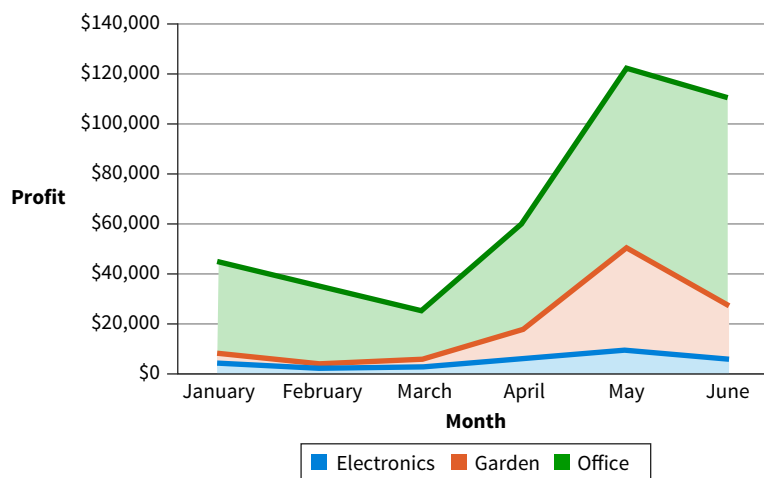
Which is the most accurate description of the accounting information captured by this chart?

- The average revenue generated by category.
 - The total revenue generated.
 - How much revenue each category has generated (inflow).
 - How much revenue each category has generated (outflow).
18. (LO 3) Which of the following analyses *cannot* be explored with this information model?



- The number of different products ordered by region.
- The total quantity ordered by female customers from female employees.
- Monthly changes in quantities ordered.
- The relative contribution of each region to the total quantity ordered.
- The top three customers per state based on dollar amount ordered.

19. (LO 3) What type of analysis is performed with this stacked area chart?



- a. Who-what
- b. Who-when
- c. What-when
- d. Who-what-when

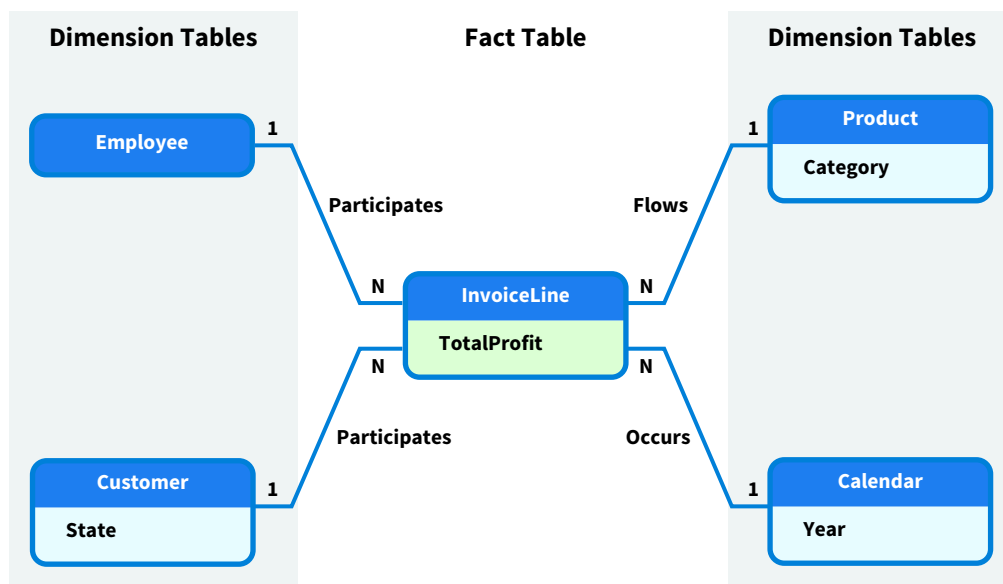
Review Questions

- (LO 1) Describe the goal of information modeling.
- (LO 1) What are the two main tasks involved in information modeling?
- (LO 1) What is the difference between a data model and an information model?
- (LO 1) What are the two different types of information fields and how do they differ?
- (LO 1) What are the two goals of star schema information models?
- (LO 1) What is an algorithm? Provide an example of an accounting algorithm.
- (LO 2) Provide two examples of text calculations.
- (LO 2) What is Boolean logic and how can it be used for classification purposes?
- (LO 2) Calculations might require information from multiple tables. Describe the mechanism that is used to connect tables.
- (LO 2) Describe the one measure, multiple analyses principle.
- (LO 2) For each operator, describe whether you most likely would use it to create a calculated column or a measure.

Mathematical Operator	Type of Information Field
Sum	
Multiplication	
Count	
Concatenation	
Average	
Division	
Exponentiation	
Standard Deviation	

- (LO 2) What are filtered measures?
- (LO 2) For what purpose is a measure hierarchy used?
- (LO 3) What are the who, what, and when dimensions of accounting transactions?
- (LO 3) What accounting information is captured by a flows relationship?
- (LO 3) What is the difference between physical and monetary flow information?

17. (LO 3) Examine this model.



1. What are you comparing?
 2. What field do you use for comparison purposes?
 3. What fields are you using for filtering purposes?
18. (LO 3) What kind of information do calendar tables contain?
19. (LO 3) Describe the difference between internal and external agents. For purchase transactions, who are the internal and external agents?
20. (LO 3) What information is captured by the occurs relationship, and why is such information crucial in an accounting context?
21. (LO 3) For the Customer table, what relationships between the dimensions can you identify? Describe how such relationships can be used for analysis purposes by identifying a few questions that can be answered using these relationships.

Customer	
	CustomerID
	CustomerName
	State
	LoyaltyGroup
	LoyaltyProgramEnrollmentDate

Brief Exercises

BE 6.1 (LO 1) Use the following terms to complete the sentence.

- | | |
|--------------|----------------------|
| a. algorithm | c. information model |
| b. data | d. star schema |
1. A(n) _____ is a data model extended with calculated columns and measures.
 2. The formula that determines the break-even point is an example of a(n) _____.
 3. A(n) _____ is the preferred data structure for data analysis.
 4. _____ is the input of an information model.

BE 6.2 (LO 1) Match the appropriate term to each definition.

- | | |
|----------------------|----------------------|
| a. Algorithm | d. Information model |
| b. Calculated column | e. Measure |
| c. Data model | f. Star schema |

Definition	Term
1. An aggregate that is calculated from a set of data that can be used for analytical purposes.	
2. A specification of the structure of a data set showing what concepts are described as tables and what characteristics are used to describe the concepts.	
3. An extension to the data model that consists of calculated columns and measures.	
4. A set of instructions that transform data into information.	

BE 6.3 (LO 1) The two building blocks of an information model are calculated columns and measures. Determine how the following five algorithms are best implemented: Either as a calculated column or measure.

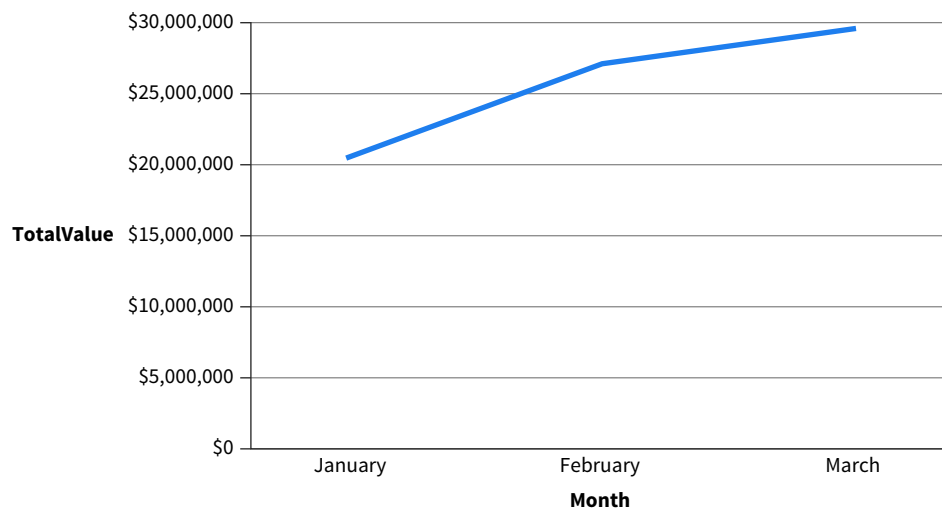
1. A publisher would like to determine, for each salesperson, the number of units sold for a specific title during the third quarter of 2025.
2. Determine the standard deviation for all prices paid in 2025 for a specific type of tire used by a bike manufacturer.
3. Given a fixed asset's net price, its number of useful years, and its salvage value, you are asked to determine its annual depreciation expense.
4. A company selling electronic gadgets would like to determine the product with the highest number of customer complaints.
5. For a company with more than a thousand employees, determine each employee's seniority starting from the hiring date.

BE 6.4 (LO 1, 3) Agree or disagree with each of the following statements. If you disagree, provide a brief explanation why.

1. Depreciation data describe a flows (inflow) relationship.
2. Data describing all the services a public accounting firm has offered to clients last month is an example of a flows relationship.
3. An audit procedure checking the vendors involved in purchases against an approved list of vendors is an analysis of a participates relationship.
4. Calendar tables are used for occurs relationships.

BE 6.5 (LO 2) **Data** **Financial Accounting** Tesla is an Austin, Texas, based electric vehicle and clean energy company. You just downloaded its inside trading report for the first quarter of 2022.

1. Build an information model for this data set that lets you compare the fluctuations in the total value of shares sold during the first three months of 2022.
2. Use your information model to create the following line chart.



BE 6.6 (LO 2) Data Managerial Accounting OCE is a large chemical company with headquarters in Birmingham (UK). One of the responsibilities of their controller is to manage projects. In January 2025, 200 projects were completed. The controller would like to know how many of them did not exceed their budgeted expenses.

1. Create the information model by writing the algorithms.
2. Create a pie chart that shows the number of projects that did and did not exceed their budgeted expenses.

BE 6.7 (LO 2) Data Financial Accounting Bellafonte is a small university with a sizeable endowment. Part of their endowment is invested in stocks. At the end of their fiscal year, they provide you the following information regarding their stocks:

- Ticker symbol
- Market Sector
- Price
- Number of stocks

They ask you to determine the total value of their stock portfolio and the relative importance of each market sector in it, presented as a pie chart.

1. Build the information model needed to generate such chart.
2. Create the pie chart.

BE 6.8 (LO 2) Data Managerial Accounting Somnum is a startup company that engineered a revolutionary mattress, the Meglio. They start with a week long sale to test the market reaction to the Meglio. A salesperson was placed in ten different malls, each with the goal of selling twenty-five mattresses per day. For each of the following questions, build an information model and perform the analysis.

1. Was the overall goal, which was selling 1,750 mattresses, achieved?
2. Which salespeople reached the goal of selling 175 mattresses?
3. Which salespeople made their target at least four out of seven days?

BE 6.9 (LO 2) Data Financial Accounting KOE is a small dairy company in southeast Canada. They track a variety of financial information on a monthly basis. They provide you with some financial information for the year 2025 and ask you to determine their 2025 net profit margin and compare the 2025 monthly net profit margins.

1. Build the information model.
2. Use the information model to perform this analysis.

BE 6.10 (LO 2) Data Managerial Accounting Charmes is a high-end clothing manufacturer. They offer three product lines: bags, coats, and shoes. Every item produced at Charmes goes through a rigorous inspection process. Last month the total inspection cost for all items was \$45,750. You are asked to determine how much of the inspection cost pool is allocated to each product line, using the number of units as the cost driver.

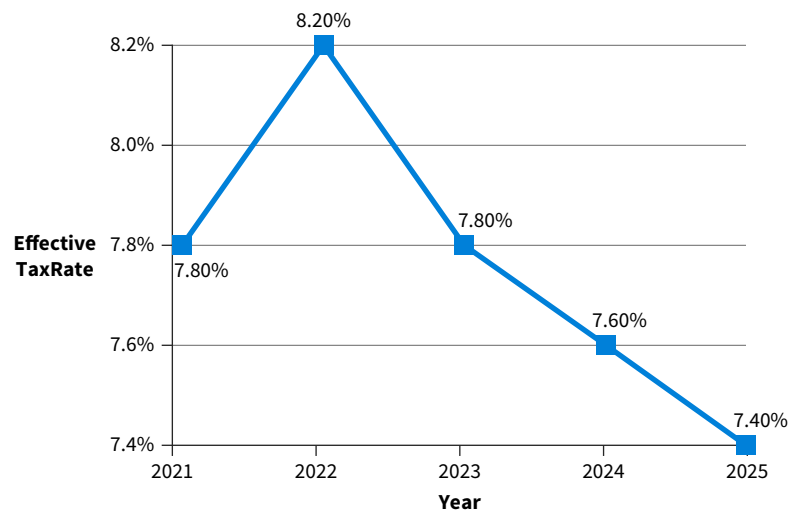
1. Build the information model.
2. Create a pie chart that shows how much of the inspection costs (in dollars) was allocated to each product line.

BE 6.11 (LO 2) Data Tax Accounting Danski is a shoe manufacturer in Oxford, PA. Five years ago, they hired a tax accountant, and the company is wondering whether the tax accountant was able to lower their effective tax rate. Danski's CEO gives you this worksheet.

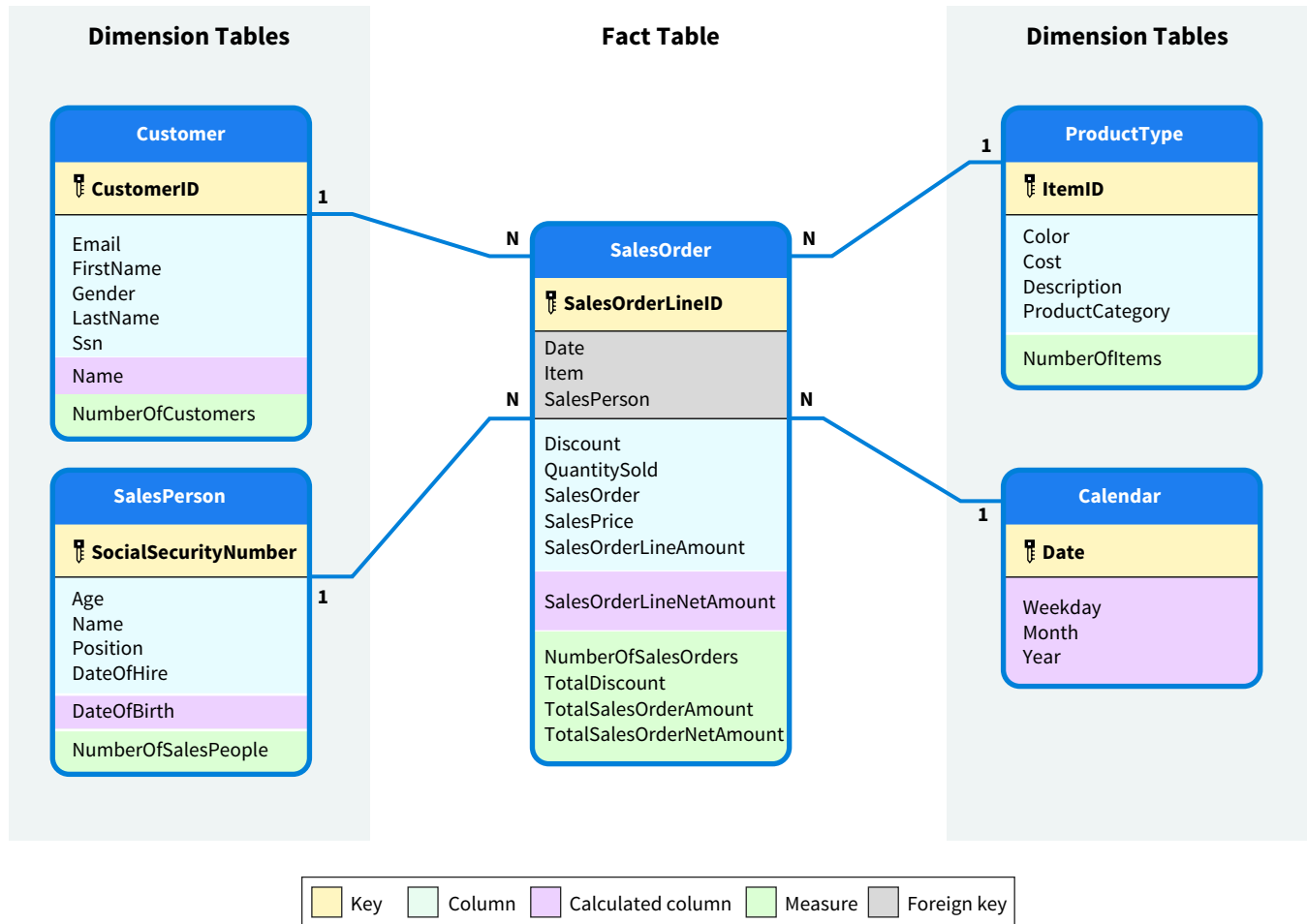
Year	Account	Amount
2021	Income Before Tax	\$ 14,365,450
2021	Income Tax Expense	\$ 1,120,505
2022	Income Before Tax	\$ 15,299,804
2022	Income Tax Expense	\$ 1,254,584
2023	Income Before Tax	\$ 165,998,110
2023	Income Tax Expense	\$ 12,947,853
2024	Income Before Tax	\$ 13,489,990
2024	Income Tax Expense	\$ 1,025,239
2025	Income Before Tax	\$ 175,009,898
2025	Income Tax Expense	\$ 12,950,732

1. Develop the information model that calculates the effective tax rate.
2. Use it to create the following report.

Year	EffectiveTaxRate
2021	7.80%
2022	8.20%
2023	7.80%
2024	7.60%
2025	7.40%



BE 6.12 (LO 3) Identify three issues with this star schema for sales orders. Briefly discuss how you would address each.

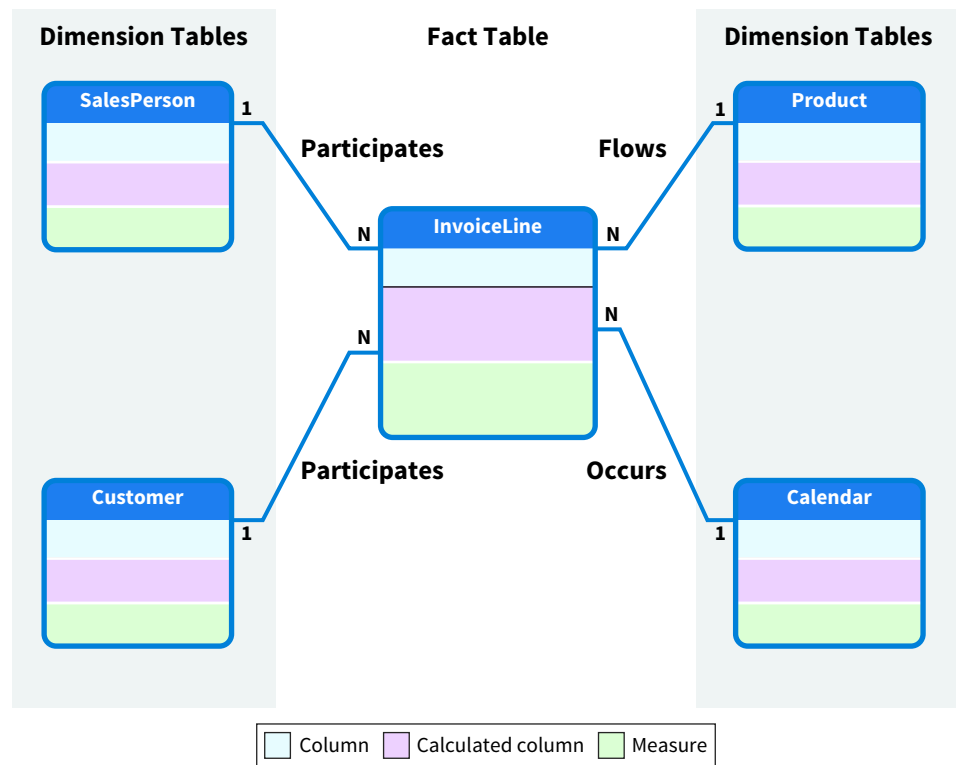


BE 6.13 (LO 3) Determine which pattern applies for each of the analyses described.

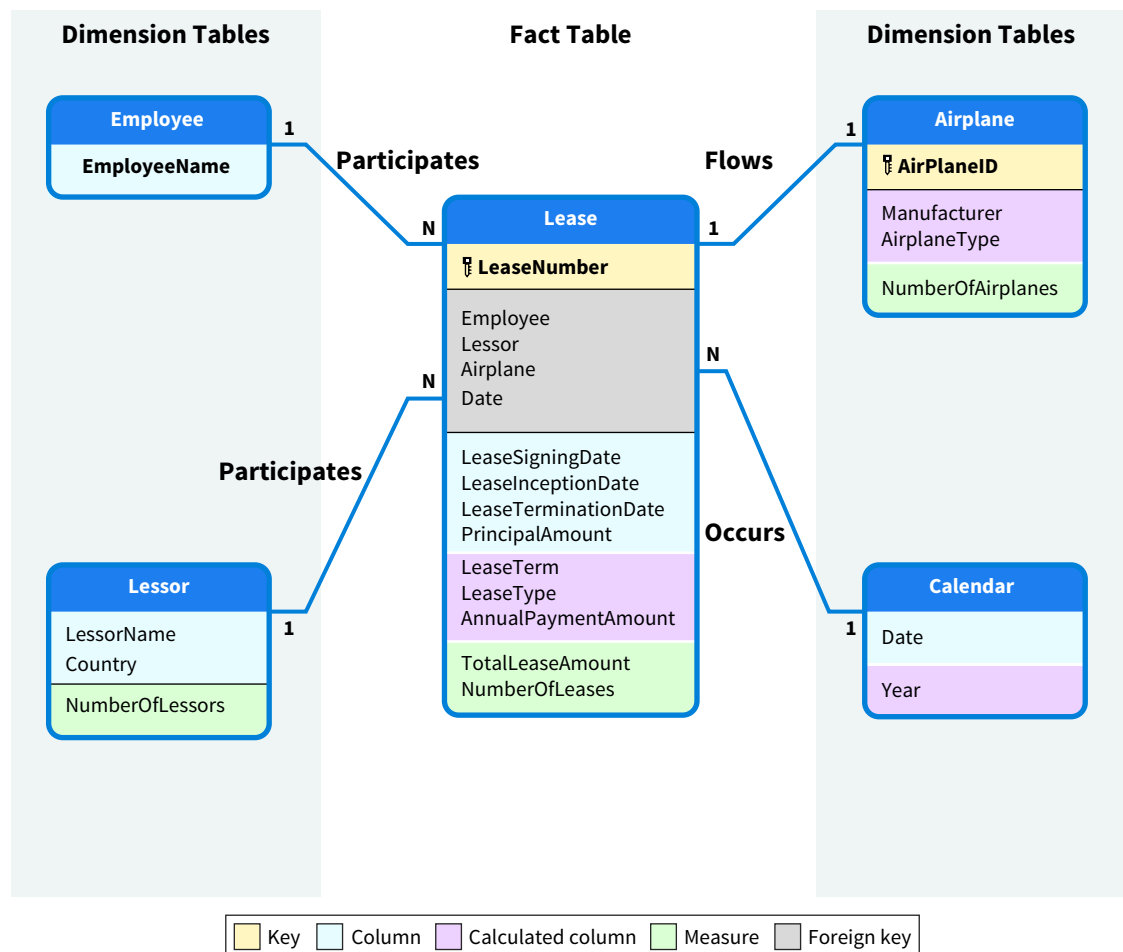
- | | |
|-------------|-----------------|
| a. flows | c. occurs |
| b. how many | d. participates |

Analysis	Pattern
1. Comparing the relative proportion of customers per sales region.	
2. An airline determining the number of hours flown on each type of airplane (e.g., Airbus 321 or Boeing 787) in use.	
3. The monthly fluctuations in the number of pesticide applications for a farm.	
4. A comparison of the number of sales completed by freelance salespeople versus full-time salespeople.	

BE 6.14 (LO 3) Assume you would like to explore how the total revenue generated by sales from senior salespeople to U.S. customers in 2025 differs across product categories. Salespeople are considered senior when they are with the company for at least 10 years after their hiring date. Complete the star schema.



BE 6.15 (LO 3) Mediterranean is a mid-size airline with flights primarily in Europe and Africa. They lease all their planes from lessors. Part of their star schema information model is shown here.



The table is the data dictionary for this information model.

Name	Description
EmployeeName	The name of the employee at Mediterranean who is accountable for the lease.
LessorName	The lessor from whom the plane is leased.
LessorCountry	The lessor's country.
NumberOfLessors	The total number of lessors from whom Mediterranean has leased planes.
LeaseNumber	A lease's ID.
LeaseSigningDate	The date the lease agreement was signed.
LeaseInceptionDate	The date on which the lease starts.
LeaseTerminationDate	The date on which the lease ends.
PrincipalAmount	Financing amount the lessee (Mediterranean) agrees to pay to the lessor.
LeaseTerm	The number of years a plane is leased.
LeaseType	Short-term or operating lease. Long-term or finance lease.
AnnualPaymentAmount	The annual amount to be paid for the lease.
TotalLeaseAmount	Total dollar amount for all leases.
NumberOfLeases	Total number of leases.
AirPlaneID	An airplane's unique ID.
Manufacturer	The airplane's builder—Boeing, Airbus, or Embraer.
AirplaneType	The airplane's type. For example, a (Boeing) 787 model.
NumberOfAirplanes	The total number of airplanes leased.

Determine at least five questions that can be analyzed starting from this information model.

Exercises

EX 6.1 (LO 1) Identify Data Relationships For each of the scenarios described, identify which relationship or relationships they describe: participates, flows, or occurs.

1. Dismissal of fixed assets.
2. Making a cash payment.
3. Generating quarterly sales reports.
4. Return of goods to a vendor.
5. Manufacturing a custom-made electric car.

EX 6.2 (LO 2) Data Financial Accounting Tax Accounting Build an Information Model for Reporting Net Revenues and Tax Curves is a custom-design furniture store. They provide you with a spreadsheet that contains the following information regarding their Q1 2025 sales:

- InvoiceNumber: Used to uniquely identify sales transactions.
 - Amount: A sale's gross dollar amount.
 - Discount: The discount, in dollars, to be deducted from the (sales) amount.
 - Tax: The tax amount, defined as a percentage, to be paid by the customer on the sales amount after the discount has been applied.
1. Create an information model that enables you to determine the total net revenue and the total tax to be collected by Curves for the Q1 sales.
 2. Once the information model is in place, create this report.

Total Net Revenue	\$3,319,960
Total Tax	\$136,463.17

EX6.3 (LO 2) Data Managerial Accounting Build an Information Model for Reporting Standard Costs FonzieBikes is a niche Canadian manufacturer of bikes and buggies. Their engineers and accountants have worked together to specify a Bill of Materials (BoM) which determines what raw materials are needed to make a finished good (product), how many (quantity), and what the expected (standard) cost is for each of them. You are given a data set that contains a sample of the BoM data.

FinishedGood	RawMaterial	STDQuantity	STDCost
Tricycle	Trike tire, large front	1	\$ 96
Tricycle	Tricycle frame	1	\$ 150
Tricycle	4" tire	1	\$ 64
Tricycle	Tricycle pedals	2	\$ 28
Mountain bike	18" Frame	1	\$ 213
Mountain bike	26" wheels and tires	2	\$ 34
Mountain bike	Mountain Bike handle	2	\$ 44
Mountain bike	Gear system	1	\$ 81
Buggy	Nuts	2,500	\$ 0.01
Buggy	Bolts	2,750	\$ 0.01
Buggy	Radio	1	\$ 111
Buggy	Truck body	1	\$ 430
Buggy	Engine	1	\$ 1,015
Buggy	Automotive tires	4	\$ 69
Buggy	Black Paint	40	\$ 2

1. Create an information model.
2. Use the information model to generate a report that shows the total estimated standard cost for each of the three products: tricycle, mountain bike, and buggy.

FinishedGood	TotalStandardCost
Buggy	\$ 1,964.50
Mountain bike	\$ 450.00
Tricycle	\$ 366.00

EX 6.4 (LO 2) Data Managerial Accounting Create a Report for Hotel Spending Emory & Grant is a regional accounting firm in the southeastern United States. When traveling to clients, their employees can stay at any Hilton or Marriott hotel. You receive a spreadsheet with all February 2025 hotel transactions and the amount paid for each stay. You are asked to build an information model that can determine two things:

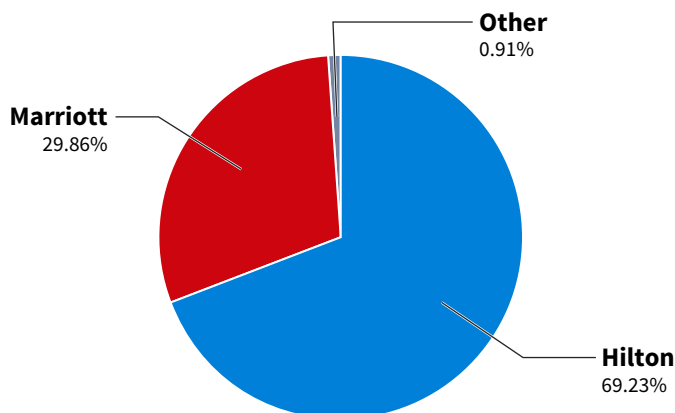
- The amount spent on Hilton hotels, the amount spent on Marriott hotels, and the amount spent on other hotels.
- The relative proportion of Hilton, Marriott, and other hotels in lodging expenses.

1. Develop the information model.
2. Then, create the following two reports:

Report 1: Amount spent per hotel chain.

Hotel	TotalAmountSpent
Hilton	\$ 170,154.52
Marriott	\$ 73,376.80
Other	\$ 2,225.38
Total	\$ 245,756.70

Report 2: Relative proportion of lodging expenses per hotel chain



EX 6.5 (LO 2) Data Create an Information Model to Support a Hiring Decision CreativeX is hiring a CFO. To be considered qualified, a candidate must meet the following three criteria:

- Between 30 and 50 years old (≥ 30 and ≤ 50).
- Expertise in either auditing or tax, or both.
- Currently employed as level 3 (manager) or higher.

The company provides you with the spreadsheet containing the information they received from the candidates.

Candidate	Age	Audit	Tax	Level
Liz Smith	33	Y	N	4
Jada Vaughn	29	Y	Y	3
Jenny Wang	52	N	N	1
Eric Kim	47	Y	Y	3
Misty Harrison	41	N	Y	4
George Star	39	Y	N	5
Elton Seger	25	N	N	3
Isabella Fernande	38	Y	N	2
Elise Jackson	41	Y	Y	2
Brenda Swift	25	N	N	5
Lindsay Lauper	56	N	Y	4
Nicole Sardo	35	N	N	2
Cindy Rafferty	40	N	N	2
Greg Cuddy	45	Y	N	3
Mateo Rodriguez	31	N	Y	4
Caleb Cooke	28	Y	N	2
Jane Harris	41	N	Y	5
Debra Pollack	29	Y	N	1
Sheldon Andrews	49	Y	Y	2

1. Create the information model.
2. Use the information model to generate a list with the names of the candidates that are qualified.

EX 6.6 (LO 2) Data Financial Accounting Build an Information Model for Reporting Net Revenues Old Marine (OM) is a Michigan clothing manufacturer. You are given a spreadsheet that contains information regarding all January 2025 orders. It has two worksheets:

- Orders: For each order, a unique ID, date, dollar amount, and shipment ID are provided.
- Shipments: For each shipment, a unique ID, the shipment type, the shipping cost (Amount), and the shipping company are provided.

There are two types of shipping:

- FOB Origin (**O**): The customer pays for the shipping costs.
- FOB Destination (**D**): Old Marine pays for the shipping costs.

The net revenue for an order is the order's amount minus the shipping cost paid by OM.

1. Create an information model that enables you to generate the total net revenue for January 2025 Orders.
2. Use the information model to create a report that shows this number.

EX 6.7 (LO 2) Data Financial Accounting Build an Information Model for Profit Margin Analysis Berok is a small investment company that wants to invest in one of the following three companies: ABC Groceries, Neighborhood Groceries, or Worldwide Groceries. They assembled a spreadsheet with revenue and profit data for each firm for the last five years. A key ratio for their financial analysis is profit margin.

1. Develop an information model that calculates the profit margin ratio.
2. Use the information model to create the following two visuals:
 - a. The profit ratios for each of the five previous years (2021–2025) for each of the three companies:

Name	2021	2022	2023	2024	2025
ABC Groceries	7.55%	7.94%	7.55%	7.14%	8.05%
Neighborhood Groceries	6.83%	6.76%	7.02%	6.67%	6.78%
Worldwide Groceries	7.62%	8.57%	7.52%	8.27%	8.59%

- b. The profit ratio based on the total revenues and profits for the 2021–2025 period for each of the three companies:

Name	Profit Margin 2021–2025
ABC Groceries	7.65%
Neighborhood Groceries	6.82%
Worldwide Groceries	8.08%

EX 6.8 (LO 2) Data Managerial Accounting Create an Information Model to Determine Employee Bonuses Ruppetware is a small manufacturer of innovative kitchen products. In addition to their fixed salaries, salespeople at Ruppetware can earn a substantial bonus at the end of the year. This is the algorithm to determine bonuses for 2025:

- The 2025 target sales amount for each salesperson is determined by adding 5% to the average 2024 sales amount for all salespeople.
 - Only salespeople that meet that target are considered for a bonus. The magnitude of the bonus is determined as follows:
 - If the target is exceeded by less than 5%, the bonus is \$5,000.
 - If the target is exceeded by at least 5% but less than 10%, the bonus is \$10,000.
 - If the target is exceeded by at least 10%, then the bonus is \$15,000.
1. Create the information model needed for such an analysis.
 2. Generate a list that shows the 2025 bonus amounts for all salespeople.

EX 6.9 (LO 2) Data Managerial Accounting Build an Information Model for Truckload Management Leno Transportation Services (LTS) is a transportation company that compiles a daily schedule linking available trucks to pallets that need to be transported that day. All transportations are local. They keep their data in three worksheets:

- The Truck worksheet contains a list of all available trucks. For each truck, LTS records a unique ID and the maximum weight it can carry (truckload).
- The Pallet worksheet contains a list of all pallets that must be transported on a given day. For each pallet, LTS records a unique ID and the total weight of the goods currently on it. Pallet weights are in kilograms (note, 1 Kilogram = 2.20462 pounds).
- The Schedule table contains the daily schedule and therefore determines which pallets must be loaded on which trucks.

Using the three worksheets, LTS asks you to create an information model that enables them to explore the following two questions:

- Is the maximum weight for any truck exceeded?
- What trucks have a truckload of less than 75%?

1. Create the information model.
2. Once your information model is in place, use it to create a table that provides a list of all trucks scheduled to go out today. For each truck, shows its ID, its maximum weight, its scheduled weight, and its scheduled load. Use a red background for trucks that exceed their maximum weight and a yellow background for trucks that have a truckload less than 75%. The report should look as follows:

TruckID	MaximumWeight	TotalWeight	Load
1	10,000	7,826.40	0.78
2	25,000	25,974.83	1.04
3	15,000	11,794.72	0.79
4	20,000	16,997.62	0.85
5	20,000	8,267.33	0.41
6	25,000	19,246.33	0.77
7	15,000	4,078.55	0.27

EX 6.10 (LO 2) Data Financial Accounting Create an Information Model to Show 401K Contributions Maurer and Cook is a small accounting firm in downtown Reno, Nevada. They offer to match \$0.75 of each dollar of their employees' 401K contributions following the graded vesting schedule shown here.

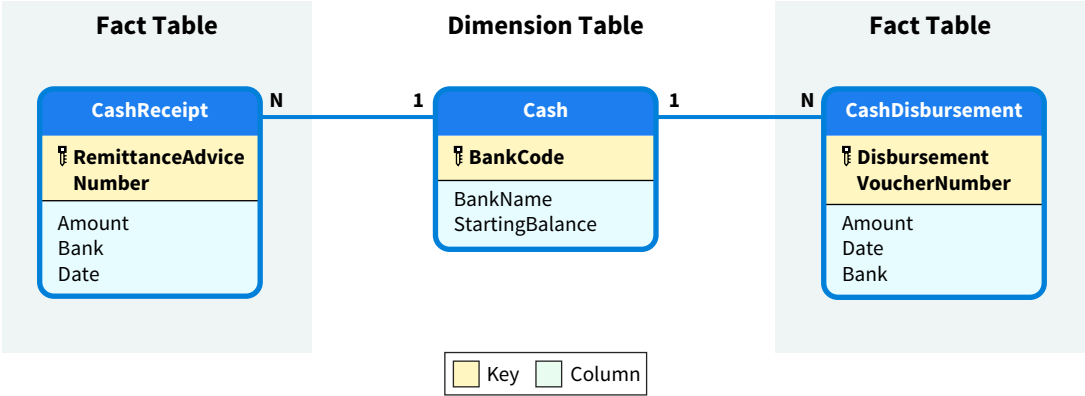
Years Employed	Vesting Schedule
0	0%
1	0%
2	20%
3	40%
4	60%
5	80%
6	100%

The managing partner gives you an Excel worksheet with the names of all employees, their years of service, and their total employee contributions.

1. Develop the information model.
2. Generate a report that shows each employee's total vested 401k contribution.

Employee	Total Vested Contribution
David Maurer	705,047.00
Mike Cook	638,946.00
Bernard Espinosa	438,663.75
Susan Hoover	437,302.25
Jamal Coleman	321,413.75
Elzbieta Jacek	234,108.00
Gail Wolcott	78,167.25
Billy Houston	50,912.40
Carlos Martinez	37,590.80
Amy Winter	30,483.70

EX 6.11 (LO 2, 3) Data Financial Accounting Create an Information Model for Cash Flow Analysis Valentina Turner owns The Blue Ballroom, which is a dance studio in Nevada. She has a spreadsheet that records incoming and outgoing payments since the beginning of the year (2025). The structure of the data set is shown here.

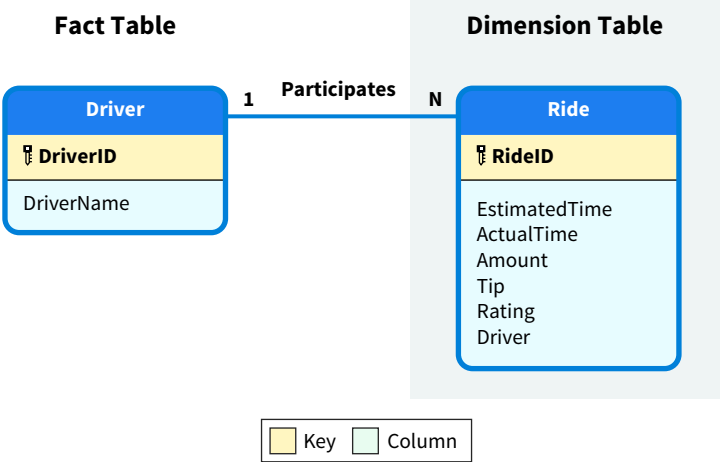


She would like to use the summary table shown here to show the overall cash flows and the cash flows for each of her three bank accounts.

BankName	StartingBalance	Inflow	Outflow	CustomerBalance
Fulton	\$ 23,009	\$ 953.34	\$ 1,148.36	\$ 22,813.98
MTB	\$ 89,012	\$ 2,116.00	\$ 900.23	\$ 90,227.77
PNC	\$ 44,099	\$ 2,408.04	\$ 2,987.36	\$ 43,519.68
Total	\$ 156,120	\$ 5,477.38	\$ 5,035.95	\$ 156,561.43

- 1. Create the information model.
- 2. Generate the report.

EX 6.12 (LO 2, 3) Data Managerial Accounting Create an Information Model for Employee Performance Analysis Carousel is a startup ride-share company that aims to compete with companies like Uber and Lyft in the U.S. Using heavy local advertisement, they ran a pilot program during the first weekend of February in the Seattle area. The diagram shows the data they collected.



The data dictionary provides a brief description for each of the fields.

Ride	
Name	Description
RideID	An ID that uniquely identifies a ride.
EstimatedTime	The estimated time of the ride in minutes.
ActualTime	The actual time of the ride in minutes.
Amount	The dollar amount paid for the ride.
Tip	The dollar amount tip received by the driver for a ride.

(Continued)

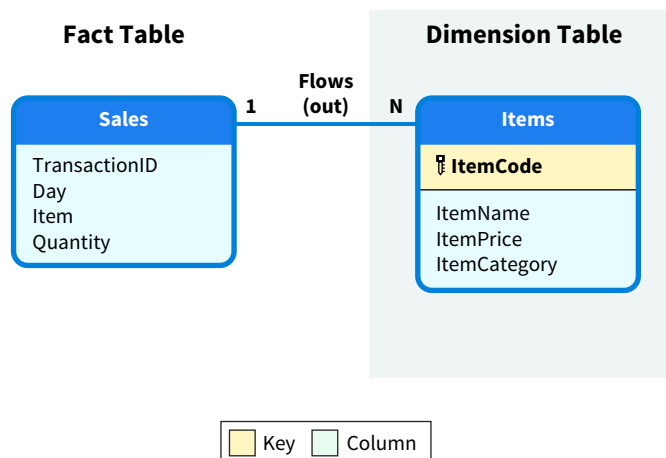
Name	Description
Rating	The customer's rating for the ride on a scale of one to five, with five being excellent.
Driver	The driver's ID.

Driver

Name	Description
DriverID	An ID that uniquely identifies a driver.
DriverName	The name of the driver.

1. Carousel's CEO asks you to develop an information model that will help analyze driver performance. Develop at least three measures.
2. Next, generate at least two reports that use the information model you developed.

EX 6.13 (LO 2, 3) Data Financial Accounting Create an Information Model for Revenue Analysis Jane Jones (JJ) just opened a small food truck, Gourmet@5, on Fifth Avenue in New York City. The menu is simple, offering hamburgers, hot dogs, water bottles, and a variety of soda and candy bars. She gives you a spreadsheet that contains all the transactions for the first week from Monday through Sunday. The diagram shows what data was collected.



The data dictionary provides a brief description for each of the fields.

Sales

Name	Description
TransactionID	An ID used to identify the transaction.
Day	The day of the week the transaction occurred.
Item	The item sold to the customer (code).
Quantity	The quantity sold of the item.

Items

Name	Description
ItemCode	A code that uniquely identifies a type of product.
ItemName	The name of the item (product type).
ItemPrice	The item's price. All items have a fixed price.
ItemCategory	The item's category.

1. Develop an information model that will help analyze the first week's sales. The owner is especially interested in knowing which products generate the most revenue.
2. Generate at least two reports using the information model you created.

EX 6.14 (LO 2, 3) Data Create an Information Model for Employee Performance Analysis
Giuseppina’s is a popular, family-owned, pizza store in Bourbon, Missouri. The store’s owner has collected the following information regarding all deliveries on February 14, 2025: order number, driver, order amount, and tip amount received by the driver. The owner would like to rank the drivers based on two criteria:

- Total tip amount received.
 - Average tip percentage received.
1. Develop an information model for such an analysis.
 2. Use the information model to create a table for each ranking that shows the names of the drivers.

Professional Application Case: D*Tunes



D*Tunes is a ballroom studio in Pocatello, Idaho. The studio’s owner has collected data regarding the studio’s sales for lessons in January 2025. You are interning with them as an accountant. The owner has a series of questions that can be answered using data that has already been given to you. You have already developed the data dictionary and the data model for the data set.

D*Tunes Data Dictionary

Instructors

Column Name	Field Description
InstructorID	An ID that uniquely identifies an instructor.
InstructorName	An instructor’s name.
NumberOfHoursTaught	The number of hours an instructor has taught for D*Tunes.
NumberOfAwards	The number of local and regional awards an instructor has received during the last three years.
NumberOfNationalAwards	The number of national recognitions an instructor has received.
Salary	For full-time employees, the salary the instructor earns.
JobTitle	An instructor’s job title.

Classes

Column Name	Field Description
ClassCode	A code that uniquely identifies a class.
ClassDescription	A brief description of the class.

Students

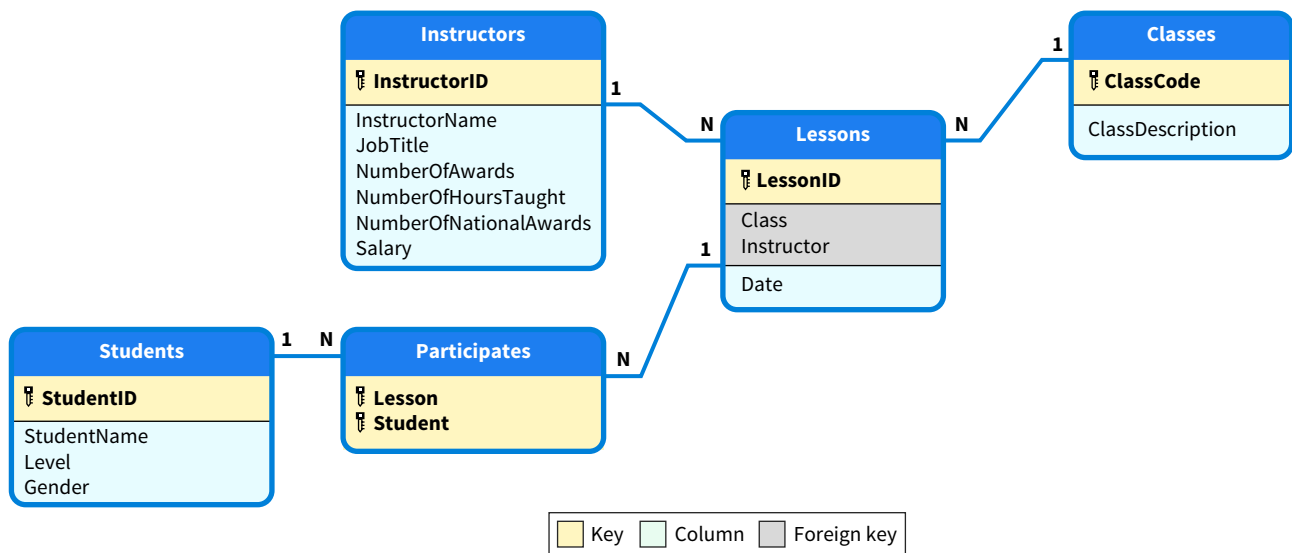
Column Name	Field Description
StudentID	An ID that uniquely identifies a student.
StudentName	A student’s name.
Level	Student’s Skill Level: B = Beginner I = Intermediate A = Advanced
Gender	A student’s gender.

Lessons

Column Name	Field Description
LessonID	An ID that uniquely identifies a lesson.
Date	The date on which the lesson occurred.
Class	The type of class taught during this lesson.
Instructor	The ID of the instructor teaching this class.

Participates

Column Name	Field Description
Lesson	The ID of the lesson taken by the student.
Student	The ID of the student taking the lesson.



PAC 6.1 Financial Accounting: Analyze D*Tunes' Profitability

Data **Financial Accounting** D*Tunes has four categories of instructors:

- Apprentice.
- Intermediate.
- Advanced.
- Champion.

The following rules are used to assign instructors to their category.

- Instructors who have taught less than 250 hours are apprentices.
- Instructors who have taught more than 250 hours are considered intermediate.
- Intermediate instructors who have won at least three awards, from a list of approved local and regional tournaments, during the last three years, are considered advanced.
- Instructors who have earned national recognition, from a list of highly prestigious awards, are classified as champions.

Instructors' pay is determined by their category:

Category	Hourly Rate
Apprentice	\$ 45
Intermediate	\$ 65
Advanced	\$ 85
Champion	\$ 110

This hourly rate applies to any class an instructor teaches: initiation, private, or group. The only exception is the Friday evening parties, for which an instructor is paid \$250. The price paid by students is the instructor's hourly rate plus \$30. So, for a one-hour lesson with an apprentice, the student pays \$75. For a one-hour lesson with a champion, the students pays \$140. The standard rate for any group lesson (swing, cha-cha, etc.) is \$40 per person. Admission to the weekly Friday party is \$25. Every student receives a free initiation lesson.

1. Develop an information model that enables you to determine the gross profit for last month. When calculating gross profit, the only cost to consider is what the owner pays the instructors. Do not include the salaries paid to full-time employees.
2. Create a report that shows the total gross profit generated, gross profit generated per class, and gross profit generated per instructor.

PAC 6.2 Auditing: Analyze Potential Fraud

Data **Auditing** D*Tunes' policy is that potential customers can have a free initiation lesson. However, instructors are still paid for these lessons. The owner does not want to change this policy, but understands that it creates an opportunity for fraud. You are asked to analyze the following four questions.

Build an information model to explore each question. Then, create a report for each of them.

1. What is the cost of the initiation lessons. What investment has been made thus far?
2. Are there students that took more than one initiation lesson?
3. Are there instructors with an unusually high number of initiation lessons?
4. What is the conversion rate? That is, how many students who signed up for an initiation lesson also eventually paid for a lesson or a Friday night party?

PAC 6.3 Managerial Accounting: Create a Breakeven Analysis of Group Classes

Data **Managerial Accounting** D*Tunes policy is to keep group classes only when there is a net revenue of at least \$150 after paying the instructor. Currently, the price paid by students for a group class is \$40. Explore the following two questions.

1. Create an information model to determine how many students are required per lesson to breakeven, and which lessons have not yet reached that goal.
2. Generate a report that shows which classes are not meeting the breakeven point. What classes would you suggest canceling?

PAC 6.4 Tax Accounting: Determine Withholdings for Salaried Employees

Data Tax Accounting For the two salaried employees, D*Tunes must withhold the employer portion of FICA taxes every month from their wages. FICA tax is a combination of Social Security tax and Medicare tax. The 2025 FICA tax rate is 7.65%, which includes Social Security tax of 6.2% and Medicare tax of 1.45%. The owner does not need to withhold Social Security tax for any earnings above \$168,900. Therefore, the studio must only withhold Medicare tax on wages above this amount. Create an information model that will allow you to design a worksheet similar to the one shown that will help the studio determine the FICA tax payments for each month.

		Month	Deduction
Salary	200,000	January	1,275.00
FICA%	0.0765	February	1,275.00
SocialSecurity%	0.062	March	1,275.00
Medicare%	0.0145	April	1,275.00
SocialSecurityMaximum	168,900	May	1,275.00
		June	1,275.00
		July	1,275.00
		August	1,275.00
		September	1,275.00
		October	1,275.00
		November	380.13
		December	241.67

LeGrind Continuing Case: Build the Information Model for Calculating Gross Profit

Data Visit Wiley's online learning platform for the case background, additional questions, data, and more details about the continuing case.

